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BMEN502 | SUMMER 2026 | GROUP ASSIGNMENT

Smart Multimodal Flexible Prosthetic Liner for Real-Time Discomfort-Risk Monitoring

A scalable array of 5.2 mm silicone sensor slabs for spatial monitoring of pressure, shear, temperature, and humidity.



Course
BMEN502 - Concept Exploration and Testing

Academic Year
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Instructor
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QUESTIONS 1–4 | PROTOTYPE STRATEGY, VALIDATION, AND REQUIREMENTS

Q1. Questions to Be Addressed Through Prototyping

The system under development is a flexible, multimodal prosthetic liner formed from multiple 5.2 mm silicone sensor slabs. Each slab combines sensing elements for pressure, shear, temperature, and humidity. The following questions were selected because their answers would reduce the most consequential technical, clinical-use, integration, and manufacturing risks before development proceeds to a complete wearable liner.

No.	Prototype question	Rationale for selection
1	Can pressure, shear, temperature, and humidity sensors be successfully integrated into the 5.2 mm silicone slab while producing stable and interpretable signals? FOCUS: Overall technical feasibility	• Integration and signal quality: Confirms that all four sensor types can coexist within the 5.2 mm slab, while checking sensitivity, signal separation, and stability after integration. • Decision: Determines whether the overall concept is technically feasible.
2	Can the humidity sensors detect moisture changes through a protective layer without being damaged or saturated by liquid sweat? FOCUS: Environmental protection and vapour access	• Vapour access and protection: Verifies water-vapour access through the protective layer while testing resistance to sweat saturation, leakage, and permanent damage. • Stability: Measures drift and recovery after moisture exposure.
3	Can the integrated slab detect controlled conditions representing slipping, perspiration, thermal buildup, and potentially harmful localized loading? FOCUS: Bench and simulated-use detection	• Response and detection: Links each controlled condition to its expected sensor response and tests spatial and temporal detection of slip, moisture, heat, and loading. • Evidence: Builds bench evidence before human-subject testing.
4	Can the prototype be manufactured using the available materials and equipment at a reasonable and scalable cost? FOCUS: Manufacturability and cost	• Constraints and production: Identifies material, equipment, alignment, and assembly constraints while estimating fabrication cost, labour, and repeatability. • Scalability: Determines whether production can scale to a complete liner.
5	Can the software accurately combine pressure, shear, temperature, and humidity signals from multiple 5.2 mm sensor slabs, assign each measurement to its correct liner coordinates, and generate an interpretable real-time heatmap of interface-condition or discomfort-risk hotspots? FOCUS: Signal fusion, coordinate mapping, and visualization	• Spatial accuracy: Verifies sensor-to-coordinate mapping and hotspot localization. • Fusion and performance: Tests normalization and fusion across four sensing modes and multiple slabs, while assessing real-time speed and stability without hiding sensor disagreement.
6	Are the current number and locations of the sensors sufficient to detect localized pressure, shear, temperature, and moisture hotspots throughout the monitored liner area? FOCUS: Sensor density and spatial coverage	• Localization and coverage: Compares known stimulus positions with reconstructed hotspots, revealing coverage gaps, spatial blur, and location errors. • Design decision: Guides the final sensor count and placement.
7	How much calibration drift occurs after prolonged exposure to body temperature, humidity, repeated compression, shear, and bending? FOCUS: Durability and measurement stability	• Drift under stress: Quantifies changes in baseline, sensitivity, hysteresis, and repeatability while evaluating body heat, humidity, compression, shear, and bending. • Maintenance decision: Establishes stability limits and recalibration needs.

Evidence anchor. Progress-report question set. Its measurement boundaries are anchored to published prosthetic-interface sensing and review evidence [13, 19, 14]; the wording is not copied from those sources. Use *interface-condition heatmap* until the displayed score is validated against user-reported discomfort or clinically justified limits.



Figure Q1-1. Prototype evidence chain. Each question removes a defined technical risk before subsequent participant-specific or clinical interpretation.

Q2. Functional-Block Literature and Development Map

The assignment requires more than a general demonstration that the concept can work: it asks which functional blocks already have reasonable solutions that can be leveraged, which blocks still require project-specific integration, and where the strongest technical uncertainty remains. The comparison below treats published prosthetic-interface systems and printed-sensor studies as **similar or competing research solutions**. It does not claim that each paper reproduces the complete liner described in this report.

2.1 Similar and competitive solutions by functional block

Functional block	Closest published solution	Reasonable solution to leverage	Project-specific integration work
Pressure and directional shear	Four partially overlapping printed capacitors separate normal and signed shear response [1, 2]; textile arrays demonstrate distributed socket-pressure measurement [23].	Reuse the four-sector cell principle, differential capacitance readout, and calibrated directional loading method.	Print and protect the cell on the 5.2 mm PDMS slab, quantify cross-axis error, and preserve its physical coordinate in the multimodal map.
Temperature	A printed silver meander provides a repeatable resistance–temperature relationship on a flexible substrate [3].	Reuse the serpentine resistive geometry and temperature-calibration framework.	Separate temperature response from bending, compression, conductor strain, and self-heating after direct printing on PDMS.
Humidity and sweat route	An all-inkjet-printed interdigitated capacitive sensor measures relative humidity through a hygroscopic layer [5].	Reuse the IDE geometry, capacitance measurement, and vapour-step characterization.	Provide vapour access while limiting liquid sweat, leakage, saturation, slow recovery, and temperature dependence inside the liner.
Printed PDMS conductors	Plasma-assisted AgNP printing and partially embedded silver traces establish two direct-on-PDMS process routes [6, 7].	Use surface treatment, wide-trace process windows, and bending/strain characterization as starting methods.	Determine the actual print window, swelling, adhesion, geometry, curing, and durability for the project's ink, printer, sensor artwork, and molded PDMS.
Sensorized liner integration	Embedded temperature/humidity sensing and a wireless multimodal skin-interface platform show that prosthetic-liner sensing can be distributed and synchronized [13, 19].	Reuse staged integration, channel identification, synchronized acquisition, and post-assembly calibration.	Combine all four printed modalities in thin slabs, route connections through a molded liner, and retain calibration under curvature, pre-strain, donning, and repeated loading.
Spatial interpretation	Socket pressure maps and movement/pistoning systems demonstrate spatial display and independent motion ground truth [21, 20, 28].	Reuse coordinate registration, known-location stimulation, interpolation checks, and movement reference measurements.	Fuse pressure, shear, temperature, and humidity without hiding modality-specific error; report localization error before interpreting a map as discomfort risk.
Skin interface and bench model	Hydrogel–elastomer bonding, hydrogel sweat collection, and prosthetic socket phantoms provide separate material and test precedents [11, 12, 22, 25].	Reuse established bonding concepts, controlled moisture comparisons, and mechanically repeatable phantom loading.	Compare bare and patterned skin-facing structures without blocking or redistributing the signals, then transfer passing slabs to a curved liner geometry.

Evidence anchor. Functional-block comparison synthesized from printed-sensor, sensorized-liner, visualization, interface-material, and prosthetic test-rig literature. Each project-specific statement is bounded to work that remains to be demonstrated on this design.

Novelty boundary: The individual sensor principles, PDMS printing methods, hydrogel bonding concepts, and spatial displays are **not treated as independently novel**. The candidate project contribution is their coordinate-preserving integration as directly printed pressure, shear, temperature, and humidity blocks on modular 5.2 mm PDMS slabs, followed by multimodal calibration and liner-level mapping. A formal novelty claim would still require a dedicated patent and systematic prior-art search.



Figure Q2-F1. Functional-block strategy. Existing solutions are used as engineering starting points; the work concentrates on the interfaces between blocks and on the evidence required after each mechanical boundary changes.

Q2. Work Completed and Current Build

2.2 What has been made and what is active now

Work already completed includes printing the full sensor artwork on a large PET sheet, molding the 5.2 mm PDMS slab, and developing the Fusion/AutoCAD sensor layouts and slab renders. PET was the available first-stage printing substrate, not a candidate liner material. The active build transfers the same functional blocks to the molded PDMS specimen, whose CAD envelope is approximately 56 mm × 55 mm × 5.2 mm.

Artifact or workstream	Present state	Role in the development sequence
Large PET process print	The complete sensor artwork has been printed on the available large PET sheet.	Records artwork transfer, printer operation, trace routing, connection access, and first-stage electrical inspection. It does not determine the liner substrate.
Molded PDMS slab	A nominal 5.2 mm PDMS specimen has been formed for direct printing.	Provides the intended flexible substrate for recording cure state, surface preparation, printed geometry, swelling, adhesion, and sensor response.
Sensor artwork	Pressure/shear capacitor geometry, resistive temperature meander, capacitive humidity IDE, conductors, and pads are available.	Defines the four established sensing blocks transferred to the PDMS print and later checked channel by channel.
CAD slab and placement design	Fusion/AutoCAD renders define the slab envelope, relative sensor arrangement, central enclosure region, and conductor exits.	Supplies the coordinate reference for the printed build; a final dimensioned export will distinguish nominal from measured dimensions.
Acquisition and mapping algorithm	Threshold logic and spatial-map generation are under development.	Receives calibrated channels only after sensor identity, units, time synchronization, and CAD coordinates are documented; algorithmic validation is addressed later in the report.

Evidence anchor. Project status derived from the completed PET print, molded PDMS specimen, supplied Fusion/AutoCAD geometry, slab renders, and stated software work. The literature supplies transfer methods and measurement precedents, not evidence that these project artifacts have already been calibrated [2, 3, 5, 6].

Drawing requirement: The next CAD export must label the coordinate origin, active footprints, conductor widths, connection pads, edge clearances, enclosure sections, and verified versus nominal dimensions. Perspective renders must not be used as dimensional evidence. The supplied render and placement blueprint are consolidated with the dimensional requirements and published liner-thickness context in Q4 Section 4.2.

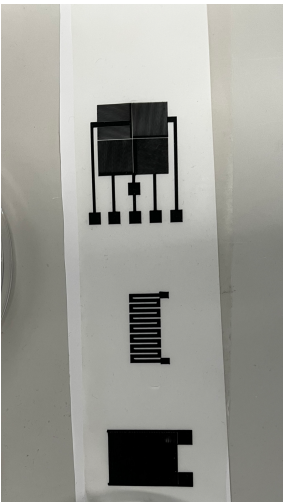


Figure Q2-W1. Sensors printed on PET during the first-stage fabrication trial. The visible layouts correspond, from top to bottom, to the multi-sector pressure/directional-shear electrode pattern, the resistive temperature meander, and the dense capacitive humidity IDE. This project photograph documents successful artwork transfer and printing on the available PET process substrate.

Evidence boundary. The photograph does not establish ink conductivity, calibration, sensitivity, channel isolation, or performance after direct printing on molded PDMS; those require electrical inspection and the modality-specific validation described later in the report.

Q2. Functional-Block Literature—Temperature and Humidity

2.4 Printed temperature sensing

Liew *et al.* printed serpentine silver resistors on PET and observed a linear resistance–temperature relationship from 30 to 100 °C [3]. Their reported design was approximately 6.9 mm × 13.2 mm, with 0.5 mm tracks and 0.4 mm gaps. Resistance also depends on printed geometry and mechanical strain, so the present meander requires temperature calibration while the slab is unloaded and under representative bending and compression.

Required outputs. Temperature coefficient, offset, response time, repeatability, strain error, self-heating check, and drift after cycling.

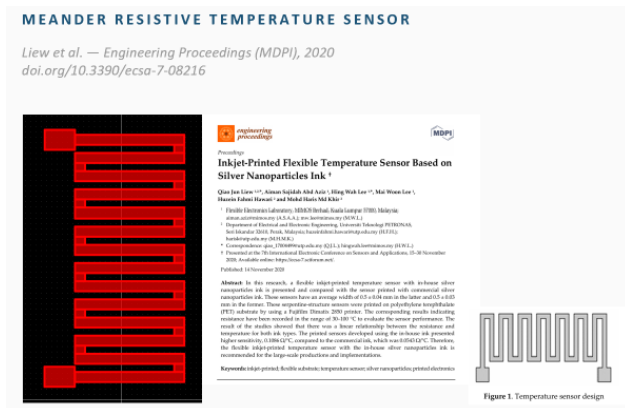


Figure Q2-S2. Temperature evidence plate. Project meander artwork and the published resistive temperature-sensor principle from Liew *et al.* [3].

2.4.1 Printed humidity sensing

Ait-Mammar *et al.* demonstrated an all-inkjet-printed capacitive relative-humidity sensor using interdigitated silver electrodes and a hygroscopic sensing layer [5]. The supplied plate connects the project IDE to the published geometry and micro-graph. The paper supports the capacitive mechanism, but it does not prove liquid-sweat resistance or the response of this device inside a prosthetic liner.

Required outputs. Vapour-response curve, temperature compensation, response and recovery time, saturation, liquid-exposure survival, hysteresis, drift, and repeatability after drying.

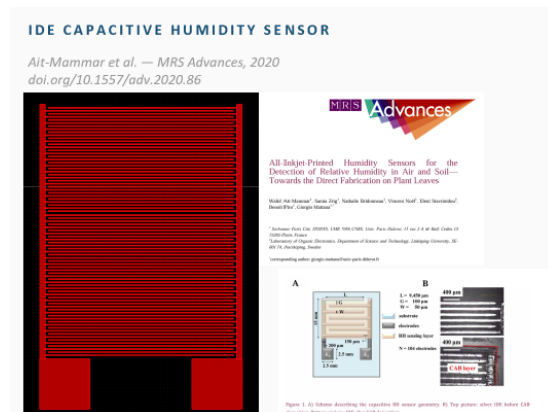


Figure Q2-S3. Humidity evidence plate. Project IDE artwork and published capacitive humidity-sensor evidence from Ait-Mammar *et al.* [5].

Consensus, difference, and gap: Printed resistive meanders and capacitive IDEs are established solutions for temperature and humidity, respectively. Their functional layers and measurands differ: the temperature trace must reject strain effects, whereas the humidity IDE requires a controlled vapour route and liquid-sweat protection. The project therefore reuses the two sensor principles while developing their PDMS-specific calibration, protection, cross-sensitivity control, and common coordinate mapping.

Q2. Direct-on-PDMS Printing Literature

2.5 Process routes, transfer limits, and project controls

Two direct-printing studies provide process solutions that can be leveraged when transferring the artwork from the successful PET print. Albrecht *et al.* used oxygen-plasma-treated silicone and inkjet-printed AgNP conductors; Sun *et al.* printed a silver precursor onto precured PDMS to create a partially embedded, semi-wrapped conductor [6, 7]. The routes agree that PDMS surface and mechanical behaviour must be designed into the print process, but they use different chemistries and do not supply calibration for the present sensor stack.

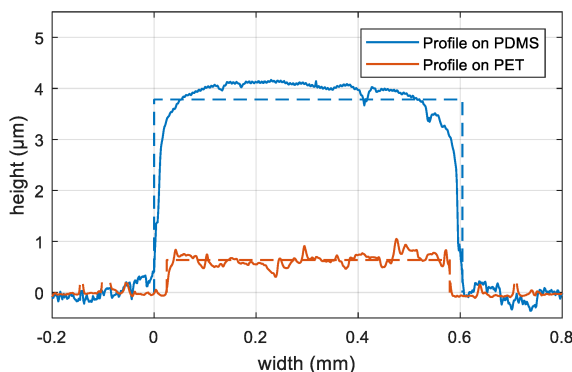


Figure Q2-P1. Same-printer height profiles. Approximate average heights were $3.8\ \mu\text{m}$ on PDMS and $0.6\ \mu\text{m}$ on PET. The apparent PDMS height includes substrate swelling, not only deposited silver [6]. CC BY 4.0.

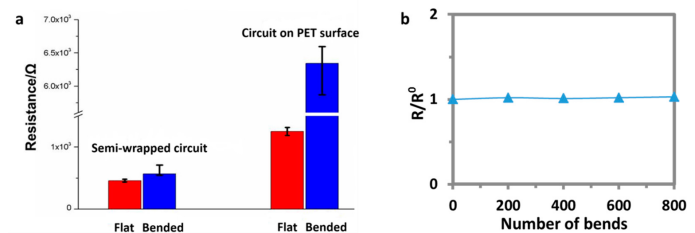


Figure Q2-P2. Bending comparison. Sun *et al.* reported a semi-wrapped PDMS conductor with substantially better resistance retention during bending than the paper's PET comparison, including repeated bending shown by the normalized resistance [7]. CC BY 4.0.

Literature finding	What it means	Project control
Surface wetting	Untreated PDMS is difficult to wet. In the Albrecht process, oxygen plasma enabled continuous lines and printing occurred within 30 min of treatment.	Record treatment power/time, contact angle if available, and the treatment-to-print interval; do not assume the paper's recipe transfers unchanged.
Print fidelity	The paper reports satellite drops from printer inaccuracy and local failures in narrow lines; 1–2 mm lines were selected for reliable stretchable conductors in that reported process.	Measure printed width, edge roughness, gaps, shorts, open traces, and yield for the project's actual sensor geometries.
Solvent interaction	The same deposited ink produced a much larger apparent profile on PDMS. The authors attributed most of the difference to solvent absorption and swelling within roughly the upper $15\ \mu\text{m}$.	Profile untreated, wet-printed, dried, and equilibrated specimens; separate conductor thickness from substrate deformation.
Mechanical survival	Printed silver remained conductive after more than 10,000 reported 20% stretch cycles, while the semi-wrapped approach maintained resistance through repeated bending.	Test the present ink and geometry under compression, bending, and shear; record in-cycle resistance, recovery time, drift, cracking, and adhesion.
Measurement accuracy	These papers demonstrate conductors and process strategies, not accurate pressure, shear, temperature, and humidity sensing in this slab.	Establish modality-specific calibration, error, hysteresis, repeatability, cross-sensitivity, drift, and localization after printing.

Evidence anchor. Literature-derived PDMS printing constraints and project controls from Albrecht *et al.* and Sun *et al.* [6, 7].

Method selected from the literature: Use direct printing on the molded PDMS slab, beginning with documented surface treatment and a print-window study. PET remains process evidence only. The Albrecht and Sun routes supply starting controls for wetting, trace width, swelling, adhesion, bending, and cycling; the final recipe and all sensor calibration are established on the project's own ink–printer–PDMS combination.

Q2. Phase 3: Hydrogel Skin-Interface Study

2.6 Literature evidence and the sensor-interference problem

This is an **evidence-led development stage following successful printing and calibration on the molded PDMS slab**. A continuous hydrogel between skin and sensors can improve softness or hold moisture, but it can also alter every quantity being measured. It is therefore treated as a separate experimental variable rather than assumed to be a neutral comfort layer.

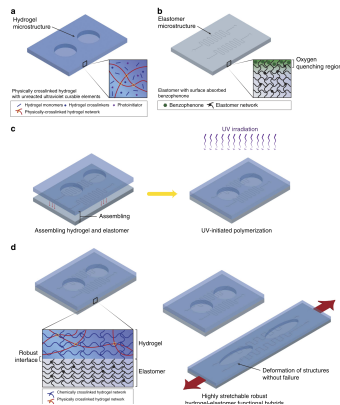


Figure Q2-H1. Hydrogel–elastomer bonding precedent. Yuk *et al.* preserved microstructures while covalently bonding tough hydrogels to elastomers including Sylgard 184 PDMS [11]. The figure supports a bonding method, not prosthetic-liner sensor accuracy. CC BY 4.0.

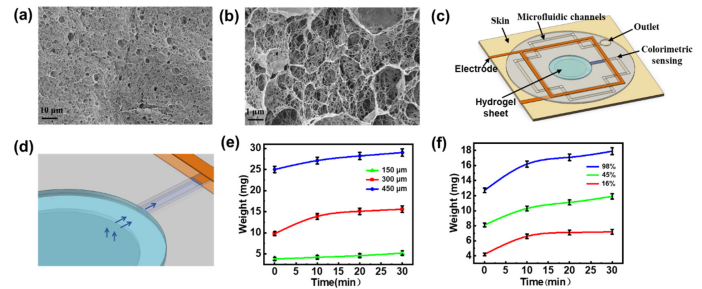


Figure Q2-H2. Hydrogel thickness and sweat uptake. Lu *et al.* compared agarose–glycerol sheets of different thicknesses; water uptake and equilibration changed with thickness [12]. This was a wearable patch, not a load-bearing prosthetic liner. CC BY 4.0.

Evidence theme	What the literature supports	What remains unproven here
Bonding to PDMS	Robust covalent interfaces and preserved microstructures are possible with specific surface chemistry and UV processing [11].	Compatibility with the printed silver, sensor films, curing history, cleaning, skin contact, and the project's fabrication equipment.
Moisture uptake	Hydrogel sheets can collect sweat, and uptake depends on thickness and time [12].	Whether uptake prevents pooling or instead stores sweat beside the skin; release, drying, hygiene, and repeat-cycle behaviour are unknown.
Sensor placement	A personalised liner study embedded temperature and humidity sensors in direct skin contact and found that integration geometry is central to interpretation [13].	The response after inserting a water-rich intermediate layer cannot be inferred from direct-contact data.
Material mechanics	Swollen hydrogels are compliant and can change dimensions; Yuk <i>et al.</i> treated interface toughness and microstructure preservation as design requirements [11].	Pressure transmission, shear transfer, friction, swelling-induced preload, spatial blur, and cyclic durability in this slab.

Evidence anchor. Literature synthesis of hydrogel–elastomer bonding, sweat uptake, and direct-contact liner sensing [11, 12, 13].

Literature conclusion: The literature supports a carefully gated hydrogel experiment, not immediate adoption of a continuous skin-facing layer. No reviewed paper validates this hydrogel arrangement over the complete combination of pressure, directional shear, temperature, and humidity sensors in a prosthetic liner.

Q2. Phase 3: Hydrogel Study—Continued

2.6.1 Sensor interactions introduced by the hydrogel

Modality	Likely interference mechanism	Required paired test
Humidity	Water absorption and diffusion can buffer a rapid humidity change, delay recovery, create hysteresis, or saturate the environment near the IDE. Liquid or ionic pathways may also increase leakage.	Compare H0 bare PDMS, H1 islands, the H2 windowed sheet, and H3 continuous hydrogel under vapour steps, controlled sweat volume, drying, and repeated wet/dry cycles.
Temperature	Added thermal mass, thermal resistance, absorbed water, and evaporation can change both the steady temperature and response time at the skin interface.	Apply identical temperature steps and heat fluxes; report offset, time constant, spatial smoothing, and recovery for dry and wet states.
Pressure	A soft swollen layer can redistribute normal stress, change contact area, attenuate peaks, introduce preload, and add viscoelastic hysteresis.	Use known loads and footprints; compare sensitivity, linearity, hysteresis, drift, peak attenuation, and hotspot localization before and after swelling.
Directional shear	Friction, adhesion, slip, and strain transfer change with water content. A continuous layer may reduce, rotate, or spread the shear reaching the four-capacitor cell.	Apply signed shear in both axes at controlled normal preload; quantify vector error, cross-axis coupling, slip, and spatial blur in dry and wet conditions.
Printed silver	Water, ions, swelling strain, and bonding chemistry may affect insulation, adhesion, continuity, corrosion, or contact resistance.	Measure leakage, resistance, insulation, adhesion, cracking, and recovery through cyclic loading and wet/dry exposure.

Evidence anchor. Sensor-interference analysis derived from the material transport and bonding evidence reviewed in Section 2.6 [11, 12, 13].

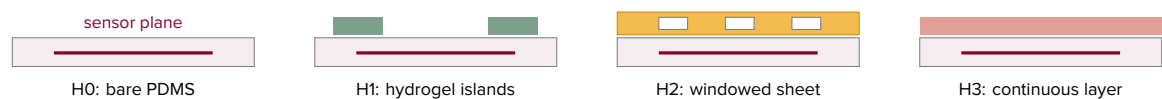


Figure Q2-H3. Evidence-informed hydrogel comparison. H0 is the required calibration baseline; H1 uses separate pads, H2 is one connected sheet with aligned sensor windows (white cut-outs), and H3 covers the sensor regions. At the humidity location, an H2 window would use a hydrophobic vapour-permeable barrier rather than an open liquid path. The diagram is not to scale.

2.6.2 Selected structures and decision gate

Option	Structure	Purpose	Decision
H0	Bare printed PDMS with minimum protective geometry.	Establish the reference calibration and the fastest possible transfer of each stimulus to its sensor.	Mandatory baseline. Test first.
H1	Patterned hydrogel islands outside active sensor footprints and conductor exits.	Test skin feel and local moisture uptake while minimizing direct interference with sensing zones.	Recommended first hydrogel experiment. Compare directly with H0.
H2	Connected windowed hydrogel sheet covering the spaces between sensors; aligned cut-outs expose pressure, shear, and temperature footprints, while a hydrophobic vapour-permeable vent protects the humidity route.	Explore broader skin coverage without placing wet gel directly over active sensing zones.	Test only if H1 passes; reject H2 if window edges redistribute load, trap liquid, or shift calibration beyond limits.
H3	Continuous hydrogel over all sensor regions.	Maximum continuous soft contact, but maximum risk of buffering and cross-sensitivity.	Not recommended for the first build. Activate only after H0–H2 evidence.

Evidence anchor. Selected H0–H3 implementation sequence. A bonded hydrogel/PDMS interface is literature-supported [11]; its effect on this multimodal slab remains experimental.

Phase 3 pass condition: Select a hydrogel configuration only if it has a repeatable PDMS bond, acceptable swelling and drying, no damaging liquid path to the silver conductors, and predefined limits for humidity delay, temperature lag, pressure attenuation, shear-vector error, hysteresis, drift, and localization. Otherwise retain the bare calibrated PDMS interface.

Q2. Whole-Liner Integration Pathway

2.7 From the validated slab to a molded silicone liner

The whole liner is the subsequent systems-integration stage. Printing and calibrating the flat PDMS slab first isolates substrate and sensor problems before curvature, pre-strain, cable routing, user fit, and repeated donning are introduced. The complete liner will be assembled from validated sensor regions rather than assuming that a successful flat print can be copied directly onto a curved surface.

Integration route	Advantage	Main risk and gate
Validated PDMS sensor tiles placed during molding	Preserves a tested printing surface and allows defective tiles to be rejected before liner assembly.	Tile edges, bonding, local stiffness, wiring exits, and strain transfer must be tested under liner-like curvature and compression.
Printed flat PDMS sheet formed or laminated into the liner	Keeps printing on a planar surface while enabling a larger connected array.	Forming can impose pre-strain, wrinkle conductors, move coordinates, and change calibration.
Direct printing on a curved cured liner	Reduces assembly interfaces in principle.	Most difficult first route: printhead clearance, drop landing, wetting, focal distance, curing, and inspection vary with curvature.
Recommended sequence	Calibrate flat sensor slabs or tiles, test them under known curvature, then integrate only passing units into a molded liner.	Recalibrate after integration and compare with the original flat baseline.

Evidence anchor. Integration routes selected for comparative implementation. Published personalized-liner work supports staged sensor integration and post-assembly testing, not any one of these manufacturing routes [13].



Figure Q2-L1. Whole-liner evidence ladder. Coordinate accuracy and calibration are rechecked whenever the mechanical boundary condition changes.

FIGURE PLACEHOLDER

Dimensioned printed-slab section

Add the final orthographic plan and cross-section. Label printed layers, active sensor footprints, protective PDMS, conductor exits, coordinate origin, nominal 5.2 mm thickness, and verified dimensions.

FIGURE PLACEHOLDER

Molded-liner integration design

Show validated sensor slabs or tiles placed at defined liner coordinates, routing toward the acquisition module, curvature, molding boundaries, and the sequence of post-integration calibration tests.

Stage boundary: The CAD liner render defines the integration-stage design. It becomes works-like evidence after the flat printed PDMS slab passes its electrical, sensor, environmental, mechanical, and coordinate gates.

Q2. Literature Synthesis and Implementation Roadmap

2.8 Reused solutions, completed work, and integration focus

Literature-backed block	Project evidence available now	System-specific implementation output
Printed sensor topologies	Pressure/shear, temperature, and humidity artwork has been created and printed together on the large PET process sheet.	Transfer the established geometries to PDMS and produce separate calibration, cross-sensitivity, repeatability, and drift records for every channel.
Direct PDMS conductors	A 5.2 mm PDMS slab has been molded; two published printing routes define starting controls for wetting and mechanical survival [6, 7].	Record the project's surface treatment, print window, profile, continuity, adhesion, curing, swelling, and cyclic electrical response.
Coordinate-aware sensor layout	Fusion/AutoCAD slab renders and a placement blueprint define relative sensor positions; the acquisition and mapping algorithm is under development.	Export a dimensioned coordinate table, bind each electrical channel to one sensor and position, and verify the mapping with known-location stimuli.
Sensorized liner systems	Embedded and wireless prosthetic-interface studies provide precedents for synchronized distributed sensing [13, 19].	Preserve channel identity through flat-slab, curvature, molding, connection routing, and post-integration recalibration.
Skin-facing material system	Hydrogel bonding and sweat-collection studies identify workable material methods and important thickness/time effects [11, 12].	Compare bare PDMS with patterned structures only after the bare slab is calibrated; retain the structure that preserves mechanical, thermal, and humidity measurements.
Spatial interpretation and bench model	Published socket maps, movement references, and mock limbs provide visualization and controlled-test precedents [21, 28, 22, 25].	Quantify localization, spatial blur, update stability, and modality disagreement on a repeatable phantom before assigning a user-facing discomfort-risk interpretation.

Evidence anchor. Synthesis of the functional-block literature in Sections 2.3–2.7 and the project artifacts documented in Section 2.2. The right-hand column states implementation outputs rather than completed results.

2.9 Research-backed implementation roadmap

Stage	Build or study	Evidence transition	Status
A	Functional-block and closest-solution map	Identify the established solution used for each block and bound the integration gap.	Completed in this review
B	Full artwork printed on PET	Preserve the print record, continuity results, defects, routing, and connection access as process evidence.	Completed build
C	Molded 5.2 mm PDMS slab and direct print	Establish the actual surface treatment, printed geometry, swelling, adhesion, electrical integrity, and yield on the intended substrate.	Active build
D	Separate sensor calibrations	Generate pressure/shear matrices, temperature calibration, humidity response/recovery, strain effects, and cross-sensitivity before signal fusion.	Follows the stable PDMS print
E	Multimodal slab on controlled phantom	Apply known single and combined stimuli; measure repeatability, drift, localization, spatial blur, and incorrect channel activation.	Follows separate calibration
F	Patterned skin-interface comparison	Compare H0 bare PDMS with H1 islands and the H2 windowed sheet for moisture handling, transfer error, drying, adhesion, and cycling.	Follows the bare baseline
G	Curved liner and spatial software integration	Recheck coordinates and calibration after curvature/molding, then connect the verified channels to the threshold and spatial-map pipeline.	System-integration stage

Evidence anchor. Roadmap ordered from completed artifacts to substrate transfer, modality calibration, controlled multimodal testing, skin-interface comparison, and liner/software integration. Phantom and localization methods are informed by [22, 25, 21, 28].

Q2 conclusion: The literature does not justify re-inventing or claiming novelty for the individual sensor mechanisms. It supports a build strategy that leverages established pressure/shear, temperature, humidity, PDMS-printing, liner-sensing, and spatial-interpretation solutions. The work specific to this project is the measured integration of those blocks on the 5.2 mm PDMS slab and then in a coordinate-preserving molded liner. The completed PET print, molded PDMS specimen, sensor artwork, and CAD layouts are reported as completed work; later roadmap stages are not presented as measured results.

Q3A. Selection of the Critical Functional Blocks

Concept-exploration question: Can the integrated slab detect controlled conditions representing slipping, perspiration, thermal buildup, and potentially harmful localized loading?

A functional block was retained when it (1) transforms a physical input into a distinct and testable output, (2) contains uncertainty that is not resolved by the literature for this directly printed 5.2 mm PDMS slab, (3) can fail independently of the other blocks, and (4) would invalidate a downstream claim if it failed. This method keeps the decomposition tied to technical feasibility rather than to arbitrary component names. Published solutions are leveraged where they already establish a mechanism; project-specific proof is reserved for transfer to the present geometry, material stack, fabrication process, and coordinate system [2, 3, 5, 19].

ID	Functional block	Why it is isolated	First technical-feasibility question
FB1	Interface mechanics and contact transfer	Every sensor depends on how the compliant slab transfers concentrated and distributed loading.	Does the 5.2 mm PDMS geometry preserve a usable local response without unacceptable spatial spreading or mechanical cross-coupling?
FB2	Normal-pressure and directional-shear transduction	Force magnitude and direction arise from the same deformable interface but require separable outputs [2].	Can normal and two-axis tangential loading be recovered over the selected range after integration?
FB3	Temperature transduction	Printed silver resistance is temperature sensitive, but bending, strain, encapsulation, and thermal mass can alter the response [3, 19].	Does the embedded element measure local temperature with adequate accuracy, response, and recovery?
FB4	Vapour humidity and liquid-sweat interface	Vapour access is required, whereas direct liquid exposure can saturate or damage the sensing route [5, 13].	Can RH change and liquid exposure be distinguished while the protected route remains recoverable?
FB5	Printed-PDMS circuit integrity and encapsulation	Wetting, swelling, adhesion, cracking, leakage, and cyclic strain are material/process risks [6, 7].	Do the printed conductors and insulation remain continuous and stable after environmental and mechanical exposure?
FB6	Signal conditioning, calibration, and synchronization	Multimodal channels require independent calibration and a common time base before their outputs can be compared [16, 19].	Can raw channels be converted into traceable physical units without timing or cross-talk errors?
FB7	Slip and relative-motion discrimination	Shear is a force condition; slip is relative motion and therefore needs an independent displacement reference [28, 16].	Can true translation be distinguished from high shear without motion?
FB8	Coordinate registration and multimodal interpretation	Measurements become useful only when mapped to the correct slab location while retaining modality-specific evidence [21].	Can event type and location be recovered without misleading interpolation or averaging?

Evidence anchor. Functional blocks selected from the concept's physical signal path and from unresolved transfer risks identified in the sensor, prosthetic interface, and printed-PDMS literature [2, 3, 5, 6, 16, 28, 21].

Selection result: Eight blocks are retained, exceeding the six required by the assignment. FB1–FB6 establish whether the slab can produce traceable measurements; FB7–FB8 establish whether those measurements support the stated condition and coordinate. A failure in an upstream block is not averaged into a downstream software score.

Q3B. Definition and Boundaries of the Functional Blocks

Each block below has an explicit input, transformation, and output and is tied to the engineering discipline that governs its dominant uncertainty. The boundaries prevent one successful subsystem from being used as evidence for a different subsystem. The validation fixtures and reference instruments are not product blocks; they are the independent means used later to examine the eight product blocks.

Block	Input → transformation → output	Dominant discipline	Boundary of responsibility
FB1: interface transfer	Contact force, heat, and moisture → PDMS deformation/diffusion → condition delivered at an active region.	Mechanical engineering; transport phenomena.	Ends at the local stimulus reaching the sensor plane; it does not include electrical conversion.
FB2: pressure/shear	Normal and tangential stress → four-capacitor geometry → calibrated F_z , F_x , F_y or stresses.	Mechanical and electrical engineering.	Includes axis separation and mechanical cross-talk; excludes deciding whether motion occurred.
FB3: temperature	Local heat flux/temperature → resistance change → calibrated temperature trace.	Thermal and electrical engineering.	Includes thermal lag and strain sensitivity; excludes clinical interpretation of temperature.
FB4: humidity/sweat route	Water vapour or liquid exposure → protected dielectric response → RH-related signal plus liquid-state flag.	Materials science; electrical engineering.	Includes vapour access, saturation, drying, and protection; RH is not treated as sweat rate.
FB5: printed hardware	Printing, curing, bending, wetting, and cycling → conductor/insulator state → continuity, leakage, adhesion, and baseline stability.	Materials science; manufacturing.	Covers the physical circuit stack, not sensor calibration or software compensation.
FB6: measurement chain	Raw capacitance/resistance and timestamps → conditioning, calibration, filtering, and synchronization → physical-unit channels with quality flags.	Electrical, instrumentation, and data engineering.	Ends before event classification; retains raw data and calibration provenance.
FB7: slip discrimination	Synchronized pressure/shear patterns plus an independent displacement reference → event comparison → slip/no-slip, direction, and onset error.	Mechatronics and signal processing.	Requires motion evidence; high shear alone remains a force event.
FB8: spatial interpretation	Calibrated channels plus CAD coordinates → registration/localization → modality-specific event location and bounded interface-condition display.	Computer science/software engineering.	Includes coverage and confidence; excludes a validated clinical discomfort or injury score.

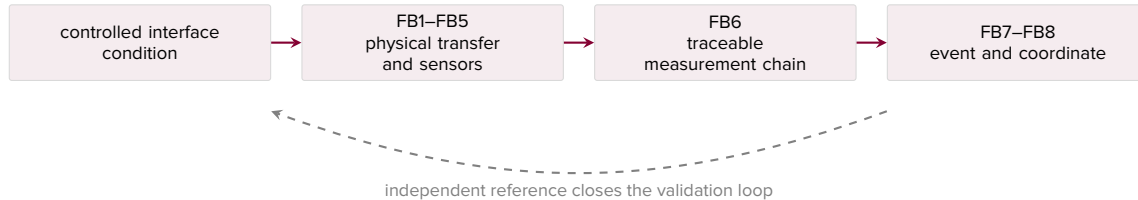


Figure Q3-0A. Functional-block signal path. Physical transduction, measurement conversion, and interpretation remain separate claim layers. The external reference evaluates the chain but is not itself part of the product.

Q3C. Literature Evidence and Project-Specific Proof

The literature establishes that the individual sensing and interpretation mechanisms are plausible, but no reviewed study implements this exact set of directly printed sensors in the present 5.2 mm slab. The matrix therefore separates **knowledge that can be leveraged** from the limited set of **project-specific relationships that must be established**. The detailed literature plots, fixtures, equations, protocols, and decision metrics in the remainder of Q3 provide the evidence route for these relationships.

Functional block	Known and leveraged from prior work	Relationship established for this prototype
FB1: interface transfer	Prosthetic-interface loading is cyclic, multidirectional, location-dependent, and sensitive to contact geometry; PDMS also introduces viscoelastic recovery [15, 16, 2].	Transfer from a known contact area and coordinate through the 5.2 mm slab, including spatial spread, recovery, and coupling between adjacent regions.
FB2: pressure/shear	A four-capacitor printed cell can separate common-mode normal loading from directional shear over a demonstrated force range, while also exhibiting priming, drift, and hysteresis [2].	The specimen-specific sensitivity matrix, usable range, axis error, hysteresis, and cross-sensitivity after printing and integration.
FB3: temperature	Printed silver meanders show an approximately linear resistance–temperature relationship, and soft prosthetic sensors can track skin-interface temperature [3, 19].	Calibration near the skin-relevant range, thermal response/recovery, spatial leakage, and separation of temperature from force- and strain-induced resistance changes.
FB4: humidity/sweat route	Printed capacitive RH sensors respond to vapour; prosthetic-liner RH can continue rising after temperature begins to recover [5, 13].	Vapour response through the selected protection, liquid-sweat saturation and drying behaviour, temperature coupling, and post-exposure drift or damage.
FB5: printed hardware	Oxygen-plasma treatment enables AgNP printing on PDMS; solvent absorption can swell PDMS, and partially embedded conductors improve mechanical retention [6, 7].	Continuity, insulation, adhesion, resistance, cracking, and calibration stability for the actual ink–printer–PDMS process under bending, heat, moisture, and cycling.
FB6: measurement chain	Published systems synchronize stress with motion and multiple skin-interface modalities while retaining independent signals [16, 19].	Channel calibration, common-clock accuracy, sampling and packet-loss behaviour, filtering delay, and traceability from raw channel to physical unit.
FB7: slip discrimination	Relative limb–socket motion can be measured against an independent displacement reference, and stress and motion are related but not interchangeable [28, 16].	Separation of translated contact from high-shear/no-motion trials, including direction, onset time, preload dependence, and forward/reverse consistency.
FB8: spatial interpretation	Socket pressure data can be registered to sensor coordinates and displayed using several interpolation methods [21].	Correct sensor-to-CAD mapping, localization error at sensor centres and gaps, validated coverage, combined-event separation, and a display that preserves disagreement between modalities.

Evidence anchor. Known mechanisms and transferable solutions are drawn from the printed-sensor, prosthetic-interface, multimodal monitoring, relative-motion, and spatial-visualization literature. Only the right-hand column is specific to the present prototype [2, 3, 5, 6, 7, 19, 28, 21].

Risk-order consequence: FB1–FB6 are examined before FB7–FB8. This follows the assignment's feasibility logic: an event classifier or smooth map cannot rescue inaccurate transduction, damaged conductors, unsynchronized channels, or an unprotected humidity route.

Q3C. Operational Definitions and Claim Boundary

Q3 is a **bench-validation question**. A known physical condition is applied at a known time and coordinate, and the integrated 5.2 mm PDMS slab is evaluated for the correct sensing modality, direction where applicable, location, magnitude, and duration. This stage addresses bench performance; participant discomfort and skin-injury thresholds belong to a subsequent clinical-validation stage. Accordingly, the admissible claim is detection of *simulated interface-condition events*; a clinical discomfort-risk claim requires comparison with user-reported discomfort, skin observations, and anatomically justified thresholds [14].

3.1 Operational meaning of each condition

Reported condition	Controlled ground truth	Required evidence from the slab	Evidence boundary at this stage
Localized loading	Known normal force, contact area, coordinate, and loading time.	Pressure channel responds at the correct coordinate; magnitude follows calibration; neighbouring and non-pressure channels remain bounded.	The load is painful, unsafe, or tissue-damaging.
Directional shear	Known tangential force in $+x$, $-x$, $+y$, $-y$ under a documented normal preload.	Correct shear direction and magnitude are recovered without being confused with normal compression.	A shear peak by itself proves that the liner slipped.
Slipping or pistoning	Known relative translation measured independently by a stage, displacement gauge, encoder, or video reference.	A time-linked change in shear, pressure distribution, or detected coordinate agrees with the reference displacement.	Slip is inferred from one high shear reading without independent motion evidence.
Thermal buildup	Known temperature step, ramp, or sustained heated contact at a known coordinate.	Temperature follows the reference with measured error, response time, recovery time, and limited mechanical cross-sensitivity.	The temperature constitutes a clinical burn or injury threshold.
Perspiration-related condition	Vapour exposure expressed as relative humidity (RH), tested separately from a known liquid artificial-sweat dose.	The humidity channel detects vapour; liquid exposure, saturation, drying, recovery, and permanent drift are separately reported.	RH is treated as sweat rate, or liquid presence is assumed from RH alone.

Evidence anchor. Operational definitions used for the current validation, anchored to published prosthetic-interface pressure, shear, temperature, and microclimate evidence [15, 16, 13, 19].



Figure Q3-0. Claim-control chain. The sensor output cannot label its own test; every event requires an external reference and a limited claim.

Central distinction: **Shear is a force condition; slip is a motion condition.** Slip can generate shear, but slip detection requires a displacement ground truth. Likewise, humidity describes water vapour, whereas perspiration can also produce liquid sweat. These conditions are therefore labelled and tested separately.

Q3C. Quantitative Literature Evidence

3.2 Mechanical loading and motion

Simultaneous pressure and shear measurement is necessary because interface loading is multidirectional. Sanders *et al.* measured resultant shear of 5.6–39.0 kPa at the tibial flares and 5.0–40.7 kPa posteriorly during walking; the waveforms were commonly double-peaked, with peaks occurring at 25–40% and 65–85% of stance [15]. Tang *et al.* combined multidirectional stress sensing with three-dimensional motion capture and reported posterior-proximal peak pressure of 55–59 kPa and shear of 12–19 kPa across nine sessions over 12 months in one transfemoral participant [16]. Figure Q3-1 summarizes these reported ranges. They are **context values, not universal pass/fail thresholds**.

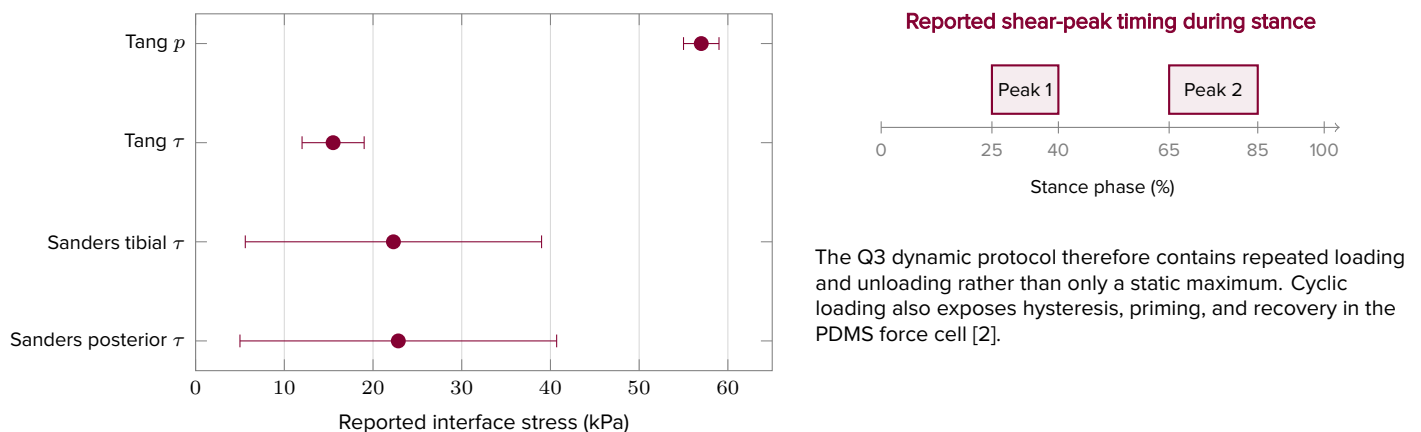


Figure Q3-1. Literature-derived mechanical context. Left: reported min–max ranges from Tang *et al.* and Sanders *et al.*; right: reported stance-phase windows for the two common shear peaks [15, 16]. Values describe their participants, sensor locations, and protocols and are not transferred as clinical thresholds.

Zhang and Roberts reported a 226 kPa finite-element peak pressure at the patellar tendon and a 50 kPa peak shear stress at the anterolateral tibia in their participant-specific model and measurements [17]. The large difference between reported studies reinforces the need to record anatomical location, activity, contact area, and duration and to avoid a single global pressure threshold. The 2026 systematic review by Armitage *et al.* similarly found no consistent pressure threshold for comfort or discomfort across regions and activities [14].

Q3C. Thermal and Moisture Literature

3.3 Expected time-dependent microclimate behaviour

Paternò *et al.* instrumented a personalized transtibial liner with 12 temperature/RH loggers and used a protocol comprising 50 min seated rest, 30 min treadmill walking, and 50 min recovery [13]. Temperature rose during activity and decreased after activity, while RH continued to rise. Figure Q3-2 replots their tabulated means and standard deviations. This pattern supports time-series assessment rather than reliance on a single end-point.

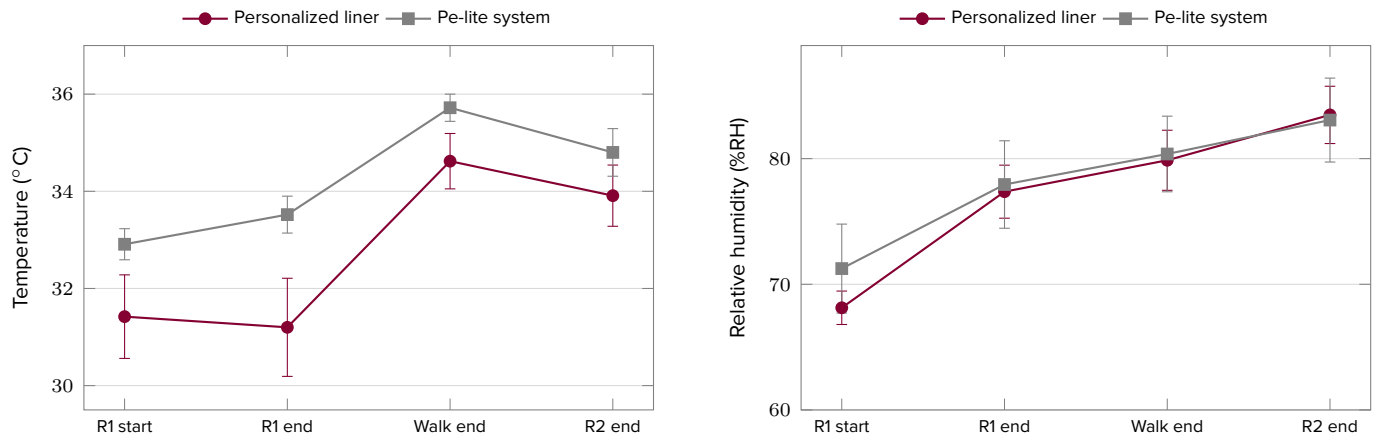


Figure Q3-2. Published temperature and RH progression. Replotted from Tables 2 and 3 of Paternò *et al.*; points are the reported mean across 12 sensors and error bars are the reported standard deviation [13]. R1 start: initial rest; R1 end: end of 50-min rest; Walk end: end of 30-min walking; R2 end: end of 50-min recovery. This was a single-participant case study and does not define universal thresholds.

Literature observation	Implication for Q3	Necessary control
Temperature plateaued after donning and rose rapidly during walking [13].	Include equilibration, a controlled heat step or ramp, and a recovery interval.	Reference thermocouple/RTD at the same contact location; record ambient temperature.
RH continued rising through activity and recovery [13].	Do not expect humidity response to follow the temperature waveform or recover at the same rate.	Reference hygrometer or calibrated chamber; document airflow and sealed volume.
Multimodal pressure, temperature, and moisture surface with independent event observations [18].	Validate event labels against external observations rather than declaring success from the sensor output itself.	Blinded event log or automated fixture log containing true event time and location.
Soft prosthetic-interface platform with concurrent pressure and temperature [19].	Time-synchronize modalities and retain separate physical-unit traces before any combined display.	Common clock, timestamp audit, and packet-loss report.

Evidence anchor. Literature-derived observations and project test controls from prosthetic-liner and multimodal interface-monitoring studies [13, 18, 19].

Q3C. Evidence Architecture and Ground Truth

3.4 Specimens, fixtures, and ground truth

The primary test article is the directly printed 56 mm × 55 mm × 5.2 mm PDMS slab defined in Q2. The PET print remains a fabrication control where a comparable electrical measurement is possible; it is not a liner-material alternative. At least three independently fabricated PDMS specimens and five repeated cycles per condition define the replication target. The specimen register records the achieved count and keeps independent slabs distinct from repeated cycles. Every file records specimen ID, channel ID, CAD coordinate, calibration version, stimulus program, reference-instrument ID, sampling rate, and ambient conditions.

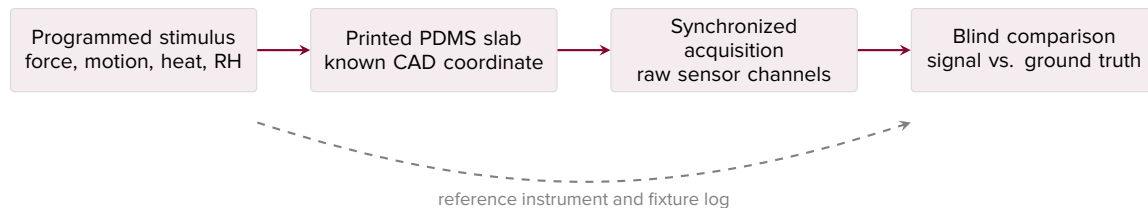


Figure Q3-3. Ground-truth validation architecture. The slab output is not used to label its own test. Stimulus magnitude, time, direction, and coordinate are measured independently and then compared with the synchronized sensor record.

ID	Condition	Programmed input	Independent reference	Primary evidence
B0	Baseline	Unloaded slab after thermal/RH equilibration; electronics on and fixture untouched.	Reference temperature/RH; zero-force check; logged DAQ offset.	Noise, zero drift, channel covariance, false-event rate.
P1	Local pressure	Normal-force staircase and cyclic load at sensor centres, between sensors, and near slab boundaries; fixed contactor area.	Calibrated load cell or force gauge plus measured contact area.	Pressure sensitivity, linearity, hysteresis, recovery, coordinate response.
S1	Directional shear	Constant normal preload followed by signed tangential force in both slab axes.	Two-axis force reference or tangential force gauge; stage direction log.	Shear magnitude/direction, cross-axis error, pressure cross-talk.
SL1	Slip/pistoning	Programmed translation under contact, initially spanning 0.5–4.5 mm as a literature-anchored pilot range [20].	Encoder, LVDT/digital indicator, or calibrated video displacement.	Event detection, direction, onset-time error, and displacement-linked spatial change.
T1	Thermal buildup	Equilibrated baseline followed by 25–45 °C steps or a body-relevant subset; localized and uniform heating.	Contact thermocouple or traceable RTD at the same coordinate.	Temperature RMSE, response/recovery time, spatial leakage, force sensitivity.
H1	Vapour humidity	20–95%RH chamber steps or the available validated subset, following the calibration range used by Paternò <i>et al.</i> [13].	Calibrated chamber or reference hygrometer positioned beside the slab.	RH RMSE, response/recovery, saturation, temperature coupling.
W1	Liquid sweat	Known artificial-sweat dose normalized by exposed area; separate deposition, dwell, and drying stages.	Calibrated pipette or gravimetric mass; image/time log of wetted area.	Detection, saturation, leakage current, recovery, permanent drift, damage.
C1	Combined events	Pressure+shear, temperature+RH, pressure+liquid, then all available modalities at one or separated coordinates.	All corresponding references synchronized to one clock.	Missed events, false events, cross-sensitivity, and degradation from isolated-condition performance.

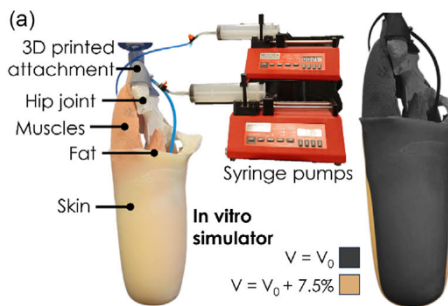
Evidence anchor. Implementation-ready test matrix. Reference methods and pilot ranges are anchored to published force, motion, and liner microclimate studies [15, 20, 13]; they are not clinical thresholds.

Q3C. Published Phantom Precedents and Resources

3.4.1 Three literature precedents retained as build options

No single phantom reproduces pressure, shear, liquid sweat, vapour humidity, temperature, and residual-limb volume change. The three candidates below are therefore retained as complementary, published build options. The photographs are reproduced from the cited papers, and every unchecked box means *availability at AUB Biomedical Engineering has not yet been confirmed*.

Candidate A — Paternò variable-volume transfemoral simulator

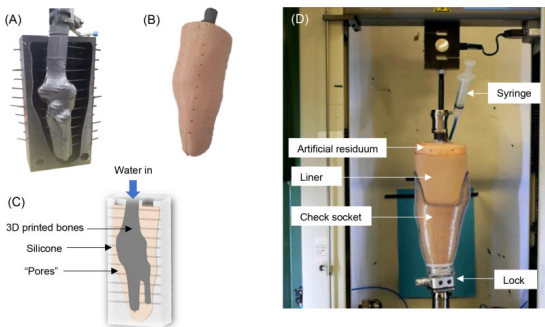


Best-matched role. Controlled whole-limb volume/preload change and pressure-map response in transfemoral geometry. The published chamber produced reversible changes up to approximately $\pm 7.5\%$, with volume checked by 3D scanning [24].

Subsystem	Required items	AUB?
Geometry	Functional hip-joint model or equivalent; 3D-printed proximal attachment; rigid mounting frame.	☐
Tissue body	Molds and graded silicone layers representing muscle, fat, and skin; release agent and casting tools.	☐
Hydraulics	Sealed internal water chamber, tubing, fittings, valves, and leak-test supplies.	☐
Actuation	Two motorized syringe pumps (published: NE-1010) or validated equivalent; syringes and water reservoir.	☐
Ground truth	3D scanner or validated geometric-volume method; reference pressure and displacement instruments.	☐

Figure Q3-4A. Published transfemoral simulator. Cropped from Paternò *et al.*, Figure 5a: anatomical hip model, silicone tissue layers, hydraulic chamber, and two syringe pumps [24]. CC BY 4.0; only the panel boundary was cropped.

Candidate B — McGrath sweating residuum/socket simulator



Best-matched role. Liquid-moisture and moisture-induced axial slip testing. The paper applied 20 mL of water through molded pores and compared dry and wet cyclic displacement with standard and perforated liners [26]. This is not a vapour-RH calibration chamber.

Subsystem	Required items	AUB?
Geometry	Cast/scan-derived transtibial shape; printed bone model; two-part negative mold.	☐
Porous body	Silicone soft-tissue body; 3-mm pore-forming straws/rods; silicone adhesive and seals.	☐
Water delivery	Graduated syringe, rubber tubing, proximal inlet, measured water dose, and drying arrangement.	☐
Socket interface	Pin-lock check socket, lock, standard liner, and perforated comparison liner.	☐
Cyclic reference	Universal test machine or controlled axial load fixture; force and displacement logging.	☐

Figure Q3-4B. Published sweating-interface simulator. McGrath *et al.*, Figure 1: printed bones, porous silicone residuum, water inlet, liner/check socket, and cyclic test-machine mounting [26]. CC BY 4.0.

Q3C. Phantom Selection Logic

Candidate C — Phillips dual-hardness, water-bladder mock limb

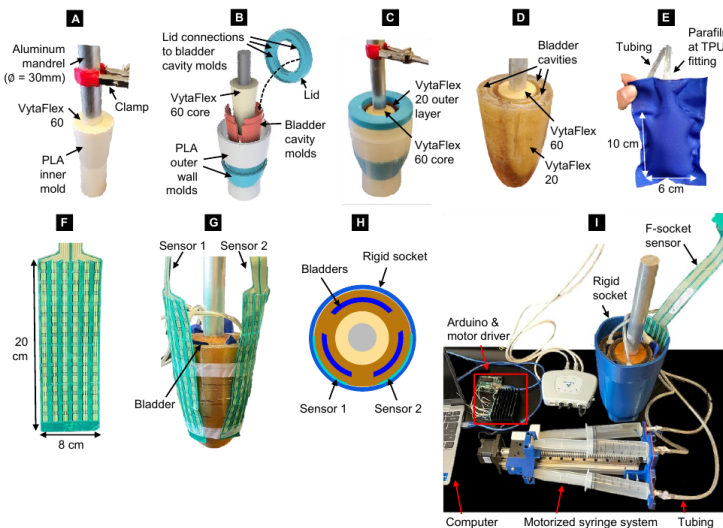


Figure 1: Design overview. **A)** Step one: casting the core with VytaFlex™ 60 around an aluminum mandrel. **B)** Mold set-up for casting the outer layer. The bladder cavity molds are held in place by clipping into the lid. The wall molds were separated to ease mold removal post-curing. **C)** Step two: casting the outer layer with VytaFlex™ 20. **D)** Final casted limb. **E)** TPU bladder. **F)** F-socket sensor, 0.15 mm thin. **G)** Sensor placement on the limb before donning the rigid socket. **H)** Top view schematic of sensor placement on the limb. **I)** Full system with the biofidelic mock limb, rigid socket, and motorized syringe system. Only one F-socket sensor is shown for clarity.

Best-matched role. A reproducible, relatively accessible adjustable limb for pressure/preload and volume-change studies. The published build used three water-filled bladders, reached controlled changes up to $\pm 5\%$, and cost less than CAD 400 [25].

Subsystem	Required items	AUB?
Core/body	30-mm aluminum mandrel; VytaFlex 60 core and VytaFlex 20 outer layer or characterized equivalents.	☐
Molds	Printed PLA inner, outer, lid, and bladder-cavity molds; epoxy seal, release agent, and clamp stand.	☐
Bladders	Three heat-sealed 6×10-cm TPU bladders; 6.4-mm fittings, tubing, Parafilm/seals, and water.	☐
Actuation	Motorized syringe system, linear actuator, Arduino, motor driver, power supply, and computer.	☐
Interface / ref.	5-mm PLA socket; reference pressure film/system; volume and circumference measurements.	☐

Figure Q3-4C. Published low-cost adjustable mock limb. Phillips *et al.*, Figure 1: urethane casting sequence, TPU bladder, sensor placement, rigid socket, Arduino, and motorized syringe system [25]. CC BY 4.0; only page margins were cropped.

3.4.2 Evidence-based selection for the project tests

Published option	Use it for	Evidence already demonstrated	Boundary requiring a separate fixture
Paternò 2025	Transfemoral volume/preload steps and sensor-map response to global fit change.	Hydraulic chamber, two pumps, repeated 3D volume measurement, and approximately $\pm 7.5\%$ range.	Does not generate liquid sweat or controlled vapour RH; comparatively complex fabrication.
McGrath 2021	Liquid-sweat exposure and wet-interface slip under axial cyclic loading.	Porous silicone body, measured 20-mL water dose, dry/wet liner comparison, and displacement reference.	Transstibial geometry; no controlled vapour RH or temperature field; liquid distribution was not perfectly uniform.
Phillips 2025	Low-cost volume/preload and localized pressure changes using independent bladders.	Dual-hardness body, three water bladders, Arduino actuation, $\pm 5\%$ change, and build cost below CAD 400.	No sweating pores or RH control; published pressure film exhibited long-duration drift.

Evidence anchor. The three options are published implementations, not speculative phantom sketches [24, 26, 25]. Their functions are complementary: volume/preload, liquid-moisture slip, and low-cost localized adjustability, respectively.

Implementation rule: Retain all three as possible constructions, then select the first build through the AUB inventory register. Use the Paternò or Phillips architecture for pressure/volume localization and the McGrath architecture for liquid-sweat/slip. Vapour humidity and contact temperature still require the separate calibrated chamber and reference probes defined in Test Matrix H1/T1; no phantom substitutes for those ground truths.

Safety and scope: Use guards, hard stops, and leak checks; begin with small loads and place no person in the fixture. These phantoms validate sensor response, coordinate mapping, and controlled interface conditions. They do not establish human comfort, tissue equivalence, or clinical safety.

Q3C. Measurement Models

3.5 Converting raw channels into testable measurements

All comparisons use baseline-corrected raw measurements. For channel j of slab i , the baseline-corrected signal is

$$\tilde{s}_{ij}(t) = s_{ij}(t) - \bar{s}_{ij,0}, \quad \bar{s}_{ij,0} = \frac{1}{N_0} \sum_{k=1}^{N_0} s_{ij}(t_k), \quad (1)$$

where the baseline window N_0 is recorded before the stimulus. Filtering may remove electrical noise, while both raw and filtered data are retained and the filter type, cutoff, phase delay, and missing samples reported. Thus, Equation (1) makes the zeroing operation explicit and reproducible.

Pressure and shear. The four-capacitor force cell is calibrated as a multivariable system because normal and tangential loads can affect more than one capacitor. For the calibration vector $\Delta \mathbf{c} = [\Delta C_1, \Delta C_2, \Delta C_3, \Delta C_4]^T$,

$$\Delta \mathbf{c} = \mathbf{K} \begin{bmatrix} F_x & F_y & F_z \end{bmatrix}^T + \varepsilon, \quad \hat{\mathbf{F}} = \mathbf{K}^+ \Delta \mathbf{c}, \quad (2)$$

where $\mathbf{K}$ is fitted from signed, known loads and $\mathbf{K}^+$ is its pseudoinverse. This empirical matrix accommodates unequal printed channels and is safer than assuming perfect geometric symmetry. Equation (2) is therefore the primary force-recovery model. The engineering stresses under a documented contact area A are

$$p = \frac{F_z}{A}, \quad \tau_x = \frac{F_x}{A}, \quad \tau_y = \frac{F_y}{A}, \quad \tau = \sqrt{\tau_x^2 + \tau_y^2}. \quad (3)$$

Albrecht *et al.* found linear response between 0.1 and 8 N in their printed cell, but also reported priming, drift, and shear hysteresis linked to the viscoelastic PDMS spacer [2]; the project's own $\mathbf{K}$ and working range are therefore obtained from the present slab before applying Equation (3).

Temperature and humidity. Calibration coefficients are fitted against reference instruments rather than copied from the source papers:

$$\hat{T} = a_0 + a_1 R_T + a_2 R_T^2, \quad \widehat{RH} = b_0 + b_1 C_H + b_2 C_H^2 + b_3 \hat{T}, \quad (4)$$

where R_T is the printed meander resistance and C_H is the humidity-IDE capacitance. Equation (4) allows curvature and thermal compensation to be tested rather than presumed. The temperature term in the RH equation is retained only if model comparison shows that it reduces reference error; otherwise it is removed to avoid unnecessary fitting. Calibration, cross-validation, and Q3 testing data are held separate.

Exposure over time. A peak alone cannot describe a sustained event. For a window $[t_0, t_1]$, report the pressure-time and shear-time integrals

$$PTI = \int_{t_0}^{t_1} \max[p(t) - p_0, 0] dt, \quad STI = \int_{t_0}^{t_1} \max[\tau(t) - \tau_0, 0] dt, \quad (5)$$

along with peak, mean, and dwell time. Here p_0 and τ_0 are engineering noise or event-detection levels established during B0, not clinical injury thresholds; Equation (5) therefore describes exposure above a measured detection level, not biological damage.

Slip label. A trial is labelled as true controlled slip only when the independent reference satisfies

$$y_{\text{slip}}(t) = \begin{cases} 1, & |d_{\text{ref}}(t) - d_{\text{ref}}(t - \Delta t)| \geq d_{\text{min}}, \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

where d_{min} is the verified resolution of the displacement fixture. Sensor-based detection is then compared with the external label in Equation (6). Without d_{ref} , the result is reported as a *shear event*, not slip.

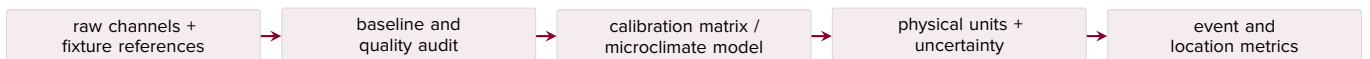


Figure Q3-5. Calibration-to-decision pipeline. Raw channels remain traceable to the external reference before any event label or map is produced.

Evidence anchor. The equations are project calibration definitions. Sensor-specific transfer functions are estimated from the present slab; published devices provide mechanisms and candidate forms, not transferable coefficients [2, 3, 5].

Q3C. Condition-by-Condition Evidence Protocol

3.6 Defined execution sequence

Protocol	Stimulus sequence	Control and negative test	Required plots
P1: localized normal loading	Apply an unloaded baseline, increasing force steps, dwell, decreasing steps, recovery, and repeated cyclic loading. Repeat at every active footprint, at inter-sensor midpoints, and near slab edges.	Repeat the fixture motion without contact; rotate or replace the contactor while keeping force and area documented.	Raw and calibrated traces; applied versus estimated force; residual plot; hysteresis loop; coordinate-response map.
S1: directional shear	Establish constant F_z , then apply $+x, -x, +y, -y$ force staircases and cycles. Include at least one priming cycle before the reported cycles because PDMS viscoelasticity can cause first-cycle differences [2].	Normal preload without tangential motion; tangential fixture motion without contact; repeat after rotating the slab 180 degrees.	$\hat{F}_x, \hat{F}_y, \hat{F}_z$ versus references; vector-angle error; cross-axis matrix; loading/unloading curves.
SL1: controlled slip	Translate the contactor by known distances at several speeds under low, medium, and high normal preload. Program forward, reverse, stop-start, and no-slip high-shear trials.	High shear with no displacement is the critical negative control; free motion without contact checks fixture artefacts. Yu <i>et al.</i> support direct displacement characterization [20].	Reference displacement and sensor traces on one time axis; detected versus true event raster; onset-time and distance error.
T1: thermal buildup	Equilibrate, apply temperature steps or a controlled ramp, hold, then remove heat and continue through recovery. Test localized contact and whole-environment heating separately.	Reference sensor at the active location; heater motion with heater off; force-only contact at constant temperature.	Estimated versus reference temperature; RMSE; response and recovery curves; spatial thermal spread.
H1: vapour humidity	Equilibrate at low RH, apply increasing chamber steps, decreasing steps, and recovery. Repeat at two or more temperatures to expose temperature coupling.	Sealed dry-air step; chamber program with unpowered sensor; reference probe placed beside the active region.	Capacitance and calibrated RH; reference residuals; hysteresis; response/recovery time.
W1: artificial sweat	Deposit a recorded areal dose, hold, remove surface liquid, dry, and repeat after recovery. Inspect protection, conductors, leakage, and permanent offset before the next dose.	Equal-volume deionized-water test only if chemically relevant and approved; dry pipette contact; protected versus unprotected route from Q2.	Wetting-area image; raw signal; saturation duration; drying/recovery; pre/post calibration and leakage.
C1: combinations	Begin with pressure+shear and temperature+RH; progress to events applied at the same coordinate, then separated coordinates, and finally all available modalities. Randomize trial order.	Repeat each constituent as an isolated event on the same specimen and day.	Synchronized small multiples; isolated-versus-combined error; confusion matrix; missed and false event map.

Evidence anchor. Execution protocol anchored to published printed force-cell behaviour, direct displacement validation, and liner microclimate measurements [2, 28, 13].

Order of testing: Run B0, isolated P1/S1/T1/H1, and the protected liquid test W1 before slip and combined-condition trials. This order identifies individual calibration and damage before several sources are superimposed. Randomize trials within each completed block and do not tune thresholds on the same repetitions used for final reporting.



Figure Q3-6. Fixture-to-evidence trace. Each validation branch records a separately measured reference channel, synchronizes it with the raw slab output, and carries both into the final plots.

Q3C. Evidence Metrics and Decision Gates

3.7 Measurement quality, detection, and localization

For each condition, report the calibration model, full test range, number of specimens, repetitions, mean, standard deviation, raw traces, and rejected trials with reasons. The core accuracy and localization quantities are

$$RMSE_m = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\hat{x}_{m,k} - x_{m,k}^{\text{ref}} \right)^2}, \quad NRMSE_m = \frac{RMSE_m}{x_{m,\max}^{\text{ref}} - x_{m,\min}^{\text{ref}}}, \quad (7)$$

$$\hat{\mathbf{r}}(t) = \frac{\sum_i w_i(t) \mathbf{r}_i}{\sum_i w_i(t)}, \quad e_{\text{loc}} = \|\hat{\mathbf{r}} - \mathbf{r}_{\text{true}}\|_2, \quad (8)$$

where m denotes modality, $\mathbf{r}_i$ is the CAD coordinate of sensor i , and w_i is a non-negative baseline-corrected response for the modality under test. Equation (7) evaluates reference agreement, while Equation (8) evaluates spatial recovery. The centroid is an engineering localization estimator used for Q3; Q5 evaluates and selects the final interpolation algorithm.

$$H_m = \frac{\max_x |y_{\uparrow}(x) - y_{\downarrow}(x)|}{y_{\text{FS}}}, \quad XT_{m \rightarrow n} = \frac{\max |\Delta y_n|}{\max |\Delta y_m|}, \quad (9)$$

Equation (9) defines full-scale-normalized hysteresis and cross-sensitivity from stimulated modality m into non-target modality n . Response time t_{90} is the time to reach 90% of the final step response; recovery time is defined analogously after removal. For event labels, report sensitivity, specificity, precision, and F1 score from the complete confusion matrix rather than accuracy alone.

Gate	Provisional engineering target	Report required	Interpretation if failed
Calibration fit	$R^2 \geq 0.95$ and $NRMSE \leq 10\%$ of calibrated full scale for each modality.	Per-specimen fit and held-out-test error; residuals versus input and time.	Revise geometry, electronics, range, or calibration model; do not hide nonlinear regions.
Repeatability	Coefficient of variation $\leq 10\%$ at repeated levels after the stated priming procedure.	Within-cycle, between-cycle, and between-specimen variation.	Manufacturing or viscoelastic response is not yet sufficiently controlled.
Hysteresis	$H_m \leq 10\%$ full scale over the declared operating range.	Full loading/unloading loops at slow and dynamic rates.	Restrict the range or use history-aware calibration; report the limitation.
Cross-sensitivity	Non-target response $\leq 15\%$ of the target full-scale response in isolated tests.	Complete modality-by-modality and axis-by-axis cross-talk matrix.	Independent interpretation requires redesign or multivariate compensation when this gate is not met.
Event detection	Sensitivity and specificity each $\geq 90\%$ on held-out repetitions.	Confusion matrix with event counts, not percentages alone.	Thresholds or features are not transferable to unseen repetitions.
Localization	Median error no greater than half the nearest-neighbour sensor pitch; 95th percentile also reported.	Error by coordinate, boundary location, and isolated/combined condition.	Sensor density or coordinate mapping is insufficient; this result feeds Q6.
Combined-condition robustness	No more than 10 percentage-point loss in event sensitivity relative to isolated tests.	Paired isolated/combined comparison on the same specimens.	Cross-sensitivity or signal fusion masks a modality.
Damage and recovery	No open/short circuit, leakage, delamination, permanent saturation, or greater than 10% full-scale baseline shift after recovery.	Electrical inspection and pre/post calibration for every specimen.	Stop progression to combined tests or liner integration.

Table note. These numerical gates are the selected engineering targets for preregistration before data collection. They are not published clinical safety or discomfort thresholds and may be revised only before the held-out Q3 evaluation, with the revision documented.

Evidence anchor. Preregistration gates selected for this development stage. They are not presented as clinical limits; the literature review reports substantial participant, location, and protocol variability [14].



Figure Q3-7. Sequential engineering gates. Failure at an earlier gate blocks the more complex claim rather than being averaged into a total score.

Q3C. Validation Outputs and Interpretation

3.8 Evidence output formats and interpretation criteria

The assignment asks what remains to be learned or proven and how the prototype is evaluated, including tools, rationale, expected outcome, and design impact. The panels below consolidate the reporting outputs paired with the project-specific proof relationships in the Q3C matrix. They define the evidence format and are not presented as completed measurements.

EVIDENCE OUTPUT FORMAT

Calibration and transient plots

Required contents: reference-versus-estimate plots with held-out points, raw and filtered traces, hysteresis, response/recovery markers, error bars, and specimen IDs for pressure, shear, temperature, and RH.

EVIDENCE OUTPUT FORMAT

Known versus detected event locations

Required contents: CAD sensor coordinates, true stimulus coordinates, detected centroids, error vectors, sensor footprints, and a confidence/coverage mask. Do not interpolate outside validated coverage.

EVIDENCE OUTPUT FORMAT

Slip ground-truth comparison

Required contents: synchronized reference displacement, normal pressure, signed shear, detected event state, onset-time error, and forward/reverse trials.

EVIDENCE OUTPUT FORMAT

Combined-condition small multiples

Required contents: separate physical-unit traces and maps before any combined score; show disagreement, false events, missed events, and isolated-condition baselines.

Evidence field	Recorded evidence	Design interpretation
Test population	Number of independently fabricated slabs, cycles, stimulus coordinates, and accepted/rejected trials.	Do not treat repeated cycles on one slab as independent specimens.
Reference agreement	R^2 , RMSE/NRMSE, bias, 95% limits of agreement where sample size supports them, and residual plots.	A high R^2 does not remove systematic bias; always report an error metric.
Dynamic response	Peak, mean, integral, t_{90} , recovery, hysteresis, and drift for each modality.	Compare with the condition's reference timescale; do not require humidity to respond as fast as pressure.
Detection performance	Full confusion matrix, sensitivity, specificity, precision, and F1 by event type.	Results are engineering event recognition, not diagnostic sensitivity for injury or discomfort.
Spatial performance	Median, interquartile range, and 95th-percentile localization error; boundary and inter-sensor errors shown separately.	A visually smooth map is not evidence of correct localization.
Combined-condition effect	Difference between isolated and combined trials in error, sensitivity, response time, and baseline.	If modalities disagree, retain separate outputs; do not average away the disagreement.
Post-exposure integrity	Continuity, insulation/leakage, visible damage, baseline and sensitivity changes after liquid, heat, shear, and cyclic load.	Passing detection before damage does not establish reusability.

Evidence anchor. Evidence reporting template aligned with the external reference architecture in Figure Q3-3 and the published distinction between force and relative motion [16, 28].

Connection to Q5 software work: Q3 uses the CAD coordinates only to verify whether a known event was detected at the correct location. Detailed interpolation, threshold fusion, real-time rendering, and the selected interface-condition heatmap architecture are developed in Q5. Turner *et al.* provide a key reference for comparing radial-basis, Clough–Tocher, nearest-neighbour, and linear interpolation on a prosthetic socket model [21].

Q3C. F = Interim Functional-Block Verdict



Figure Q3-8. Claim maturity at the present project stage. Q3 ends at controlled phantom detection; the dashed clinical step is outside the current evidence.

F = FUNCTIONAL BLOCKS DEFINED; LITERATURE BASIS ESTABLISHED

- **Leveraged knowledge:** prior studies support the underlying mechanical, thermal, humidity, printed-PDMS, relative-motion, and spatial visualization mechanisms [15, 16, 13, 19, 14].
- **Prototype-specific proof:** the right-hand column of the Q3C matrix identifies the relationships that are not transferable from those studies: calibration, separation, localization, recovery, and integrity for this slab and sensor placement.
- **Decision rule:** report “yes,” “partial,” or “no” from the preregistered gates in Sections 3.6–3.8. A partial result names the passing and failing modalities; an overall average cannot conceal failed localization, separation, recovery, or integrity.
- **Clinical boundary:** even a complete engineering pass supports only detection of controlled interface conditions. The words *discomfort*, *harmful*, or *skin risk* require subsequent participant-specific validation and clinical oversight.
- **Evidence sequence:** B0 and isolated-condition calibration precede blinded slip, combined-condition, heatmap, and whole-liner claims.

Current verdict: Q3A identifies the eight critical blocks, Q3B defines their engineering boundaries, and Q3C distinguishes leveraged literature from the specimen-level evidence required for the present prototype.

Claim-evidence statement: This is the appropriate boundary for an open-ended project progress report: the functional architecture, literature basis, candidate fixtures, ground-truth comparisons, metrics, and evidence formats are established. A claim that the slab **successfully detects** the four conditions is reserved for specimen-level experimental results.

Q4. Detailed Design Requirements and Technical Specifications

4.1 Requirement basis and evidence boundary

Q4 converts the functional blocks isolated in Q3 into testable engineering requirements. A specification is included only when its evidence status is clear: **P** denotes a result demonstrated in a published precedent, **D** denotes a value selected for the present design, and **V** denotes a value that must be verified on the fabricated slab. A paper can support a mechanism or a starting process without proving the performance of the present ink–printer–PDMS–sensor combination.

Critical evidence boundary: No reviewed paper reports the exact $56 \times 55 \times 5.2$ mm slab with directly printed silver pressure/shear, temperature, and humidity sensors. The 5.2 mm thickness is a project CAD and molding requirement. Published work separately supports molded PDMS, direct Ag printing, multimodal prosthetic sensing, and prosthetic-liner thickness effects; Q5 must verify their integration.

Q1 origin → linked Q2 evidence	Requirement purpose	Q4 specification families
Q1-1 → Q2 §2.1 / §2.8	Four-mode integration in the 5.2 mm slab	GEO, MAT, PRT, SYS
Q1-2 → Q2 §2.4.1	Humidity vapour access and liquid protection	HUM, ENV, SAFE
Q1-3 → Q2 §2.3 / §2.4 / §2.4.1	Detection of controlled interface conditions	MECH, TEMP, HUM
Q1-4 → Q2 §2.2 / §2.5	Manufacturability, available equipment, and cost	MFG
Q1-5 → Q2 §2.7 / §2.8	Synchronized coordinate mapping and interpretation	DAQ, MAP, ALG
Q1-6 → Q2 §2.2 / §2.7	Sensor density, coverage, and localization	MAP
Q1-7 → Q2 §2.5 / §2.6.1	Drift, environmental exposure, and durability	DUR, ENV

Table Q4-1. Clickable Q1–Q2–Q4 traceability. Q1 identifies the requirement origin; each underlined red Q2 reference opens the detailed design, literature, or implementation evidence used to define its Q4 specification.

ID	Project requirement	Basis and status	Acceptance evidence
GEO-01	Active slab envelope shall be 56×55 mm with nominal total thickness 5.2 mm.	D: fixed by current CAD/mold. Published 4 and 6 mm liner models provide context only [10].	Dimensioned CAD export; caliper thickness at centre and four corners.
GEO-02	Total thickness shall be 5.2 ± 0.2 mm; five-point range shall not exceed 0.4 mm.	D/V: preliminary manufacturing tolerance, not a literature-derived clinical limit.	Five recorded measurements for every slab; reject visible voids, incomplete fill, or out-of-tolerance specimens.
GEO-03	Baseline stack shall allocate 0.20 mm skin-side cover, 3.50 mm sensor carrier, and 1.50 mm back encapsulation, summing to 5.20 mm.	D/V: current engineering allocation; layer transfer and local stiffness require verification.	Cross-sectional coupon or mold witness; report each layer and sensor-plane depth rather than total thickness alone.
MAT-01	The first calibrated build shall use bare PDMS as H0; hydrogel is not part of the baseline specification.	D: Q2 Section 2.6 evidence gate . Hydrogel changes moisture transport and stress transfer [11, 12].	H0 calibration completed before any H1/H2 material comparison.

Evidence anchor. Thickness context from Adda Cherif *et al.* [10]; dimensions and tolerances are explicitly identified as project decisions rather than published outcomes.

Q4. Implementation Geometry and Liner Context

4.2 Project slab, sensor placement, and thickness precedent

The two supplied CAD views are implementation inputs: the render communicates the physical arrangement, while the blueprint supplies the coordinate and enclosure reference used by MAP-01. They are shown here with the published liner-thickness precedent because Q4 defines how the geometry becomes a measurable specification. Neither perspective view replaces a dimensioned manufacturing drawing.

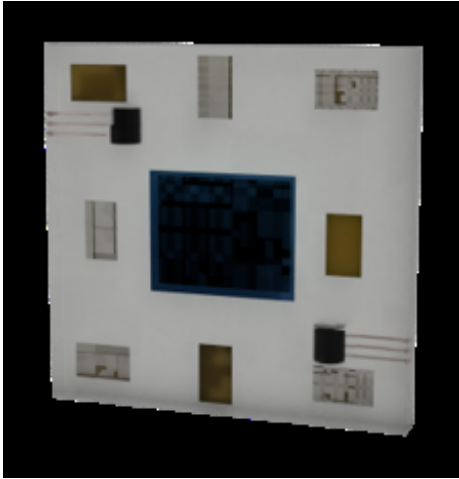


Figure Q4-1A. Slab design render. Supplied three-dimensional view of the distributed sensing elements and central enclosure region.

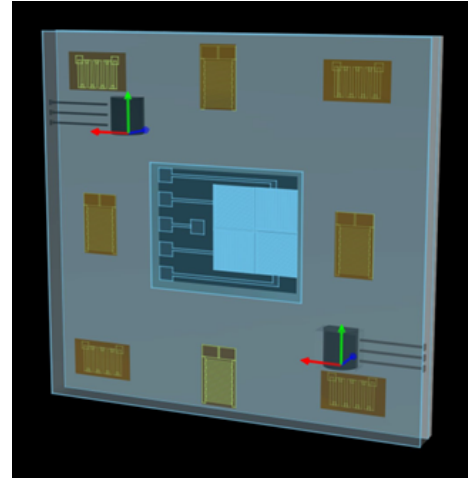


Figure Q4-1B. Sensor-placement blueprint. Supplied CAD view showing relative sensor placement, the enclosure principle, and conductor exits.

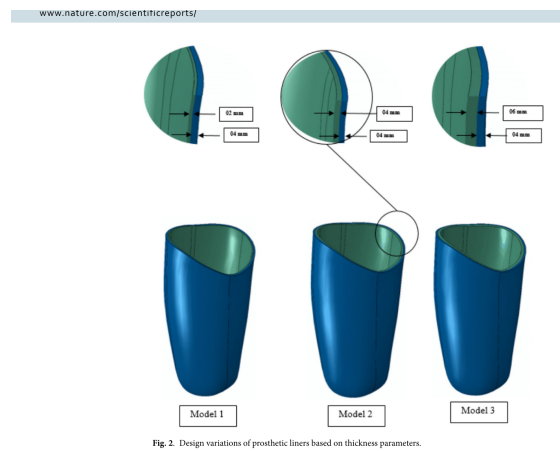


Fig. 2. Design variations of prosthetic liners based on thickness parameters.

Figure Q4-1C. Published liner-thickness context. Adda Cherif *et al.* compared 2, 4, and 6 mm transfemoral liner models [10]. The study brackets the project's 5.2 mm value but does not validate the printed sensor stack or the exact selected thickness. CC BY 4.0.

Required drawing output: Release to fabrication requires one orthographic plan and one cross-section labelled with the 56×55 mm envelope, 5.2 mm total thickness, layer depths, sensor IDs and footprints, coordinate origin, conductor widths, pad positions, edge clearances, enclosure boundary, and nominal versus measured dimensions.

Q4. PDMS Molding and Direct Silver Printing

4.3 A literature-backed fabrication window

The closest direct precedent is not a 5.2 mm prosthetic slab. Abu-Khalaf *et al.* molded 1 mm Sylgard 184 specimens in acrylic molds, using a 10:1 base-to-curing-agent ratio, 30 min vacuum degassing, and 70 °C curing for 2 h before direct AgNP printing [8]. Albrecht *et al.* established a separate low-temperature printing route using oxygen-plasma-treated PDMS, printing within 30 min, and drying at 60 °C for 30 min [6]. These are *separate published starting recipes*; their parameters must not be combined silently.

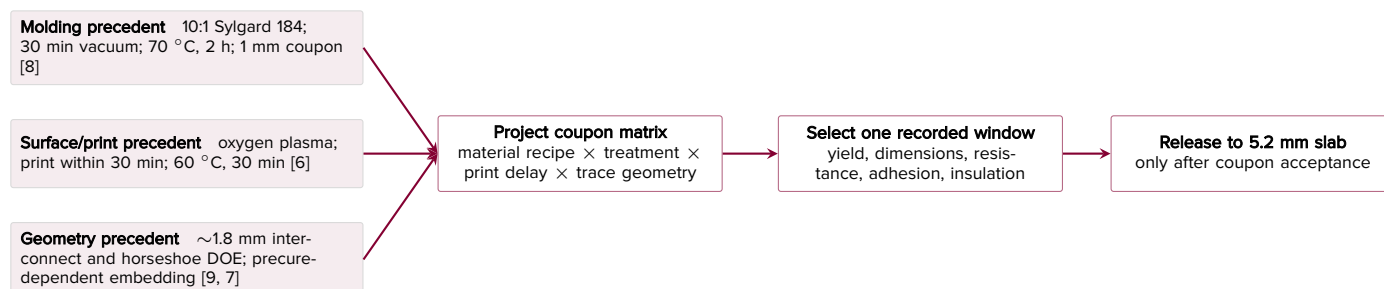


Figure Q4-2. Evidence-to-process decision route. The published routes remain separate inputs. A project coupon matrix selects one compatible process before the full slab is printed; this is an implementation diagram, not a reproduction of the paper figures already shown in Q2.

ID	Published precedent	Project specification	Pass evidence
MFG-01	Sylgard 184, 10:1, 30 min vacuum, 70 °C for 2 h on 1 mm specimens [8].	Use as the documented candidate recipe if the current PDMS is Sylgard 184; otherwise record material, ratio, degassing, cure, and hardness as a separate recipe.	Complete fill; no visible bubbles in active areas; GEO-01/02 measurements pass.
PRT-01	Continuous AgNP lines required plasma; printing occurred within 30 min and drying was 60 °C for 30 min [6].	Run untreated and treated coupons with fixed treatment-to-print delay before committing the slab.	Printed width/profile, open/short count, sheet resistance, and yield reported by treatment.
PRT-02	Reliable low-cost-printer interconnects were 1–2 mm; an independent DOE found a 1.78 mm straight trace optimum and 4 mm-amplitude horseshoe traces for strain tolerance [6, 9].	Retain the functional sensor fingers from their source designs; use ≥ 1.8 mm where space permits for long interconnects.	100% intended paths continuous, no adjacent shorts, and printed dimensions within $\pm 10\%$ of artwork.
PRT-03	Bare nanostructured Ag traces failed tape adhesion; thin coating or a PDMS sandwich was recommended [6].	No skin-contacting silver. Encapsulate all traces except the controlled RH vapour window.	Visual inspection, dry/wet insulation test, adhesion test, and pre/post resistance.

Evidence anchor. Direct molding and Ag-printing parameters are traceable to Abu-Khalaf *et al.* and Albrecht *et al.* [8, 9, 6]; the full 5.2 mm process remains a project verification item.

Q4. Pressure and Directional-Shear Requirements

4.4 Mechanical sensing envelope, accuracy, and separation

Albrecht *et al.* demonstrated the selected four-capacitor topology on 170 μm PET with a 40 μm PDMS dielectric—not on a 5.2 mm molded PDMS slab. The reported cell separated normal and two shear directions over 0.1–8 N, with approximately 5.2 fF/N normal and 13.1 fF/N shear sensitivity, but also showed viscoelastic drift and shear hysteresis [2]. The topology is therefore reusable; its calibration is not.

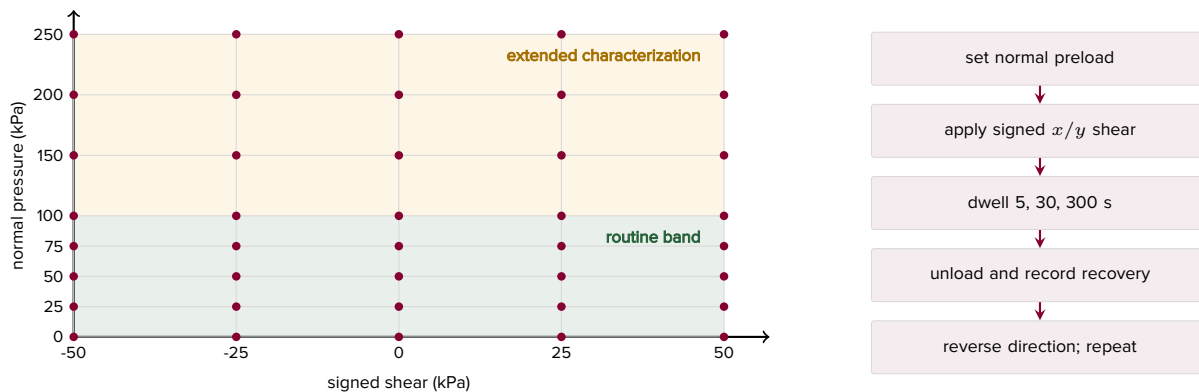


Figure Q4-3. Planned mechanical characterization matrix. Red points are required pressure–shear combinations, not measured results. The envelope is selected from the interface ranges reviewed in Q3 [15, 16, 17]; the sequence exposes hysteresis, cross-axis response, and PDMS recovery.

ID	Requirement	Why this value was selected	Verification and gate
MECH-01	Characterize normal pressure from 0 to 250 kPa; designate 0–100 kPa as the routine operating band.	D: published interfaces include measured tens-of-kPa values and a reported 226 kPa finite-element peak [15, 16, 17]. These are context, not universal injury limits.	Dead weights/load cell and known footprint; report full-range and routine-band calibration, residuals, saturation, and overload recovery.
MECH-02	Characterize signed shear from –50 to +50 kPa in both slab axes under at least three normal preloads.	D: selected to include published shear values up to roughly 40–50 kPa [15, 17].	Independent tangential force/displacement reference; report vector magnitude, angle, cross-axis coupling, and slip onset.
MECH-03	Within each calibrated band: $R^2 \geq 0.95$, $\text{NRMSE} \leq 10\%$, $\text{hysteresis} \leq 10\%$ full scale, $\text{repeatability CV} \leq 10\%$.	D/V: predeclared engineering gates; not clinical limits and not inherited from the published PET cell.	At least three independently fabricated slabs, increasing and decreasing steps, three cycles per level, and held-out verification points.
MECH-04	Non-target axis response shall be $\leq 15\%$ of target full-scale response; baseline shall recover to within 10% full scale after unloading.	D/V: matches Q3 separation and recovery gates. PDMS viscoelasticity makes both checks necessary [2].	Modality-by-modality matrix; 5, 30, and 300 s dwell/recovery records; failure cannot be averaged into a combined score.
MECH-05	Shear signals may indicate a possible slip event but shall not label slip without an independent relative-motion reference.	P/D: force and pistoning are related but distinct measurements [16, 28].	Synchronized camera/encoder trace; onset-time error and confusion matrix on held-out repetitions.

Evidence anchor. Mechanical envelope drawn from prosthetic-interface literature; sensor topology and its transfer limitation from Albrecht *et al.* [2, 15, 16, 17].

Q4. Temperature and Humidity Requirements

4.5 Separate measurands, separate protection, common coordinates

The source temperature meander and humidity IDE were demonstrated on PET and Kapton, respectively—not on molded PDMS. Their geometry and response ranges are starting specifications; transfer, protection, and calibration remain project verification items [3, 5].

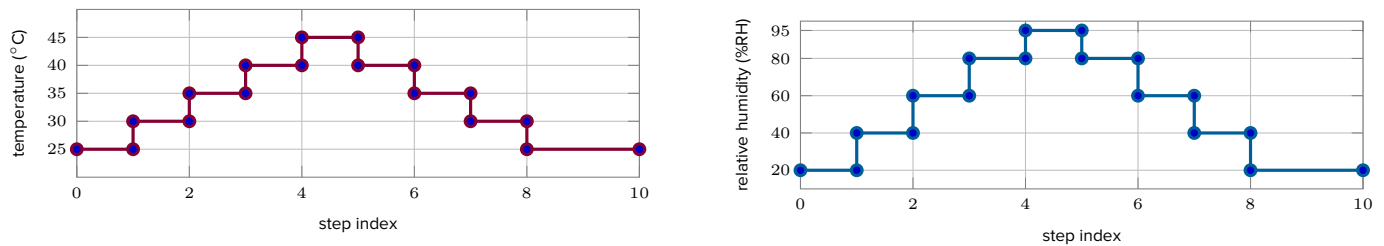


Figure Q4-4. Planned chamber profiles. Left: increasing, decreasing, and recovery temperature levels. Right: the corresponding RH sequence, repeated at two or more temperatures. These are prescribed reference inputs, not project measurements; the levels follow the calibration ranges reviewed in Q2 [3, 5, 13].

ID	Project specification	Published anchor	Verification
TEMP-01	Calibrate every temperature channel from 25 to 45 °C at 5 °C steps; target error ≤ 0.5 °C and displayed resolution ≤ 0.1 °C.	Liew used 30–100 °C; Paternò calibrated liner sensors over 25–45 °C and demonstrated substantially smaller post-correction error using different devices [3, 13].	Co-located traceable RTD/thermocouple; increasing, decreasing, and held-out temperatures; RMSE, bias, t_{90} , and recovery.
TEMP-02	Use the project meander geometry (6.9 × 13.2 mm, 0.5 mm track, 0.4 mm gap) and quantify strain cross-sensitivity.	Geometry taken from Liew <i>et al.</i> ; their sensitivities were 0.0543 and 0.1086 $\Omega/^\circ\text{C}$ for two inks [3].	Compare resistance under temperature only, strain only, and combined loading; published sensitivity is not the project calibration coefficient.
HUM-01	Calibrate vapour response from 20% to 95%RH at two or more temperatures; target error $\leq 5\%$ RH, hysteresis $\leq 10\%$ full scale, and vapour $t_{90} \leq 100$ s before final protection selection.	Published CAB IDE covered 20–90%RH and reported response below 100 s and recovery below 5 s; prosthetic liner calibration extended to 95%RH with different devices [5, 13].	Reference hygrometer; increasing/decreasing RH, held-out levels, temperature compensation, and dry recovery.
HUM-02	The humidity region shall receive vapour through a hydrophobic, vapour-permeable barrier; liquid shall not directly contact Ag traces or electrical joints.	Paternò protected non-waterproof sensors with breathable waterproof fabric; the printed IDE paper did not test prosthetic sweat [13, 5].	Vapour steps followed by controlled artificial-sweat droplets; inspect leakage, saturation, baseline recovery, and post-exposure calibration.
HUM-03	Vapour RH and liquid sweat shall remain separate software states; a saturated or nonrecovering IDE shall be flagged, not converted to a higher “discomfort” score.	D/V: measurement-quality requirement arising from the different transport mechanisms.	Quality flag triggered by out-of-range capacitance, recovery failure, leakage, or reference disagreement.

Evidence anchor. Temperature and RH geometry/performance precedents from Liew, Ait-Mammar, and Paternò *et al.* [3, 5, 13].

Q4. Acquisition, Coordinates, and Interface-Condition Mapping

4.6 Data integrity is part of the sensor specification

The physical slab does not produce a meaningful map by itself. Every channel must retain its units, calibration version, timestamp, and CAD coordinate before interpolation. Kwak *et al.* demonstrated a wireless prosthetic interface at 30 Hz; Paternò *et al.* logged slower temperature/RH channels every 5 s; Turner *et al.* compared interpolation methods on a three-dimensional prosthetic socket [19, 13, 21]. These precedents justify a multirate pipeline rather than forcing every modality to the same physical response speed.

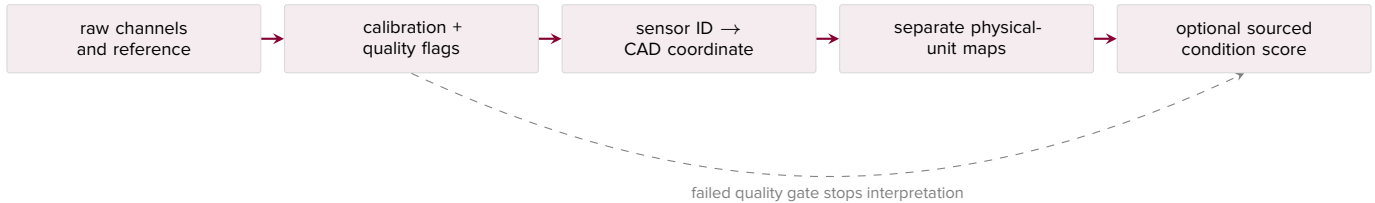


Figure Q4-5. Traceable mapping pipeline. A smooth image is the last step, not the evidence source; failed calibration or coordinate registration cannot be hidden by interpolation.

ID	Requirement	Literature/design basis	Acceptance evidence
DAQ-01	Pressure/shear shall be sampled at ≥ 50 Hz; temperature/RH at ≥ 0.2 Hz. All records shall share one monotonic clock.	D: 50 Hz gives margin over a published 30 Hz prosthetic system; 0.2 Hz follows a 5 s liner microclimate precedent [19, 13].	Timestamp audit; channel rate, jitter, and lost samples reported during a 30 min run.
DAQ-02	Inter-channel time skew shall be ≤ 20 ms for pressure/shear; packet loss shall be $< 1\%$; raw and filtered values shall both be retained.	D/V: engineering synchronization and traceability targets.	Inject a common electrical/mechanical event; verify timestamps, sample count, and filter delay.
MAP-01	Each channel shall have a unique ID, modality, unit, slab ID, CAD (x, y, z) coordinate, active footprint, orientation, and calibration version.	Personalized liner work demonstrates the importance of registered placement [13]; this schema is project-specific.	Automated manifest-to-CAD check: zero duplicate IDs, missing units, or unassigned coordinates.
MAP-02	Known hotspot median localization error shall not exceed half the nearest-neighbour sensor pitch; the 95th percentile and boundary errors shall also be reported.	D/V: Q3 preregistered gate; Turner demonstrates interpolation choices but not this array's accuracy [21].	Blinded stimulus positions; true/detected coordinates, error vectors, and confidence/coverage mask.
MAP-03	Display pressure, signed shear, temperature, and RH separately before any fusion. Show sensor markers and measured coverage; do not extrapolate a clinical conclusion outside validated coverage.	Prosthetic maps can be smoothly interpolated, but interpolation is not measurement [21].	Screenshot plus numerical test: known inputs, correct units, correct coordinates, no value outside coverage without a flag.
ALG-01	Any combined interface-condition score shall publish its equation, normalization range, threshold source, version, and missing-channel rule. It shall not be labelled a validated discomfort-risk score without participant evidence.	D: claim boundary established in Q1–Q3 and supported by the variability reported in interface-pressure reviews [14].	Software unit tests, versioned configuration, and a display disclaimer until participant validation is completed.

Evidence anchor. Sampling precedents from Kwak and Paternò; spatial mapping precedent from Turner *et al.* [19, 13, 21].

Q4. Durability, Safety, Manufacturability, and Traceability

4.7 Master design gate before Q5 testing



Figure Q4-6. Build-and-test release sequence. A failed process coupon or flat-slab calibration blocks curved integration. Published benchmarks include 1,000-cycle prosthetic-sensor and > 10,000-cycle printed-PDMS conductor tests [19, 6].

ID	Requirement	Evidence basis	Release criterion
DUR-01	Stage cyclic characterization at 1,000 then 10,000 cycles using compression, signed shear, and liner-relevant bending.	1,000-cycle soft prosthetic-sensor and > 10,000-cycle printed-PDMS conductor precedents [19, 6].	No open/short, delamination, exposed silver, or leakage; report $\Delta R/R_0$, calibration shift, hysteresis, and 24 h recovery. Do not substitute continuity for sensor accuracy.
ENV-01	Repeat calibration after 35–37 °C conditioning, 95%RH, artificial sweat, and combined cyclic loading.	Prosthetic microclimate and biofluid exposure are established integration concerns [13, 19].	No damage and no > 10% full-scale baseline shift after defined recovery; failing modalities are reported separately.
SAFE-01	No conductive silver, sharp transition, or fluid path shall contact skin. Skin-side materials, cleaning method, and adhesives shall be identified before participant use.	Silicone encapsulation and protected sensor integration are published precedents [19, 13]; this is an engineering prototype, not yet cleared for human use.	Dry/wet insulation, visual/tactile inspection, documented material list, and institutional safety review before any participant protocol.
MFG-02	Every slab shall carry a traveler recording material batch, mix ratio, degassing, cure, plasma settings, print delay, ink batch, print passes, drying, encapsulation, and inspection.	Direct-PDMS studies show sensitivity to surface treatment, geometry, precure, and solvent interaction [6, 7, 9].	No missing process fields; specimen ID links traveler, calibration files, and CAD coordinate manifest.
MFG-03	A bill of materials shall list quantity, supplier, unit cost, reusable equipment, AUB availability, and fabrication time; cost is reported, not guessed.	D/V: directly answers Q1 manufacturability; literature cannot establish local inventory or 2026 pricing.	Completed department availability table, receipts/quotes, labour log, yield, and cost per passing slab.
SYS-01	A slab passes only if geometry, electrical continuity, modality calibration, cross-sensitivity, coordinate registration, environmental recovery, and data integrity all pass.	D: system-level gate derived from Q1 and the eight Q3 functional blocks.	Signed requirements matrix with P/D/V status, raw evidence path, pass/partial/fail result, and corrective action.

Evidence anchor. Durability precedents from printed PDMS conductors and soft prosthetic-interface sensors [6, 19]; process traceability is required because those source processes are not identical to the present build.

V = REQUIREMENTS DEFINED; COMPLETE 5.2 mm SYSTEM NOT YET VERIFIED

The literature supports every major functional mechanism and supplies concrete starting processes, geometries, ranges, and validation methods. It does **not** supply a completed equivalent of the present slab. The final Q4 decision is therefore to fabricate the 5.2 mm H0 PDMS slab under a recorded process window, apply the modality and system gates above, and advance to curved phantom or whole-liner integration only after the flat specimen passes. Hydrogel remains a later controlled comparison; the map remains an *interface-condition map* until participant evidence supports a discomfort or clinical-risk interpretation.

Q5. Bench and Simulated-Use Testing

5.1 Literature-backed test programme and evidence boundary

Q5 converts the literature synthesis, functional blocks, and specifications into a controlled test programme. The cited papers establish sensing principles, reference apparatus, loading sequences, environmental ranges, and reporting metrics. They do not establish the performance of the present ink–printer–PDMS–sensor combination. The project must generate its own raw records, calibration models, uncertainty estimates, plots, and pass/partial/ fail decisions. This structure follows the assignment requirement to state the test, materials and tools, rationale, expected outcome, and impact on the overall design.

Interpretation rule: An “expected outcome” below is a predeclared engineering expectation, not a measured result. A literature value is marked as precedent; a Q4 value is a project specification; only a recorded project measurement can close a verification item.

Origin	Clickable evidence chain	Q4 requirement family	Q5 test IDs
Q1-1	Q2 2.1 → Q3A	GEO, MAT, PRT, SYS	F0, X1, SYS1
Q1-2	Q2 2.4.1 → Q3 3.3	HUM, ENV, SAFE	H1, H2, D1
Q1-3	Q2 2.3 / 2.4 → Q3 3.1	MECH, TEMP, HUM	M1–M3, T1, H1
Q1-4	Q2 2.2 → Q2 2.5	MFG, PRT, MAT	F0, R1
Q1-5	Q2 2.7 → Q3 3.5	DAQ, MAP, ALG	X1, MAP1, SYS1
Q1-6	Q2 2.2 → Q3 3.7	MAP, GEO	MAP1, S1
Q1-7	Q2 2.5 → Q3 3.8	DUR, ENV, SAFE	D1, D2, R1

Table Q5-1. Clickable Q1–Q5 traceability. The links open the detailed literature, operational definitions, and metrics instead of repeating those sections. The complete specification set is in [Q4 Section 4.1](#).

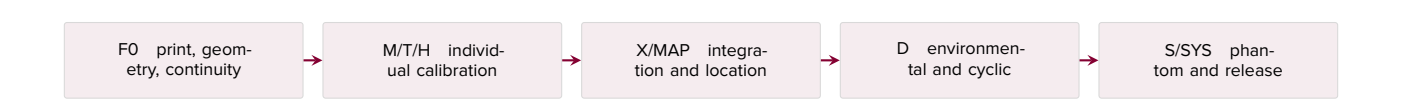


Figure Q5-1. Gated execution sequence. A failure at an earlier gate triggers rework or a documented partial result; it cannot be hidden by a later combined score. This implements the sequence defined in [Q3 Section 3.6](#) and [Q4 Section 4.7](#).

Q5. Specimen Release and Print-Quality Bench Tests

5.2 F0 — release each printed specimen before calibration

The PET sheet in [Q2 Section 2.2](#) demonstrates first-stage artwork transfer; it is not the calibration specimen. F0 applies to direct-on-PDMS coupons and then to the complete $56 \times 55 \times 5.2$ mm slab. Liew *et al.* used optical microscopy and design-to-print dimensional comparison before electrical temperature characterization [3]. Albrecht *et al.* combined optical inspection, profilometry, and electrical monitoring for printed electrodes and PDMS-based conductors [2, 6]. These methods support the inspection categories below; the acceptance limits remain those declared in [Q4 Section 4.3](#).

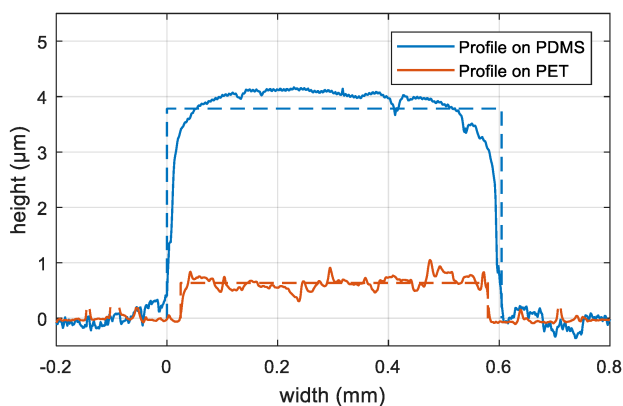
Step	Materials and tools	Recorded evidence and rationale	Expected outcome and design impact
F0-1	Printed coupon/slab, CAD export, caliper, optical microscope or USB microscope, scale target.	Photograph every modality and pad; measure line width, gap, active footprint, edge clearance, slab dimensions, and layer thickness. This detects print spreading, nozzle loss, mold error, and misregistration before electrical fitting.	Complete image set and dimension table. Out-of-tolerance geometry changes the print file, surface treatment, drop spacing, mold, or alignment method before calibration.
F0-2	Four-wire meter where available; otherwise calibrated DMM, guarded LCR meter, probes, documented cables.	Record conductor resistance, open/short status, pad-to-pad isolation, and the zero-load baseline of every channel. Repeat after 10 min to identify unstable contacts.	Every channel has a unique ID and stable baseline. An open, short, or drifting contact is repaired or the specimen is rejected; missing channels are not interpolated away.
F0-3	Profilometer where available; alternatively a sacrificial step coupon, microscope focus method, or cross-sectional witness.	Record printed height and PDMS swelling/relief. The same-printer study reported a much larger apparent profile on PDMS than PET, partly from solvent-induced swelling [6].	Geometry is reported as measured, not inferred from deposited ink volume. Excess swelling changes drying, surface treatment, or the chosen ink route.
F0-4	Adhesive-tape screen, gentle bend mandrel, insulation probe, camera.	Inspect adhesion, cracking, exposed silver, encapsulation discontinuity, and continuity before and after a non-destructive handling screen.	Only intact, insulated, electrically continuous specimens advance. Failure changes encapsulation, trace geometry, or handling fixtures.
F0-5	Batch traveller, printer log, ink lot, PDMS ratio/cure record, specimen label.	Preserve printer settings, substrate preparation time, cure, drying, inspection, and operator. Direct-on-PDMS performance is sensitive to surface treatment and process history [8, 9].	A traceable build record links every later result to one specimen and prevents selective reporting of only successful prints.



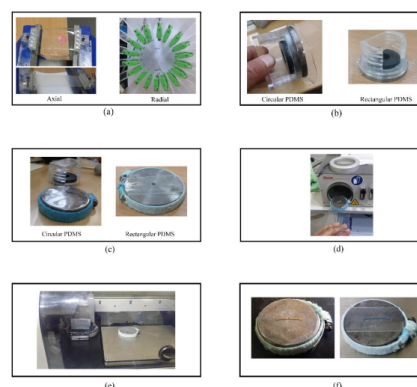
Figure Q5-2. F0 specimen traveller. The released specimen ID becomes the key used by every calibration file, CAD coordinate, photograph, and durability record.

Q5 Literature Evidence Plate: Direct Printing and Release

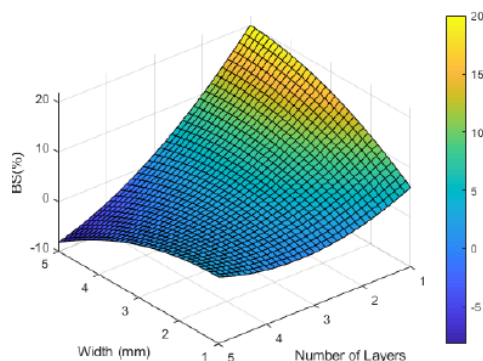
These four source figures are the visual basis for F0. They show the substrate effect, an implemented PDMS-printing workflow, a geometry/process response, and the electrical calibration obtained from an inkjet-printed meander. They support the methods in [Q5 Section 5.2](#) and the material route in [Q2 Section 2.5](#); they are not results from the present 5.2 mm slab.



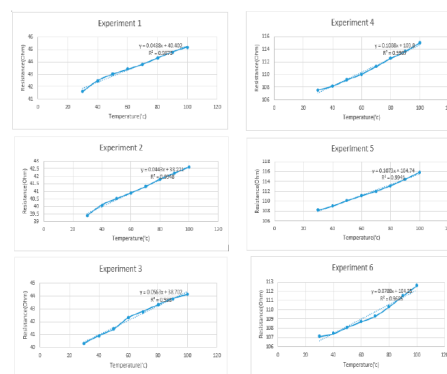
(a) Substrate-dependent printed profile. Original Fig. 2 in Albrecht *et al.*: silver printed with the same settings produced an apparent mean height of $3.8 \mu\text{m}$ on PDMS and $0.6 \mu\text{m}$ on PET, with PDMS swelling identified as a major contributor [6]. Open DOI.



(b) Implemented direct-on-PDMS process. Original Fig. 4 in Abu-Khalaf *et al.*: pre-stretching and clamping, surface treatment, inkjet printing, and oven sintering of silver traces on PDMS [8]. Open DOI.



(c) Geometry–process trade-off. Original Fig. 4 in Abu-Khalaf *et al.*: response-surface prediction of breakdown strain versus printed line width and number of layers. It supports recording geometry and print history before durability interpretation [9]. Open DOI.



(d) Printed-meander calibration precedent. Original Fig. 5 in Liew *et al.*: six resistance–temperature series for commercial and in-house AgNP inks. F0 first verifies the printed geometry and continuity; T1 then fits this type of calibration [3]. Open DOI.

Q5. Pressure and Directional-Shear Bench Testing

5.3 M1–M3 — calibration, separation, and spatial coverage

The closest direct force-cell precedent used a voice-coil actuator with an integrated force sensor, an Agilent E4980A LCR meter at 100 kHz and 1 V AC, and 0.1–8 N loading in 20% increments every 5 s, followed by unloading and three repeated cycles [1, 2]. Q5 reuses the stepped loading, unloading, directional reversal, and independent force reference. It does not copy the paper's calibration coefficients because the present sensor is printed and enclosed in a different 5.2 mm PDMS structure. The project envelope and gates are clickable in [Q4 Section 4.4](#).

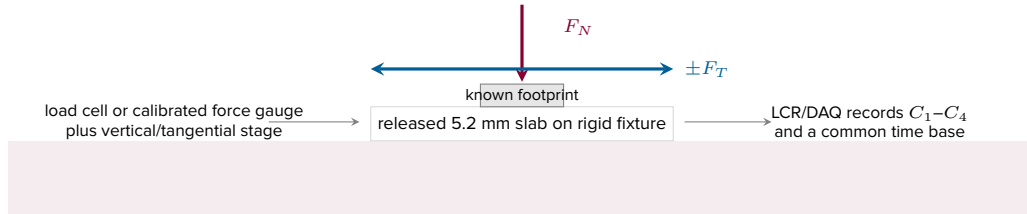


Figure Q5-3. Mechanical bench arrangement. A motorized actuator is preferred for controlled ramps; calibrated masses, a force gauge, and a low-friction translation stage are acceptable lower-cost implementations if force and displacement are recorded independently.

Test	Procedure and tools	Rationale and expected outcome	Output and design impact
M1	Apply increasing/decreasing normal pressure from 0 to 250 kPa through a measured footprint; use at least three cycles per level and held-out points. Record force, displacement, C_1 – C_4 , dwell, and recovery.	Establishes the usable range, saturation, viscoelastic hysteresis, and baseline recovery. All four capacitors should change coherently under normal loading.	Calibration curve, residuals, R^2 , NRMSE, hysteresis, coefficient of variation, and recovery. Failure changes footprint, spacer/cover thickness, preload, electrode overlap, or fitting model.
M2	At three or more normal preloads, apply signed shear from -50 to $+50$ kPa in both slab axes; reverse direction and include 5, 30, and 300 s dwells. Record an independent tangential force/displacement reference.	Tests direction separation and normal-load cross-coupling. Opposed capacitor pairs should change with opposite signs for the driven axis, while the non-target pair remains bounded [2].	Vector sensitivity, angular error, cross-axis response, hysteresis, and recovery. Failure changes alignment, four-sector geometry, dielectric thickness, or compensation matrix.
M3	Apply M1/M2 stimuli at registered sensor centres, between adjacent cells, and near slab boundaries. Randomize positions and blind the software operator.	Determines whether spacing and placement detect localized conditions rather than only centre loads.	Coverage mask, missed-event rate, true/detected coordinate pairs, and edge error. Failure changes sensor density, placement, interpolation boundaries, or the claimed coverage region.

$$P = F_N/A,$$

$$C_x = C_1 - C_3,$$

$$\tau = F_T/A,$$

$$C_y = C_2 - C_4,$$

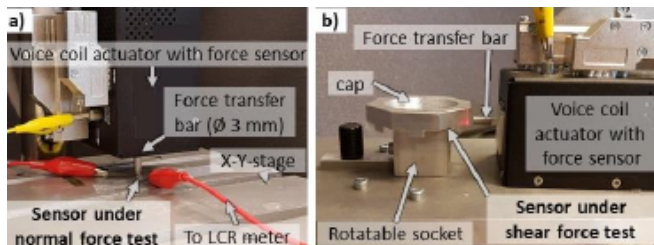
$$C_z = \frac{1}{4}(C_1 + C_2 + C_3 + C_4),$$

$$H_{FS} = \max |y_{\uparrow} - y_{\downarrow}|/FS.$$

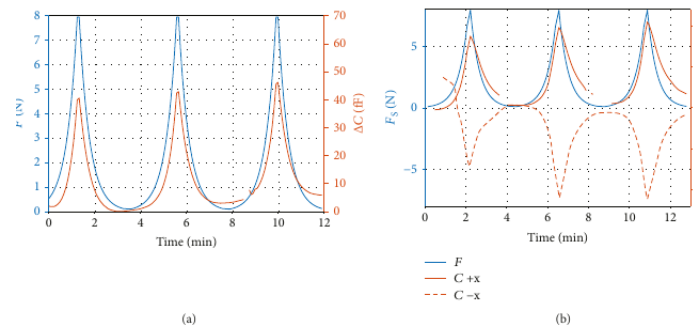
Pressure and shear are assigned only after the reference footprint, force, sign convention, calibration version, and uncertainty are recorded. A force signal alone is not proof of slip; relative motion requires the independent reference described in [Q3 Section 3.4](#) [16, 28].

Q5 Literature Evidence Plate: Force and Motion Benches

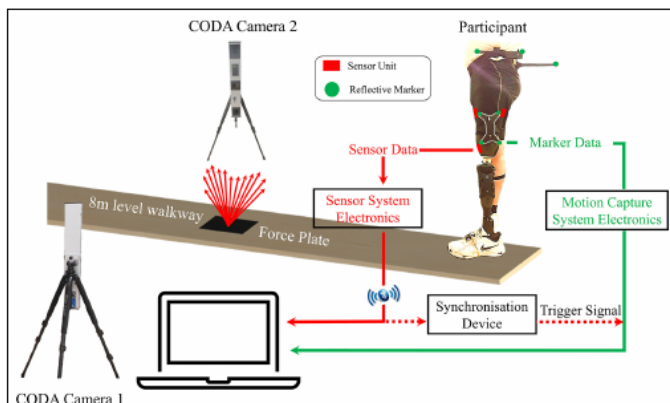
The closest published apparatus separate controlled normal force, controlled shear, and relative limb–socket motion. This plate therefore supports the independent references and signed loading sequence in [Q5 Section 5.3](#), not a claim that the present slab will reproduce the published transfer functions.



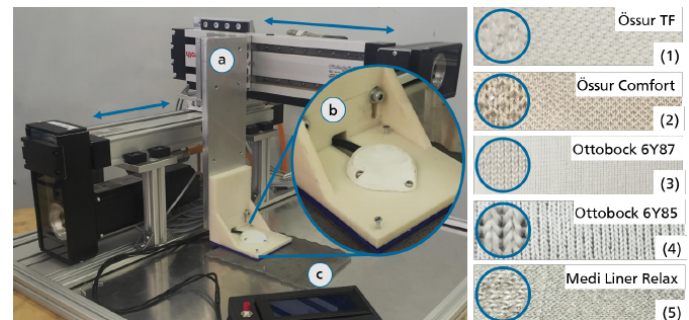
(a) Controlled normal/shear fixtures. Original Fig. 1 in Albrecht *et al.* (ALLSENSORS 2018): voice-coil/force-sensor arrangements for normal loading and rotatable signed shear [1]. The image directly motivates separate F_N and F_T references in M1–M2.



(b) Normal/shear channel separation. Original Fig. 9 in Albrecht *et al.*: time-domain normal-force response and opposite capacitance signs under reversed shear direction in the printed sensor [2]. Open DOI.



(c) Synchronized interface stress and motion. Original Fig. 1 in Tang *et al.*: pressure/shear acquisition synchronized with a 3D motion-capture system and force plate [16]. This is the precedent for treating slip or pistoning as an independently labelled event. Open DOI.



(d) Low-motion test rig and liner surfaces. Original Fig. 2 in Noll *et al.*: a movable cantilever, fixed 2D-motion sensor, and five liner/reference textures [28]. It supports an independent motion check under controlled surface and direction changes. Open DOI.

Q5. Temperature, Humidity, and Sweat-Protection Tests

5.4 T1, H1, and H2 — separate thermal, vapour, and liquid evidence

Liew *et al.* inspected the printed meander and calibrated resistance from 30 to 100 °C in 10 °C increments with a digital multimeter [3]. Jäger *et al.* later characterized inkjet-printed resistive sensors through temperature cycles and reported TCR, hysteresis, non-repeatability, drift, and maximum error following an IEC-based method [4]. Ait-Mammar *et al.* swept 20–90%RH at 25 °C in a climatic chamber, used increasing and decreasing steps, and separately applied 20–80%RH transients while measuring with a reference hygrometer [5]. Q5 adopts these measurement patterns within the prosthetic-interface ranges specified in [Q4 Section 4.5](#); it keeps vapour calibration and liquid-sweat protection as different tests.

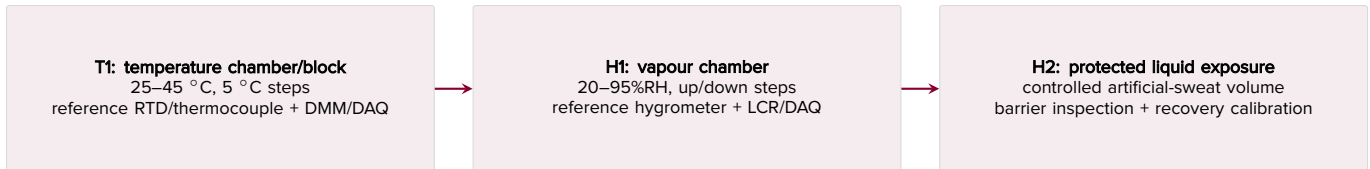


Figure Q5-4. Ordered microclimate evidence. T1 and H1 establish dry calibration before H2 challenges the selected protective structure. Saturated salt solutions provide literature-defined RH fixed points when a programmable chamber is unavailable [27]; a calibrated reference hygrometer must still be logged beside the specimen.

Test	Materials, tools, and sequence	Expected evidence	Design impact
T1	Released slab, temperature chamber/block, co-located calibrated RTD or thermocouple, DMM/DAQ. Apply 25–45 °C at 5 °C steps up and down; stabilize, hold out intermediate temperatures, and repeat after bending.	Resistance-temperature model, TCR, sensitivity, bias, RMSE, t_{90} , hysteresis, non-repeatability, and drift. Expected: monotonic repeatable response with Q4 error and recovery gates met.	Revise meander geometry, readout current, thermal isolation, encapsulation, compensation, or sampling rate. A wide 30–100 °C coupon screen is not substituted for the 25–45 °C integrated-slab calibration.
H1	Vapour chamber or sealed salt-reference vessels, calibrated hygrometer, temperature reference, LCR/DAQ. Apply 20–95%RH up/down at two or more temperatures, with held-out RH levels and dry recovery.	Capacitance-RH model, temperature compensation, hysteresis, repeatability, $t_{10 \rightarrow 90}$, $t_{90 \rightarrow 10}$, saturation, and dry baseline. Expected: stable vapour response through the selected vent/barrier.	Revise IDE spacing/coating, barrier area, vent geometry, temperature compensation, equilibration time, or the stated RH range.
H2	Artificial sweat of documented composition, micropipette or syringe, hydrophobic vapour-permeable barrier, absorbent containment, camera, leakage inspection, and H1 references. Apply controlled volumes outside the protected region; dry and repeat held-out H1 points.	Liquid ingress, leakage current, short/open status, peak saturation, recovery time, and post-exposure calibration shift. Expected: vapour access without liquid contact with silver or joints.	Reject the barrier, increase sealed perimeter, move the sensor, separate the liquid state in software, or add a replaceable protective layer. Do not interpret saturation as higher discomfort.

$$\alpha = \frac{R(T) - R_0}{R_0(T - T_0)},$$

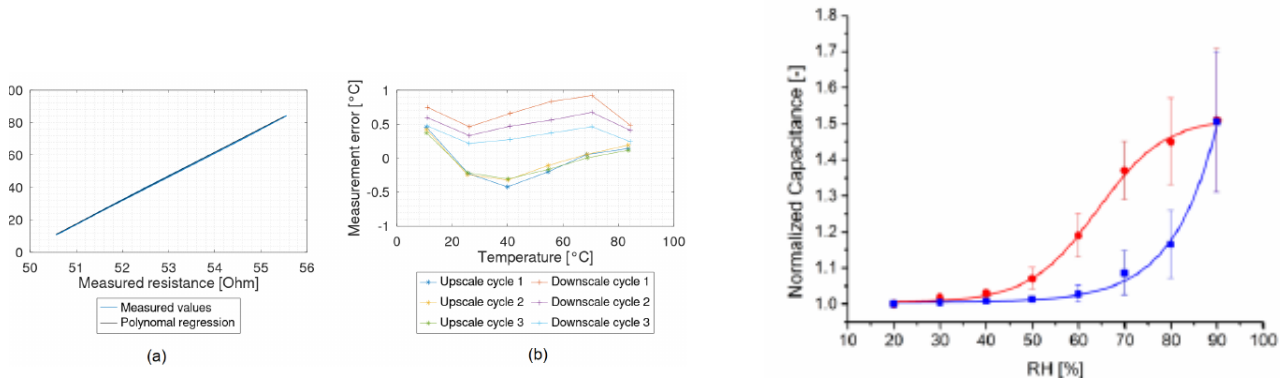
$$S_T = \frac{\Delta R}{\Delta T},$$

$$D_{post} = \frac{y_{post} - y_{pre}}{FS} \times 100\%.$$

The published humidity response below 100 s and recovery below 5 s belong to the CAB-coated Kapton device, not to the protected PDMS slab [5]. The barrier and PDMS enclosure require new project response-time measurements.

Q5 Literature Evidence Plate: Temperature and Humidity

These figures show the actual curve shapes and reference values behind T1 and H1. They justify regression, error, hysteresis, and fixed-RH checks in [Q5 Section 5.4](#); H2 remains a separate liquid-sweat protection test because neither dry calibration plot proves liquid exclusion.



(a) Standardized temperature error reporting. Original Fig. 5 in Jäger *et al.*: a fitted characteristic curve paired with the cycle-by-cycle measurement-error curve. The same data are used to quantify TCR, hysteresis, non-repeatability, and maximum error [4]. Open DOI.

(b) Humidity upswEEP/downswEEP. Original Fig. 4 in Ait-Mammar *et al.*: normalized capacitance versus RH for three devices, including the separation between adsorption and desorption curves [5]. This is the visual precedent for the H1 hysteresis metric. Open DOI.

TABLE 2. Equilibrium Relative Humidity of Selected Saturated Salt Solutions from 0 to 100 °C										
T °C	Relative Humidity, %									
	Cesium Fluoride	Lithium Bromide	Zinc Bromide	Potassium Hydroxide	Sodium Hydroxide	Lithium Chloride	Calcium Bromide	Lithium Iodide	Potassium Acetate	Potassium Fluoride
0		7.75 ± 0.03				11.23 ± 0.34				
5	5.32 ± 1.9	7.43 ± 0.76	8.86 ± 0.89	14.34 ± 1.7		11.26 ± 0.47		21.68 ± 0.30		
10	4.89 ± 1.6	7.14 ± 0.69	8.49 ± 0.74	12.34 ± 1.4		11.29 ± 0.61	21.62 ± 0.50	20.61 ± 0.25	23.38 ± 0.33	
15	4.33 ± 1.4	6.86 ± 0.63	8.19 ± 0.61	10.68 ± 1.1	9.57 ± 2.8	11.30 ± 0.35	20.20 ± 0.50	19.57 ± 0.20	23.40 ± 0.32	
20	3.83 ± 1.1	6.61 ± 0.58	7.94 ± 0.49	9.32 ± 0.90	8.91 ± 2.4	11.31 ± 0.31	18.50 ± 0.50	18.56 ± 0.16	23.11 ± 0.25	
25	3.39 ± 0.94	6.37 ± 0.52	7.73 ± 0.39	8.23 ± 0.72	8.24 ± 2.1	11.30 ± 0.27	16.50 ± 0.20	17.56 ± 0.13	22.31 ± 0.32	
30	3.01 ± 0.77	6.16 ± 0.47	7.62 ± 0.31	7.38 ± 0.56	7.58 ± 1.7	11.28 ± 0.24		16.57 ± 0.10	21.61 ± 0.33	30.85 ± 1.3
35	2.69 ± 0.63	5.97 ± 0.43	7.53 ± 0.25	6.73 ± 0.44	6.92 ± 1.5	11.25 ± 0.22		15.57 ± 0.08		24.59 ± 0.94
40	2.44 ± 0.52	5.80 ± 0.39	7.54 ± 0.20	6.26 ± 0.35	6.26 ± 1.2	11.21 ± 0.21		14.55 ± 0.06		22.66 ± 0.81
45	2.24 ± 0.44	5.65 ± 0.33	7.59 ± 0.17	5.94 ± 0.29	5.60 ± 1.0	11.16 ± 0.21		13.49 ± 0.05		21.46 ± 0.70
50	2.11 ± 0.40	5.53 ± 0.31	7.70 ± 0.16	5.72 ± 0.27	4.94 ± 0.85	11.10 ± 0.22		12.38 ± 0.05		20.80 ± 0.62
55	2.04 ± 0.38	5.42 ± 0.28	7.67 ± 0.17	5.58 ± 0.28	4.27 ± 0.73	11.03 ± 0.23		11.22 ± 0.05		20.60 ± 0.56
60	2.03 ± 0.40	5.33 ± 0.25	8.09 ± 0.19	5.49 ± 0.32	3.61 ± 0.65	10.95 ± 0.26		9.98 ± 0.06		20.77 ± 0.53
65	2.08 ± 0.44	5.27 ± 0.23	8.38 ± 0.24	5.41 ± 0.39	2.95 ± 0.60	10.86 ± 0.29		8.65 ± 0.07		21.18 ± 0.53
70	2.20 ± 0.52	5.23 ± 0.21	8.72 ± 0.30	5.32 ± 0.50	2.29 ± 0.60	10.75 ± 0.33		7.23 ± 0.09		21.74 ± 0.56
75	2.37 ± 0.62	5.20 ± 0.19			1.63 ± 0.64	10.64 ± 0.38				22.33 ± 0.61
80	2.61 ± 0.76	5.20 ± 0.18				10.51 ± 0.44				22.85 ± 0.69
85		5.22 ± 0.17				10.38 ± 0.51				23.20 ± 0.80
90		5.26 ± 0.17				10.23 ± 0.59				23.27 ± 0.93
95		5.32 ± 0.16				10.07 ± 0.67				
100		5.41 ± 0.17				9.90 ± 0.77				

TABLE 2. Equilibrium Relative Humidity of Selected Saturated Salt Solutions from 0 to 100 °C—Continued										
T °C	Relative Humidity, %									
	Magnesium Chloride	Sodium Iodide	Potassium Carbonate	Magnesium Nitrate	Sodium Bromide	Calcium Chloride	Potassium Iodide	Sodium Chloride	Sodium Nitrate	Sodium Chloride
0	33.66 ± 0.33		43.13 ± 0.66	60.35 ± 0.55						75.51 ± 0.34
5	33.60 ± 0.28	42.42 ± 0.99	43.13 ± 0.50	58.86 ± 0.43	63.51 ± 0.72		73.30 ± 0.34	77.19 ± 0.12	78.57 ± 0.52	75.65 ± 0.27
10	33.47 ± 0.24	41.83 ± 0.83	43.13 ± 0.39	57.36 ± 0.33	62.15 ± 0.60		72.11 ± 0.31	75.66 ± 0.06	77.53 ± 0.45	75.67 ± 0.22
15	33.30 ± 0.21	40.86 ± 0.70	43.15 ± 0.33	55.87 ± 0.27	60.68 ± 0.51		70.98 ± 0.28	74.13 ± 0.06	76.46 ± 0.39	75.61 ± 0.18
20	33.07 ± 0.18	39.63 ± 0.59	43.16 ± 0.33	54.38 ± 0.23	59.14 ± 0.44		69.80 ± 0.26	72.52 ± 0.05	75.36 ± 0.35	75.47 ± 0.14
25	32.78 ± 0.16	38.17 ± 0.50	43.16 ± 0.39	52.89 ± 0.22	57.57 ± 0.40	64.92 ± 3.5	68.86 ± 0.24	70.85 ± 0.04	74.25 ± 0.32	75.29 ± 0.12
30	32.44 ± 0.14	36.15 ± 0.43	43.17 ± 0.50	51.49 ± 0.24	56.03 ± 0.38	61.88 ± 2.8	67.89 ± 0.23	69.12 ± 0.03	73.14 ± 0.31	75.09 ± 0.11
35	32.05 ± 0.13	34.73 ± 0.39		49.91 ± 0.29	54.55 ± 0.30	58.63 ± 2.2	66.96 ± 0.23		72.06 ± 0.32	74.87 ± 0.12
40	31.69 ± 0.13	32.88 ± 0.37		48.42 ± 0.37	53.17 ± 0.41	55.48 ± 1.8	66.09 ± 0.23		71.00 ± 0.34	74.68 ± 0.13
45	31.19 ± 0.13	31.02 ± 0.37		46.93 ± 0.47	51.95 ± 0.47	52.56 ± 1.5	65.26 ± 0.24		69.99 ± 0.37	74.52 ± 0.16
50	30.54 ± 0.14	29.21 ± 0.40		45.44 ± 0.60	50.93 ± 0.55	50.01 ± 1.4	64.49 ± 0.26		69.04 ± 0.42	74.43 ± 0.19
55	29.93 ± 0.16	27.50 ± 0.45			50.15 ± 0.65	48.02 ± 1.4	63.78 ± 0.28		68.15 ± 0.49	74.41 ± 0.24
60	29.26 ± 0.18	25.95 ± 0.52			49.66 ± 0.78	46.74 ± 1.5	63.11 ± 0.31		67.35 ± 0.57	74.50 ± 0.30
65	28.54 ± 0.21	24.62 ± 0.62			49.49 ± 0.94	46.33 ± 1.9	62.50 ± 0.34		66.64 ± 0.67	74.71 ± 0.37
70	27.77 ± 0.25	23.57 ± 0.74			49.70 ± 1.1	46.97 ± 2.3	61.93 ± 0.38		66.04 ± 0.78	75.06 ± 0.45
75	26.94 ± 0.29	22.85 ± 0.88			50.33 ± 1.3	48.80 ± 2.9	61.43 ± 0.43		65.56 ± 0.91	75.38 ± 0.55
80	26.05 ± 0.34	22.52 ± 1.0			51.43 ± 1.5	52.01 ± 3.7	60.97 ± 0.48		65.22 ± 1.1	76.29 ± 0.65
85	25.11 ± 0.39	22.63 ± 1.2					60.56 ± 0.54		65.03 ± 1.2	
90	24.12 ± 0.46	23.25 ± 1.4					60.21 ± 0.61		65.00 ± 1.4	
95	23.07 ± 0.52									
100	21.97 ± 0.60									

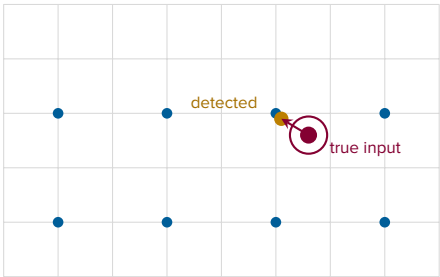
(c) Chamber-independent RH reference values. Table 2 in Greenspan lists equilibrium RH versus temperature for saturated salt solutions. Q5 may use selected salts as fixed points only when the specimen temperature and a calibrated reference hygrometer are recorded [27]. Open NIST full text.

Q5. Multimodal Cross-Sensitivity and Map Validation

5.5 X1 and MAP1 — prove that a correct map follows correct channels

Cross-sensitivity is tested by applying one controlled input while recording every channel, followed by selected combined conditions. Kwak *et al.* demonstrated synchronized multimodal prosthetic-interface acquisition at 30 Hz, Paternò *et al.* registered sensor locations and logged slower temperature/RH data, and Turner *et al.* compared spatial interpolation on a three-dimensional socket [19, 13, 21]. These precedents support synchronized acquisition and spatial visualization; they do not prove this slab’s coordinate mapping or combined score. The data schema and claim boundary are in [Q4 Section 4.6](#).

Run	Controlled input	Channels and ground truth	Required interpretation
X1-1	Normal pressure only, several levels and coordinates.	Force/load cell, C_1-C_4 , temperature, RH, timestamps.	Quantify mechanical target response and apparent thermal/RH interference; do not remove it by averaging.
X1-2	Signed shear only under fixed normal preloads.	Tangential force and displacement, four capacitances, temperature, RH.	Quantify target-axis, cross-axis, and non-mechanical response.
X1-3	Temperature only at controlled RH.	Reference temperature, meander resistance, force channels, RH channel.	Separate temperature coefficient from thermomechanical baseline shifts.
X1-4	Vapour RH only at controlled temperature, then protected liquid sweat.	Reference RH/temperature, IDE capacitance, leakage/continuity, all other channels.	Separate vapour response, swelling/load redistribution, liquid saturation, and electrical failure.
X1-5	Predeclared combinations: pressure+shear; temperature+RH; pressure+shear+heat+moisture.	All references and channels on one monotonic clock; raw and filtered records retained.	Fit compensation only on training runs; verify on held-out combinations. Report each physical-unit map before any interface-condition score.



CAD coordinate plane with measured sensor sites

For MAP1, randomize known stimulus coordinates at sensor centres, between sensors, and near boundaries. The operator viewing the software is blinded to the true coordinate. Save the stimulus coordinate, detected coordinate, nearest-neighbour pitch, interpolation method, coverage mask, and confidence flag.

$$e_{loc} = \sqrt{(x_d - x_t)^2 + (y_d - y_t)^2}$$

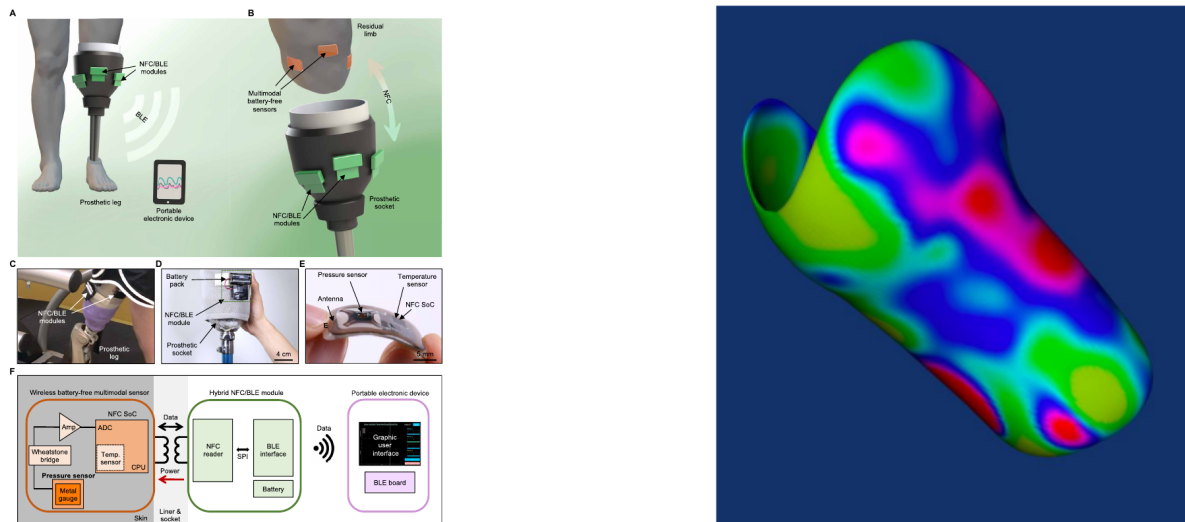
Report median, 95th percentile, and boundary error; do not report only a visually convincing heatmap.

Figure Q5-5. Blinded localization test. Blue dots are measured sensor sites, red is the applied coordinate, and yellow is the algorithm output. The Q4 gate is median error no greater than half the nearest-neighbour pitch; sensor markers and unsupported regions remain visible.

Expected outcome and impact: A successful run preserves channel identity, unit, timestamp, calibration version, and CAD coordinate; localizes held-out inputs within the Q4 gate; and leaves pressure, signed shear, temperature, and RH viewable separately. Failure changes sampling, synchronization, shielding, channel compensation, sensor density, coordinate registration, interpolation, or the displayed claim. The map remains an *interface-condition map* until participant evidence validates a discomfort or clinical-risk interpretation.

Q5 Literature Evidence Plate: Multimodal Mapping

The literature already provides reasonable solutions for synchronized multimodal acquisition, embedded liner microclimate sensing, and 3D surface visualization. X1 and MAP1 therefore test whether those functional ideas remain valid for the present printed-slab geometry and CAD manifest; see [Q5 Section 5.5](#) and [Q3 Section 3.5](#).



(a) Wireless multimodal prosthetic interface. Original Fig. 1 in Kwak *et al.*: pressure and temperature sensors, NFC/BLE modules, wearable placement, and the functional data path [19]. Open DOI. The figure is a system precedent, not evidence for the present silver/PDMS channels.

(b) Interpolated socket-surface display. Original Fig. 3 in Turner *et al.*: radial-basis interpolation applied to a 3D transtibial socket model [21]. It supports a live coordinate-registered visualization while reinforcing the need to report true-versus-detected location error. Open DOI.

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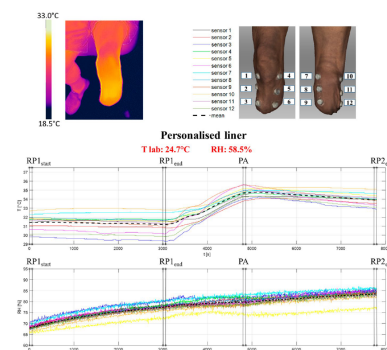


Fig. 4 Left top corner: Test image acquired by the thermo-camera at the end of the test session (the image has been recorded in the frontal view at the end of the protocol). Right top corner: Buttons positions on the residual limb during the test session (anterior and posterior views). Bottom: Temperature and relative humidity values within the prosthetic socket with the new liner. The vertical lines indicate the different 1-min stages used to analyse the data: RP1_{rest}, first minute of the 1st resting period (test minute 1); RP1_{end}, last minute of the 1st resting period (test minute 50); PA, last minute of walking on the treadmill (test minute 80); RP2_{end}, last minute of the 2nd resting period (test minute 130). Temperature and humidity room conditions are reported in red on the top of the graphs.

(c) Embedded-liner microclimate data. Original Fig. 4 in Paternò *et al.*: thermography, sensor locations, and temperature/RH traces from a personalised liner with embedded sensors [13]. It is a direct precedent for synchronized physical-unit channels and location-aware interpretation. Open DOI.

Q5. Environmental, Cyclic, and Recovery Testing

5.6 D1–D2 — measure drift instead of assuming durability

Directly printed silver conductors on PDMS have shown strong dependence on strain, recovery time, and cycle count; Albrecht *et al.* reported conductive traces after 10,000 cycles but with a large resistance increase under their specific geometry and 20% strain protocol [6]. Kwak *et al.* reported 1,000-cycle soft prosthetic-sensor testing, while Paternò *et al.* demonstrated temperature/RH operation in a liner microclimate [19, 13]. These are durability precedents, not substitutes for the present slab's post-exposure recalibration. Q5 follows the release gate in [Q4 Section 4.7](#).

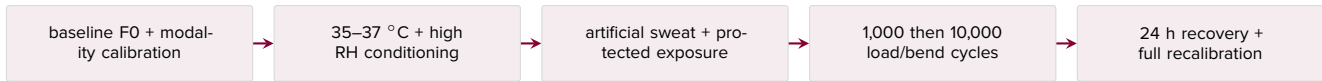


Figure Q5-6. D1–D2 exposure sequence. Intermediate inspections at defined cycle counts distinguish progressive damage from a single final failure. Conditioning, load, and recovery records share the specimen ID.

Test	Exposure and tools	Measurements	Expected outcome and impact
D1-1	Chamber or sealed enclosure at 35–37 °C and high RH; reference temperature/RH; powered slab where safe.	Baseline, resistance/capacitance, noise, leakage, drift, visual condition, and modality calibration before, during, and after conditioning.	No damage and no Q4-exceeding calibration shift after recovery. Failure changes materials, encapsulation, compensation, operating range, or recalibration interval.
D1-2	Controlled artificial sweat, selected barrier, absorbent containment, camera, electrical isolation/leakage checks.	Liquid path, saturation, recovery, open/short status, adhesion, corrosion/discolouration, and H1 recalibration.	No liquid contact with exposed silver or joints; vapour response recovers. Failure changes barrier, seal, vent position, replaceable layer, or cleaning instructions.
D2-1	Cyclic normal compression and signed shear fixture; force/displacement reference; staged 1,000-cycle screen then 10,000-cycle characterization.	Raw signals at defined cycle intervals; $\Delta R/R_0$, $\Delta C/C_0$, sensitivity, hysteresis, repeatability, baseline recovery, and inspection images.	No open/short, delamination, or unsafe exposure; performance remains inside Q4 gates. Failure changes trace geometry, neutral-axis placement, PDMS stack, bonding, or claimed service interval.
D2-2	Mandrels or curved phantom representing intended liner curvature; controlled bend radius and cycle count.	Continuity during bending, strain-only cross-sensitivity, crack location, and post-bend modality calibration.	Bending must not be interpreted as temperature, humidity, or load without compensation. Failure changes routing, strain relief, sensor orientation, or enclosure geometry.

$$D_y(\%) = \frac{y_{post} - y_{pre}}{FS} \times 100,$$

$$\Delta R/R_0 = \frac{R_N - R_0}{R_0},$$

$$CV(\%) = 100 s/\bar{y}.$$

Report raw and normalized changes because a small percentage change near zero can be misleading. Recovery at 5, 30, 300 s and 24 h separates short-term PDMS relaxation from persistent damage.

Q5. Simulated-Use Phantom Testing

5.7 S1 — controlled curved-interface scenarios before human use

The three build precedents, photographs, and material lists are retained in [Q3 Section 3.4.1](#) and are not duplicated here. Paternò *et al.* built an anatomically informed variable-volume transfemoral simulator using silicone tissue layers, a sealed hydraulic chamber, and motorized syringe pumps [24]. McGrath *et al.* built a silicone residuum with molded pores that released water inside a liner/socket under cyclic loading [26]. Phillips *et al.* reported a lower-cost dual-hardness urethane mock limb with water-filled TPU bladders and motorized syringe control [25]. Q5 uses these as selectable modules because no single phantom provides traceable normal load, signed shear, vapour RH, liquid sweat, temperature, and anatomical motion simultaneously.

Candidate module	Materials and tools	Q5 role and ground truth	Limitation and decision impact
A — variable-volume anatomical simulator	Molded silicone anatomical/tissue layers, sealed hydraulic chamber, syringe pumps, frame, volume/displacement reference, force/load references.	Apply repeatable volume changes and interface loading while checking pressure/shear localization, coverage, and map stability.	Higher fabrication complexity; it does not by itself provide controlled vapour RH. Select when anatomical loading and volume change are the dominant questions.
B — sweating residuum/socket simulator	3D-printed bone/core, silicone, molded pores, water reservoir/pump, liner/check socket, cyclic actuator, collected-volume measurement.	Challenge liquid-sweat protection and moisture-induced changes during cyclic loading; record delivered and recovered water volume.	Liquid delivery is not RH calibration. Use only after H1 dry vapour calibration; select when sweat management and wet slip are priorities.
C — low-cost bladder mock limb	Dual-hardness urethane, water-filled TPU bladders, mold/printed parts, Arduino-class control, motorized syringe, force/displacement reference.	Reproduce controlled shape/pressure change at lower reported construction cost and compare known actuation with mapped response.	Biofidelity and shear require separate validation. Select when reproducible low-cost pressure/shape testing is the priority.

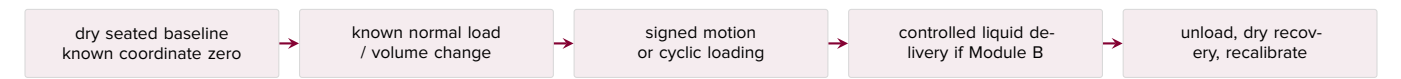
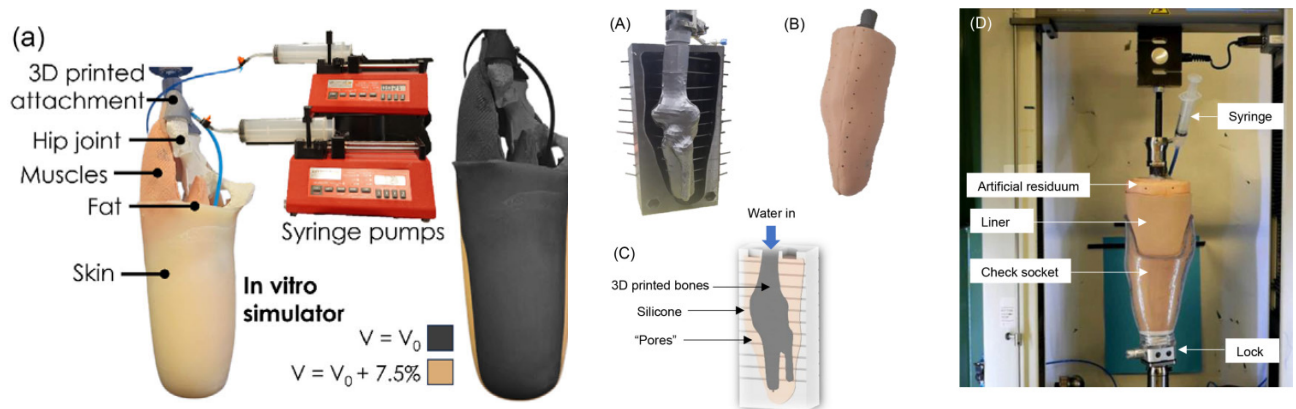


Figure Q5-7. S1 simulated-use sequence. Ground truth comes from the phantom actuator, reference load/displacement, delivered liquid volume, and known CAD locations. The sensor map is the device under test, not the source of the event label.

Expected outcome and impact: The curved assembly retains electrical continuity and coordinate identity, detects the correct imposed event and location within the Q4 gates, and returns toward baseline after unloading/drying. Failure changes slab placement, liner integration, routing, encapsulation, protection, calibration, interpolation, or the chosen phantom module. This is simulated-use evidence only; it does not authorize a human-use or clinical discomfort claim.

Q5 Literature Evidence Plate: Residual-Limb Phantoms

The three proposed modules in **Q5 Section 5.7** are all implemented research systems. Their published photographs are shown together so the project can choose a build route from demonstrated hardware rather than from a generic sketch. Their separate strengths also explain why no one module is treated as a universal ground truth.



(a) Anatomical variable-volume simulator. Original Fig. 5a in Paternò *et al.*: anatomical skeletal/tissue structure, silicone layers, sealed hydraulic chamber, and two motorized syringe pumps for controlled residual-limb volume change [24]. Open DOI.

(b) Sweating residuum/socket simulator. Original Fig. 1 in McGrath *et al.*: a 3D-printed bone/core encased in porous silicone, supplied with water and tested inside a liner/check socket under cyclic loading [26]. Open DOI.

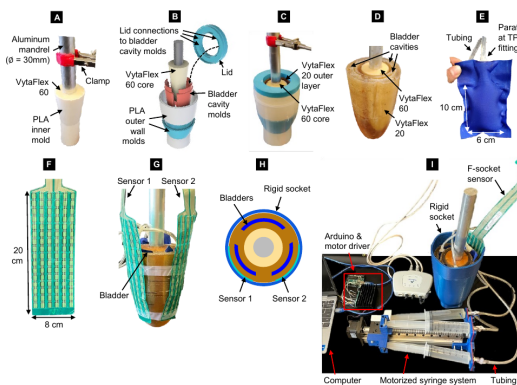


Figure 1: Design overview. A) Step one: casting the core with VytaFlex™ 20 around an aluminum mandrel. B) Mold set-up for casting the outer layer. The bladder cavity molds are held in place by clipping into the lid. The wall molds were separated to ease mold removal post-curing. C) Step two: casting the outer layer with VytaFlex™ 20. D) Final casted limb. E) TPU bladder. F) F-socket sensor, 0.15 mm thin. G) Sensor placement on the limb before donning the rigid socket. H) Top view schematic of sensor placement on the limb. I) Full system with the biofidelic mock limb, rigid socket, and motorized syringe system. Only one F-socket sensor is shown for clarity.

(c) Lower-cost biofidelic mock residual limb. Original Fig. 1 in Phillips *et al.*: dual-hardness urethane tissues, water-filled TPU bladders, mold components, and a motorized syringe/Arduino-class control route reported at below CAD 400 [25]. Open DOI.

Figure Q5-L5. Implemented phantom candidates. Paternò supplies anatomical volume change, McGrath supplies controlled liquid sweat during cyclic loading, and Phillips supplies a reproducible lower-cost pressure/shape platform. These images are duplicated here intentionally because Q5 uses them to choose test hardware; the detailed materials comparison remains in **Q3 Section 3.4.1**.

Q5. Analysis Outputs, Decision Rules, and Design Feedback

5.8 SYS1 and R1 — convert each test into auditable evidence

Every test produces a raw file, reference file, specimen/build record, analysis script or calculation sheet, figure, numerical metric table, and decision. The output formats below implement the metrics defined in [Q3 Section 3.7](#) and the planned evidence panels in [Q3 Section 3.8](#). Literature figures may explain precedent, but only project records occupy the measured-result panels.

Output	Required plot/table	Decision rule	Design feedback
F0	Annotated print photograph, CAD-versus-measured dimensions, continuity/isolation table, build traveller.	Advance only intact, identified, traceable specimens; otherwise rework/reject with cause.	Print parameters, mold, surface preparation, alignment, contacts, or encapsulation.
M1–M3	Reference input and raw channels versus time; calibration with residuals; loading/unloading hysteresis; cross-axis and recovery; localization error map.	Apply MECH-01–MECH-05 individually. Averages cannot hide a failed direction, boundary, or recovery gate.	Electrode/spacer geometry, preload, footprint, placement, compensation, or coverage claim.
T1/H1/H2	Reference and sensor versus time; fitted calibration and held-out error; t_{90} ; hysteresis; temperature compensation; pre/post-liquid shift.	Apply TEMP-01/02 and HUM-01–03 separately. Saturation or quality failure is flagged rather than converted into a condition score.	Meander/IDE geometry, barrier, vent, seal, compensation, range, or recalibration.
X1/MAPI	Modality-by-modality matrix; timestamp audit; true/detected coordinate scatter; error vectors; coverage/confidence mask; separate physical-unit maps.	Apply DAQ-01/02, MAP-01–03, and ALG-01. Combined interpretation is blocked when calibration, time, coordinate, or quality metadata are missing.	DAQ, shielding, synchronization, manifest, sensor density, interpolation, thresholds, or displayed wording.
D1/D2/S1	Drift/recovery versus exposure or cycle count; inspection images; pre/post calibration; event and location results against phantom ground truth.	Apply DUR, ENV, SAFE, and SYS gates per modality and specimen. Report pass/partial/fail/not run with the evidence path.	Materials, routing, barrier, stack, service interval, phantom module, or decision to remain at flat-slab testing.
R1	Batch yield, quantities, supplier/unit cost, reusable equipment, labour, failed-print causes, and AUB availability status.	No guessed availability or cost; attach inventory confirmation, receipts/quotes, and build time.	Process simplification, supplier choice, sensor count, reusable fixtures, or scope.

PASS all applicable gates met	PARTIAL usable block; stated limitation	FAIL requirement not met; redesign	NOT RUN no conclusion; reason recorded
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Figure Q5-8. Required verdict vocabulary. A partial result preserves a useful functional block while identifying its limitation; “not run” is not a failure and cannot be presented as evidence.

Q5 completion criterion: Q5 is complete when every planned test has a specimen ID, apparatus/reference description, controlled inputs, raw and processed outputs, metric definitions, acceptance rule, actual status, and design action. Before data collection, the section is correctly reported as a literature-backed validation plan. After testing, planned panels are replaced with measured project plots without changing the preregistered decision rules.

Evidence anchor. Bench methods and performance reporting are grounded in printed force, temperature, humidity, PDMS-conductor, prosthetic-interface, mapping, and phantom studies [2, 3, 4, 5, 6, 19, 13, 21, 24, 26, 25].

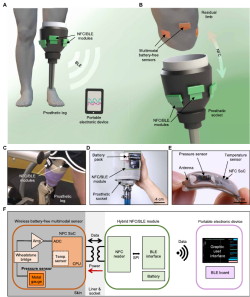
Q6. Individual Assignment – Abdulrahman Kobeissi

6.1 Three comparable state-of-the-art technologies

The comparison was restricted to systems that solve one of the three central parts of the present concept: multimodal interface sensing, physical liner integration, or spatial pressure mapping. The selected technologies are not presented as identical competitors. Together, they expose which functional blocks already have credible solutions and which combination remains specific to the proposed modular 5.2 mm PDMS liner. A commercial screen also considered the current Tekscan F-Socket system, which uses a paper-thin trimmable sensor, wired USB acquisition at up to 160 Hz, and real-time 2D/3D pressure software [31]. The pliance-PMS system was selected for the detailed commercial comparison because its official documentation discloses sensor technology, element geometry, range, calibration, channel capacity, telemetry, and mapping functions [30].

 **green check:** function demonstrated in both systems  **yellow partial:** related function, but adaptation or proof is required  **red cross:** function absent or deliberately excluded

Technology	Implemented architecture	Evidence already demonstrated	Main boundary relative to our concept
Kwak <i>et al.</i> , 2020	Millimetre-scale 3D Au/PI strain-gauge pressure sensor plus on-chip temperature sensor; battery-free inner device; paired NFC/BLE outer module and portable GUI.	Linear pressure response across the prosthetic range; real-time pressure and temperature at multiple landmarks; feasibility in transtibial and transfemoral users [19].	No humidity measurement, no directional shear output, no liner-wide interpolated map, and no printed silver-on-PDMS integration.
Paternò <i>et al.</i> , 2020	Twelve commercial Hygrochron iButtons embedded in CAD-defined blind enclosures in a personalised cryogenically machined neoprene liner.	Calibrated temperature/RH sensing, anatomical alignment, wearable liner manufacture, and a 130-min single-participant protocol [13].	Discrete 5.89-mm-thick loggers, offline retrieval, temperature/RH only, no printed electrodes, and no real-time spatial fusion.
novel pliance-PMS	Flexible capacitive pressure pads connected to a wearable analyser, SD/Bluetooth telemetry, homogeneous-pressure calibration, and 2D/3D expert software.	Up to 1024 elements; disclosed pads contain 18–90 10 × 10 mm sensels, are below 1 mm uncoated thickness, cover 10–600 kPa, and support static/dynamic socket evaluation [30].	Mature pressure-only measurement apparatus rather than a multimodal sensing liner; no printed Ag/PDMS, temperature, RH, or directional shear channel.



Kwak Fig. 1: sensors, NFC/BLE modules, and the three-block wireless data path [19].



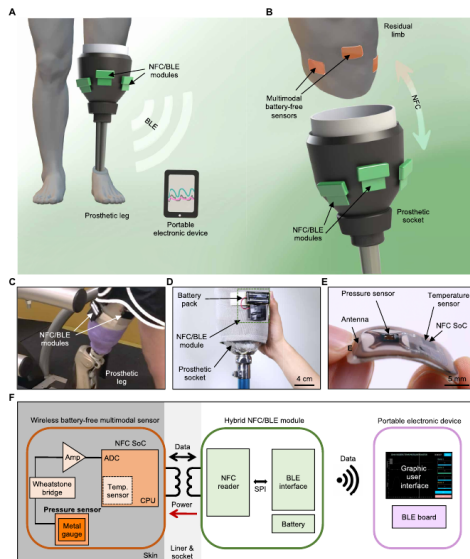
Paternò Fig. 3: CAD, machined liner halves, logger enclosures, and final liner [13].



pliance-PMS datasheet: flexible capacitive pads mounted at selected socket locations [30].

Figure Q6-1. Comparator selection. The three systems were selected by functional proximity, not visual similarity. Paper figures retain their original citations; the commercial image is an attributed product-datasheet excerpt. Detailed evidence appears on the following three pages.

Q6.1 Comparator 1 – Kwak Wireless Multimodal System



Published system. Battery-free pressure/temperature devices sit on the residual-limb skin; a battery-powered NFC/BLE module sits outside the socket and relays the data to a portable GUI. Original Fig. 1 [19].

Technical architecture.

- The pressure transducer is a buckled 3D Au/PI strain-gauge structure, approximately 1.95×1.95 mm laterally and 0.5 mm high, supported by a strain-limiting frame rather than printed directly on a liner.
- The wireless device uses an fPCB, NFC power/data at 13.56 MHz, and a 10-Hz on-chip ADC. The external module forwards data by BLE over more than 10 m; its two-AAA-cell implementation was reported for about 2 h operation.
- Reported pressure performance included $R^2 > 0.99$, hysteresis error below 0.02%, negligible static drift, and a range designed to span interface loads up to approximately 350 kPa. The design intentionally suppresses shear sensitivity rather than measuring shear.
- Multi-site pressure and temperature were demonstrated on two prosthesis users. The authors explicitly identify humidity, shear, larger participant samples, and automatic fit evaluation as extensions rather than completed functions [19].

Feature	Kwak implementation	Status	Direct comparison with our 5.2 mm slab
Normal pressure	3D metal strain-gauge transducer with silicone encapsulation and an independent calibration.	✓ shared	Our pressure block also requires normal-load calibration, but uses four printed capacitive sectors and must preserve that response after PDMS molding; see Q2 Section 2.3 .
Temperature	On-chip temperature channel on the same wireless device.	✓ shared	Both systems measure temperature, but ours uses a printed resistive meander that needs slab-specific TCR, self-heating, and bending tests in Q5 Section 5.4 .
Directional shear	Designed to be insensitive to shear; no signed x/y output.	✗ absent	Directional shear is a required independent modality in our four-capacitor design, making cross-axis separation a major additional burden in Q5 Section 5.3 .
Humidity/sweat	Short sweat/high-humidity exposure was used as a robustness check, not an RH measurement.	✗ absent	Our design includes a printed IDE humidity sensor and explicitly separates vapour calibration from liquid sweat protection.
Physical integration	Small devices adhere to skin beneath a gel liner; outer reader modules align across the socket.	⚡ partial	This minimizes in-socket batteries, but differs from sensors printed on modular PDMS slabs that become part of the liner structure.
Acquisition	Real-time, battery-free inner sensor with NFC/BLE relay and GUI.	✓ reusable	The three-block sensor–gateway–display architecture is a strong precedent. Our implementation still must prove synchronized acquisition for four signal types and multiple slabs.
Spatial interpretation	Multiple selected anatomical sites; no continuous liner-wide interpolation or displayed coverage mask.	⚡ partial	Our coordinate manifest and blinded localization test extend the multinodal idea into a spatial interface-condition map; see Q5 Section 5.5 .
Clinical evidence	Feasibility on two users, without a discomfort-score model or large statistical study.	⚡ partial	It supports feasibility of wearable measurements, not validation of our planned discomfort-risk gradient.

Evidence anchor. Kwak *et al.* is the closest comparator for wireless multimodal electronics, but not for printed manufacture, shear, humidity, or spatial fusion [19].

Q6.1 Comparator 2 – Paternò Personalised Sensorised Liner



Fig. 3 a, b CAD and Neoprene foam prototype of the new personalised liner with specific enclosures designed to embed the Hydrochron Data Loggers iButtons; c the two refined halves of the new personalised liner with the twelve embedded Hydrochron Data Loggers covered with the layer of breathable yet waterproof fabric; d the final personalised liner prototype

Published system. CAD geometry, cryogenically machined neoprene liner halves, twelve embedded iButtons, protective fabric, and the final personalised liner. Original Fig. 3 [13].

Technical architecture.

- Twelve DS1923-F5 Hydrochron data loggers measured temperature and RH. Each stainless-steel logger was 16.25 mm in diameter and 5.89 mm thick and was placed in a CAD-defined enclosure for direct skin contact.
- The non-waterproof loggers were covered by breathable waterproof fabric. Each was chamber-calibrated over 25–45 °C and 20–95%RH before use.
- Residual-limb and liner scans were landmark-aligned; the liner was split into two parts and cryogenically CNC-machined from closed-cell neoprene. The reported build took about 4 h and weighed 107.3 g.
- One transtibial participant completed 50 min rest, 30 min walking, and 50 min post-activity rest. Data were logged locally and downloaded through a 1-Wire USB interface rather than streamed as a real-time map [13].

Feature	Paternò Implementation	Status	Direct comparison with our 5.2 mm slab
True liner integration	Sensor pockets were created in a personalised liner derived from patient scans.	✓ shared	This is the closest precedent for preserving fit while embedding sensors. Our route uses repeated modular silicone slabs rather than a split CNC-neoprene liner; see Q2 Section 2.7 .
Temperature and RH	Twelve calibrated dual-function loggers at registered locations.	✓ shared	These are two of our four modalities and establish the need for chamber calibration and location identity.
Pressure and shear	Not instrumented in the reported liner.	✗ absent	Our pressure/directional-shear pair adds load transfer, cross-axis separation, and mechanical hysteresis risks that this system did not need to solve.
Sensor form factor	Rigid 5.89-mm-thick commercial logger in a blind enclosure.	⚡ partial	The thickness is close to our complete 5.2-mm slab, but our printed sensor is distributed within the slab instead of occupying a rigid local cavity.
Moisture protection	Breathable waterproof fabric protected the logger while allowing environmental exchange.	⚡ reusable	This is a functional precedent for a vapour-access/liquid-exclusion barrier, but H1/H2 must verify it for the printed IDE and exposed silver interfaces.
Manufacture	Scan alignment, CAD subtraction, split design, and cryogenic CNC machining of neoprene.	⚡ partial	The digital personalization workflow is reusable; the material, printing, cure, electrical routing, and slab-to-liner assembly are different.
Acquisition and mapping	Local logging and post-test download; graphs retain sensor location but are not a live surface reconstruction.	✗ absent	Our concept requires synchronized acquisition, a CAD coordinate manifest, and a real-time map with visible measured sites and uncertainty.
Human evidence	One participant and one liner architecture; comfort and thermal outcomes were reported.	⚡ partial	Valuable wearable-use evidence, but insufficient to validate universal thresholds or our combined discomfort-risk score.

Evidence anchor. Paternò *et al.* provides the strongest liner-integration and microclimate precedent. Its published limitations closely define what our pressure, shear, printed manufacture, and real-time mapping blocks add [13].

Q6.1 Comparator 3 – novel pliance-PMS Prosthesis System



Commercial system. Flexible socket-mounted sensor pads, acquisition software, and registered 3D pressure display from the official pliance-PMS prosthesis datasheet [30].

Disclosed technical architecture.

- Flexible capacitive pads are placed at selected anatomical regions and connected to a pliance-PMS analyser. The analyser can scan up to 1024 elements and transmit by Bluetooth or store data on an SD card.
- Listed pad examples contain 18, 45, or 90 sensels. Each sensel is 10×10 mm; the uncoated pad thickness is below 1 mm, elasticity is about 4%, and the disclosed pressure range is 10–600 kPa.
- The product literature reports acquisition up to 20,000 sensor samples per second, selectable matrix localization, 2D/3D display, and synchronization with video and EMG.
- A triblu calibration device applies homogeneous air pressure to all elements for simultaneous calibration. The 2019 datasheet also states that the product was not certified as a European medical product at that time [30].

Feature	pliance-PMS Implementation	Status	Direct comparison with our 5.2 mm slab
Pressure range	Calibrated capacitive pads covering 10–600 kPa.	✓ shared	The commercial range contains our 0–250 kPa Q4 target, but our four-sector printed cell needs its own calibration, saturation, and hysteresis evidence.
Spatial density	10×10 mm sensels; selected pads provide 18–90 elements and the analyser supports up to 1024.	✓ benchmark	This is a stronger pressure-density benchmark than the current sparse multimodal slab layout and exposes potential missed-hotspot risk.
Directional shear	No signed tangential-force channel in the disclosed PMS sensor set.	✗ absent	Our design aims to recover x/y shear from opposed capacitor pairs while maintaining normal-pressure separation.
Temperature and RH	Not measured by the PMS pads.	✗ absent	Our liner adds separate printed meander and IDE modalities, increasing clinical context but also calibration and synchronization complexity.
Geometry and comfort	Thin, flexible, elastic pads conform to curved sites but remain added measurement hardware at selected locations.	⚡ partial	Our slab is thicker but intended to become part of the liner structure, with sensor placement fixed by CAD rather than temporary pad installation.
Calibration	Homogeneous pneumatic calibration across all capacitive elements.	✓ reusable	The independent pressure reference and simultaneous channel release are directly reusable concepts for M1, although the printed stack and force-transfer geometry differ.
Mapping software	Matrix localization, 2D/3D pressure display, Bluetooth, and video/EMG synchronization.	✓ benchmark	This is the closest mature map benchmark. Our map must add four physical-unit layers, signed shear, confidence masks, and calibration metadata without averaging conflicting modalities.
Clinical Interpretation	Quantifies interface pressure for socket evaluation; the datasheet does not validate a multimodal discomfort score.	⚡ partial	Our display must remain an interface-condition map until participant-reported discomfort and clinically justified thresholds are added.

Comparison verdict: Each comparator already solves a substantial functional block. The defensible project distinction is the proposed combination of printed pressure, directional shear, temperature, and humidity on coordinate-registered 5.2 mm PDMS slabs with modality-preserving software. This is an engineering differentiation statement, not a patentability or clinical-superiority claim.

Q6.2 Challenges – Materials, Printing, and Sensor Physics

Evidence boundary. Two fabrication milestones are confirmed: the sensor artwork was printed successfully on PET as the accessible first-stage process check, and a 5.2 mm PDMS slab was molded. These achievements do not yet establish direct-on-PDMS print quality, a calibrated multimodal slab, or whole-liner durability. The challenges below are therefore stated as engineering questions with published causes, project-specific consequences, and linked validation routes—not as claims that the prototype has failed.

Challenge	What the literature establishes	Present design implication	Linked evidence route
PDMS wetting and solvent swelling	Oxygen-plasma treatment enabled continuous AgNP printing on PDMS, while the same printer settings produced an apparent profile of about 3.8 μm on PDMS versus 0.6 μm on PET; much of the difference was attributed to solvent-induced PDMS swelling [6].	PET confirms artwork transfer and printer operation only. It cannot predict trace height, spreading, resistance, or sensor calibration on molded PDMS.	Measure contact angle, line width, dry/equilibrated profile, continuity, and sheet resistance using the process controls in Q2 Section 2.5 and release gate Q5 Section 5.2 .
Silver adhesion, cracking, and stretch	Direct PDMS routes use surface activation, wider trace geometry, pre-stretching, or semi-embedded conductors to improve continuity under deformation [7, 8, 9].	Successful printing is not equivalent to a mechanically stable electrode. Compression, flexion, shear, cable pull, and cure history can change resistance or create intermittent opens.	Use witness traces and full sensor coupons; record resistance during bending and cyclic loading, adhesion, crack images, and recovery as linked in Q4 Section 4.3 and Q5 Section 5.6 .
5.2 mm force-transfer stack	The four-capacitor shear architecture obtains direction from differential capacitance; dielectric thickness, overlap, spacer geometry, normal preload, and viscoelastic hysteresis all affect the recovered vector [2].	The molded thickness is a geometric fact, not evidence that local pressure and signed shear reach the printed electrodes without attenuation or cross-axis coupling.	Calibrate normal load and signed x/y shear at several preloads, then quantify sensitivity, nonlinearity, hysteresis, drift, and axis coupling in Q5 Section 5.3 .
Temperature-channel stability	Printed Ag temperature meanders require trace-specific temperature coefficient, response-time, bending, humidity, and self-heating characterization [3, 4].	The meander may respond to temperature, strain, lead resistance, and changing thermal contact simultaneously after encapsulation.	Fit resistance–temperature calibration on the final stack; repeat it dry, humid, bent, and after cycling using Q2 Section 2.4 and Q5 Section 5.4 .
Humidity: vapour versus liquid sweat	Printed IDE sensors can measure RH, while embedded-liner work shows that a breathable waterproof barrier and chamber calibration are necessary for interpretable skin-interface microclimate data [5, 13].	An open IDE may saturate or leak under liquid sweat; an aggressive barrier may prevent liquid failure but slow or bias the vapour response.	Run controlled RH steps separately from metered artificial sweat; report response/recovery time, hysteresis, leakage, and post-drying recovery in Q2 Section 2.4 and Q5 Section 5.4 .
Hydrogel as a later interface variable	Hydrogel–PDMS bonding and sweat uptake are feasible, but water uptake depends on thickness and time and changes mechanics and transport [11, 12]. No reviewed study validates a hydrogel over this exact four-modality prosthetic slab.	A hydrogel may soften contact, but it may also store sweat, delay RH and temperature, redistribute pressure, alter friction/shear, or expose silver to water and ions. It is not the baseline architecture.	First validate bare PDMS H0; then compare H1 islands and the H2 windowed sheet against H0. The structure and rejection criteria are in Q2 Section 2.6.1 and Q2 Section 2.6.2 .

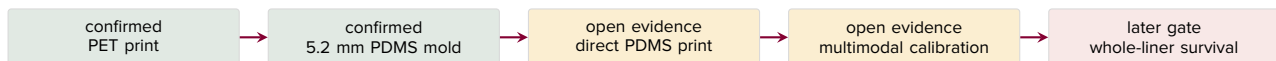


Figure Q6-2. Evidence progression for the material stack. Green marks documented project stages; yellow marks the next measurement gates; the final liner gate follows only after coupon and slab evidence. This sequence is consistent with the fabrication evidence reviewed in [Q2 Section 2.5](#).

Q6.2 Challenges – Integration, Mapping, Durability, and Use

Challenge	Evidence from comparable systems	Specific risk in this design	Required control and report link
Sensor identity and CAD coordinates	Paternò used scan-aligned anatomical landmarks and CAD-defined pockets; Kwak retained the identity of multiple selected sites [13, 19].	A correct value attached to the wrong slab, modality, or coordinate creates a convincing but false hotspot. Curvature and assembly can shift the printed site away from its nominal CAD point.	Create a permanent channel–slab–modality–coordinate manifest and perform blinded stimulus localization in Q5 Section 5.5 .
Sparse coverage and missed hotspots	Commercial pliance-PMS arrays disclose 10×10 mm sensels and up to 1024 elements; Turner used 144 pressure sensors arranged as twelve strips [30, 21].	The proposed multimodal slabs are intentionally sparser. A hotspot between sensors may be attenuated, shifted, or entirely absent, especially near slab boundaries.	Test centres, between-sensor points, and boundaries; report nearest-neighbour pitch, coverage mask, median/95th-percentile localization error, and density trade-offs in Q3 Section 3.7 and Q5 Section 5.5 .
Interpolation and false visual confidence	Turner demonstrated real-time 3D socket visualization using radial-basis interpolation, proving the display architecture but not that interpolated pixels are measured data [21].	A smooth colour gradient can hide sparse coverage and imply certainty between physical sensors. Combining different units into one colour also risks misleading averaging.	Display pressure, signed shear, temperature, and RH separately first; retain sensor markers, unsupported-region masks, method/version metadata, and confidence flags. See Q3 Section 3.5 and Q4 Section 4.6 .
Sampling and multimodal synchronization	Kwak used real-time multimodal acquisition, Turner sampled pressure at 200 Hz, while Paternò's microclimate loggers were downloaded after the activity protocol [19, 21, 13].	Mechanical events are fast; temperature and RH are slower. Unsynchronized streams can incorrectly associate a pressure/slip event with a later microclimate response.	Use one monotonic clock; preserve raw timestamps and modality-specific rates; define resampling windows explicitly and verify combined-condition timing in Q5 Section 5.5 .
Cross-sensitivity and compensation	Published sensors are normally calibrated by modality. The Kwak architecture suppressed shear sensitivity; the printed shear cell relies on differential responses; humidity and temperature outputs depend on their environments [19, 2, 5, 4].	Pressure, shear, bending, temperature, moisture, cable motion, and hydrogel swelling may change multiple electrical channels at once. A compensation model could overfit and remove real events.	Apply one controlled input at a time before combined tests; fit compensation on training runs and verify it on held-out combinations. Preserve raw and corrected values as specified in Q5 Section 5.5 .
Cyclic drift and whole-liner routing	Printed Ag/PDMS conductors can survive demanding stretch tests under published processes, but performance depends on ink, surface treatment, geometry, and loading mode [6, 7, 8].	Repeated compression, bending, donning, sweat, cure interfaces, connectors, and strain relief may change baselines even when an isolated trace remains continuous.	Repeat calibration after staged mechanical and wet/dry exposure; inspect cracks and adhesion; quantify zero shift, span change, hysteresis, and channel loss in Q5 Section 5.6 before the Q2 Section 2.7 integration step.
Phantom evidence and clinical meaning	Published systems support interface measurement and controlled socket evaluation, yet the human evidence in close comparators is small and a recent systematic review emphasizes limitations in using pressure alone for clinical decisions [19, 13, 14].	A phantom can verify loading, moisture, temperature, localization, and repeatability; it cannot prove perceived discomfort or tissue-injury risk. Thresholds based only on engineering judgment must not be presented as clinical truth.	Use the modular phantom protocols in Q5 Section 5.7 . Call the output an <i>interface-condition map</i> ; reserve <i>discomfort-risk</i> for later participant-reported and clinically reviewed validation.

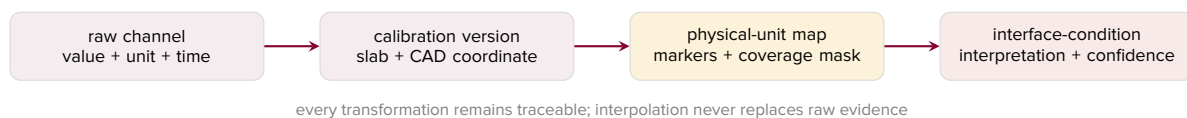


Figure Q6-3. System evidence chain. The software can only make a defensible spatial statement if channel identity, calibration, time, and CAD position survive every processing stage. The architecture synthesizes the published acquisition and mapping precedents [19, 13, 21].

Priority conclusion: The highest-risk path is not any single sensor principle. It is preserving calibrated, synchronized, correctly located four-modality information after direct PDMS printing, 5.2 mm mechanical integration, repeated loading, moisture exposure, and spatial interpolation. That path is tested progressively in Q5 rather than deferred to a complete liner.

Q6.3 Team Expertise, Ownership, and Remaining Gaps

The project already spans biomedical, chemical/material, and mechanical/ electronic work. The table records who owns each functional contribution; it does not imply that one member works in isolation. Interfaces between roles are explicit because printing, material selection, mechanics, circuitry, and software all affect the same calibrated signal.

Member	Discipline	Primary responsibility	Concrete project outputs and interfaces
Abdulrahman Kobeissi	Biomedical Engineering	System architecture; all printed-sensor CAD and print work; sensor proportions and placement; 5.2 mm silicone slab and circuit-layout design; software algorithms for calibration, interpolation, coordinate mapping, and interface-condition visualization.	PET print artwork and execution; CAD sensor/slab designs; multimodal placement logic; channel-to-coordinate manifest; processing and map pipeline. Interfaces with Hawraa on PDMS/hydrogel compatibility and Amir on electrical signals, sampling, and connector definitions.
Hawraa	Chemistry	Hydrogel and silicone material science; surface chemistry, bonding, moisture uptake, protection, and material compatibility.	PDMS and hydrogel formulation/process input; swelling, absorption, drying, bonding, and wet/dry compatibility criteria. Interfaces with Abdulrahman on sensor exposure/window geometry and with Amir on insulation, leakage, and material effects on the readout.
Amir	Mechanical Engineering	Physical sensor acquisition, mechanical test implementation, signal/data acquisition, and circuitry/readout design.	Loading and acquisition hardware; circuit topology; channel excitation and readout; sampling and synchronization implementation; force/application fixtures and connector/strain-relief input. Interfaces with Abdulrahman on the data schema and with Hawraa on encapsulation and wet-operation constraints.

Expertise not yet represented by a named specialist

Missing specialist perspective	Why it matters	How the team has compensated at this stage
Certified prosthetist/orthotist and rehabilitation clinician	Socket fit, bony landmarks, safe loading, donning constraints, clinical workflow, and interpretation of whether a measured condition is meaningful to a prosthesis user require clinical and prosthetic practice.	Placement and validation are grounded in prosthetic-interface literature and published liner systems [13, 19, 16]. Work remains on benches and phantoms; no human-safety, comfort, or clinical-risk claim is made. External clinical review is required before participant testing.
Printed-electronics process/metrology specialist	AgNP wetting, plasma activation, sintering, profilometry, adhesion, swelling, crack morphology, and production yield on PDMS are process-specific. PET success cannot certify them.	The team separated the PET print stage from direct-PDMS validation, adopted published process controls [6, 7, 8, 9], and specified coupons, witness traces, profilometry/microscopy, resistance tracking, and build records in Q5 Section 5.2 .
Human-factors, biostatistics, and regulatory/quality expertise	Discomfort thresholds, sample size, uncertainty, usability, traceability, participant protection, and the boundary between an engineering tool and a medical claim cannot be settled by sensor calibration alone.	The report predeclares test gates, uses held-out validation, retains raw and processed data with calibration versions, distinguishes pass/partial/fail/not run, and limits current language to an interface-condition map. Participant and clinical validation remains a separate reviewed stage; see Q4 Section 4.7 and Q5 Section 5.8 .

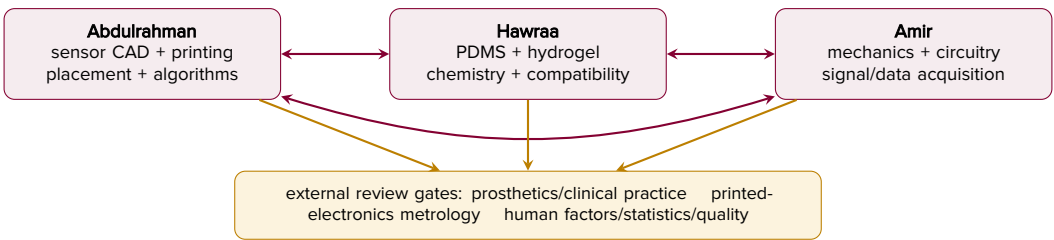


Figure Q6-4. Responsibility and review structure. Internal ownership covers the principal engineering blocks; the lower review gate identifies specialist input required before human testing or clinical interpretation.

Team conclusion: The principal prototype can be developed by the current multidisciplinary team. Literature-based controls and modular bench tests reduce the consequences of missing specialist expertise, but they do not replace external prosthetic/clinical, process-metrology, and human-validation review at the relevant design gates.

Q6.4 Integrated Liner Architecture

The consolidated design places the compact multimodal slabs exclusively on the inner, skin-contacting surface of the liner. The rolled-open view exposes the distributed sensing array while preserving a smooth silicone exterior. Signal routing remains enclosed within the liner wall, and the burgundy electronics enclosure is mounted externally on the lateral side.

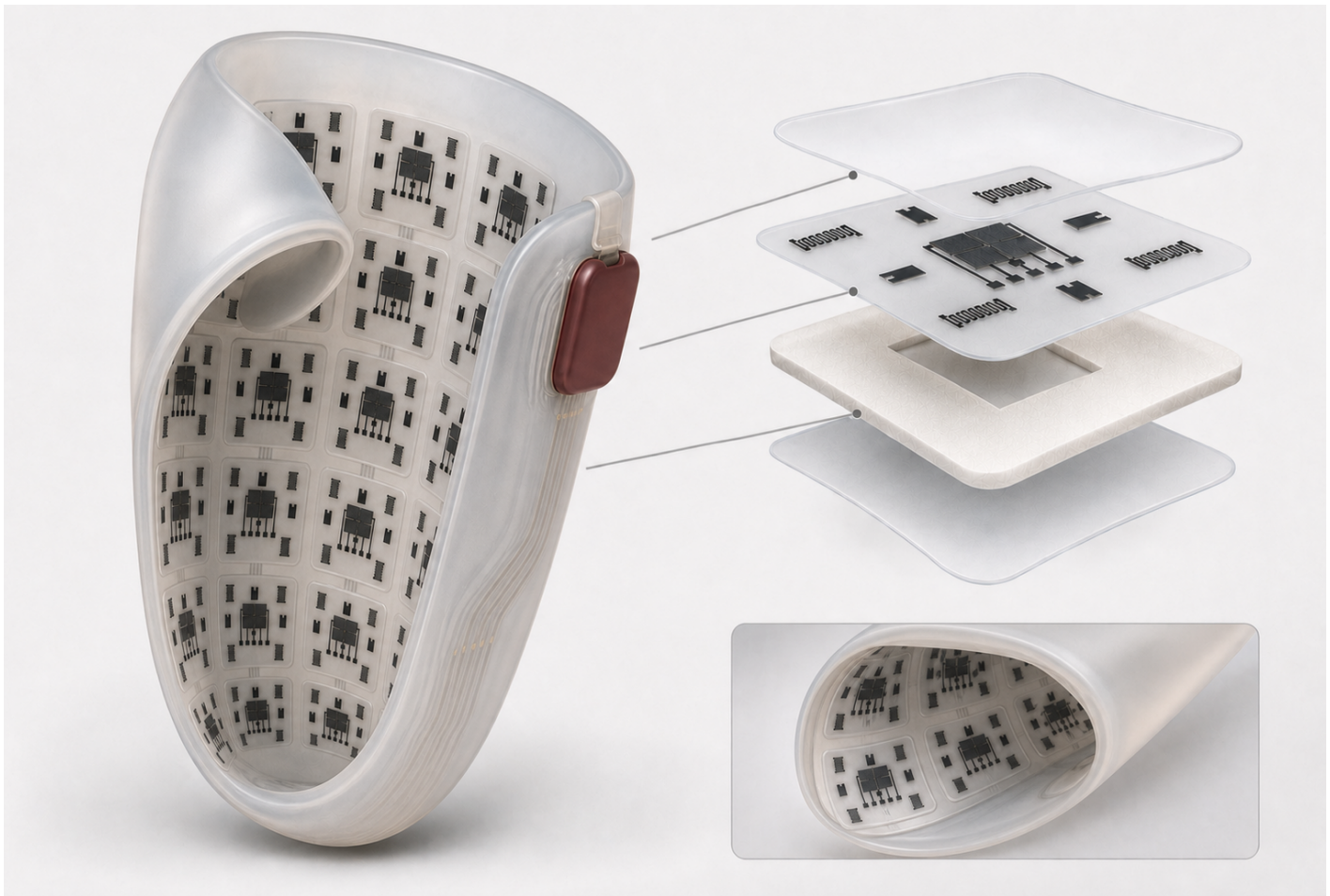


Figure Q6-5. Integrated smart-liner concept. The main rolled-open view shows the dense slab array on the skin-contacting interior and the plain silicone exterior with its side-mounted electronics enclosure. The exploded view identifies the layered slab construction, while the distal inset shows how the inner modules continue toward the closed end of the liner. This is a project design visualization rather than evidence of a completed whole-liner prototype.

Architecture represented: Inner multimodal sensing coverage; flexible spacing between complete slabs; protected in-wall signal routing; plain outer silicone; and an externally accessible lateral electronics pod.

Q6. Individual Assignment – Hawraa

Question :6
State-of-the-Art Technologies Comparable to the Proposed Design

1. Össur AeroFit™ Vented Liner–Socket System

The Össur AeroFit™ vented liner–socket system is a commercially available prosthetic interface designed to reduce moisture accumulation at the skin–liner interface. The system combines a vented silicone liner containing an integrated mesh with a vented socket that promotes both vertical and horizontal airflow during walking. This ventilation mechanism allows perspiration to escape from the socket, thereby reducing the warm and humid environment that commonly develops inside conventional prosthetic liners. In a randomized crossover clinical study involving individuals with transfemoral amputation, the vented system significantly reduced relative humidity by approximately 30% and improved users’ perceived sweating without compromising socket suspension, stability, or overall comfort. (Gnyawali et al. 2023)

Comparison with the proposed design. Unlike the AeroFit™ system, which relies solely on passive ventilation, the proposed smart prosthetic liner adopts a multifunctional approach. It combines a reinforced PAM–CMC double-network hydrogel to absorb and manage moisture, microencapsulated phase-change materials (mPCMs) to regulate temperature, and embedded flexible sensors to continuously monitor pressure, temperature, and humidity. By integrating both active microclimate regulation and real-time physiological monitoring, the proposed design aims not only to reduce moisture but also to improve thermal comfort and provide objective data for optimizing socket fit and residual-limb health.

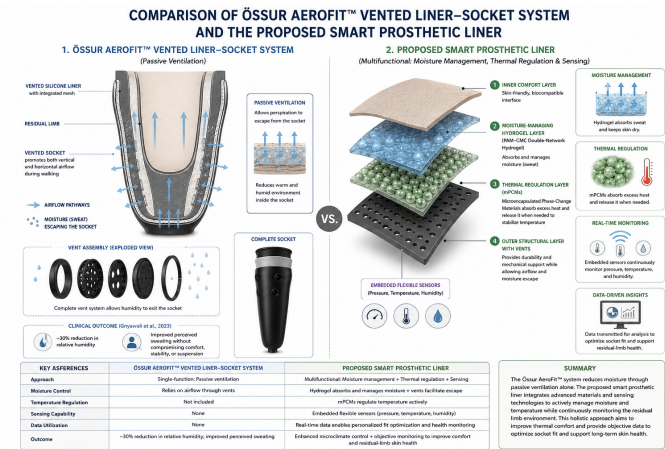
Gnyawali, Surya C., Jeffrey A. Denune, Bryce Hockman, Jóna Valgerður Kristjánsdóttir, Margrét Sól Ragnarsdóttir, Lava R. Timsina, Subhadip Ghatak, Knut Lechler, Chandan K. Sen, and Sashwati Roy. 2023. "Moisture Mitigation Using a Vented Liner and a Vented Socket System for Individuals with Transfemoral Amputation." *Scientific Reports* 13: 16557. <https://doi.org/10.1038/s41598-023-43572-2>

2. Roliner: Dynamically Adaptive Prosthetic Liner

Roliner is a novel soft metamaterial prosthetic liner designed to dynamically adapt to changes in the residual limb during daily activities. The liner is fabricated from silicone elastomers containing embedded millifluidic channels that can be pneumatically pressurized to modify the liner’s shape, volume, and mechanical properties in real time. Controlled through a portable electro-pneumatic unit and smartphone application, Roliner enables personalized adjustment of socket fit, helping accommodate residual-limb volume fluctuations caused by muscle atrophy, body weight changes, edema, or physical activity. Preclinical testing demonstrated improved comfort, prosthetic utility, residual limb health, and fitting while maintaining gait performance comparable to conventional prosthetic liners.

Comparison with the proposed design. While Roliner focuses on active mechanical adaptation through pneumatic actuation, the proposed smart prosthetic liner emphasizes passive microclimate regulation using a reinforced PAM–CMC hydrogel and microencapsulated phase-change materials (mPCMs), combined with embedded flexible sensors for continuous monitoring of pressure, temperature, and humidity. This approach avoids the need for pneumatic hardware while providing thermal regulation, moisture control, and real-time physiological monitoring within a simpler multilayer liner architecture.

Tanriverdi, Ugur, Guglielmo Senesi, Tarek Asfour, Hasan Kurt, Sabrina L. Smith, Diana Toderita, Joseph Shalhoub, Laura Burgess, Anthony M. J. Bull, and Firat Güder. 2025. "Dynamically Adaptive Soft Metamaterial for Wearable Human–Machine Interfaces." *Nature Communications* 16: 2621. <https://doi.org/10.1038/s41467-025-57634-8>



3- SmartTemp® Prosthetic Liner (by WillowWood)

The SmartTemp® prosthetic liner is a commercially available silicone liner designed to improve the thermal environment inside the prosthetic socket by incorporating phase-change materials (PCMs) into a conventional silicone liner. PCMs absorb, store, and release thermal energy during phase transitions, allowing the liner to buffer temperature fluctuations and delay heat accumulation at the skin–liner interface. In a double-blind randomized crossover clinical trial involving 16 individuals with transtibial amputations, the SmartTemp liner significantly reduced residual-limb skin temperature and perspiration compared with a conventional silicone liner following 25 minutes of cycling and a 10-minute recovery period. The reduction in heat and moisture improved the internal socket environment and may help reduce discomfort, improve suspension, and decrease the risk of skin complications associated with excessive perspiration.

Similar to SmartTemp, the proposed smart prosthetic liner incorporates microencapsulated phase-change materials (mPCMs) to regulate residual-limb temperature. However, the proposed design extends this concept by integrating mPCMs within a reinforced PAM–CMC double-network hydrogel, which provides simultaneous thermal regulation and passive moisture management. In addition, embedded flexible sensors continuously monitor pressure, temperature, and humidity, enabling real-time assessment of the skin–liner interface. Consequently, the proposed liner offers a multifunctional approach that combines climate regulation, physiological monitoring, and structural reinforcement within a single multilayer system rather than relying solely on PCM-based cooling.

Wernke, Matthew M., Ryan M. Schroeder, Christopher T. Kelley, Jeffrey A. Denune, and James M. Colvin. 2015. "SmartTemp Prosthetic Liner Significantly Reduces Residual Limb Temperature and Perspiration." *Journal of Prosthetics and Orthotics* 27 (4): 134–139.

Challenges Still Facing the Proposed Design.

Despite integrating several advanced technologies, the proposed smart prosthetic liner still faces a number of engineering challenges before clinical implementation. The successful integration of the reinforced PAM–CMC hydrogel, microencapsulated phase-change materials (mPCMs), embedded flexible sensors, and the outer structural layer into a single durable and flexible system remains a major challenge. In addition, the hydrogel must maintain its mechanical integrity under repeated compression, shear loading, and prolonged moisture exposure, while the embedded sensors must provide accurate and stable measurements during continuous bending and daily prosthetic use. Future development also requires transferring the printed sensors from the current PET substrate to PDMS to improve flexibility and wearer comfort, although reliable printing on PDMS is more challenging because of its low surface energy and reduced ink adhesion. Finally, the complete three-layer liner has not yet been fabricated or clinically validated, and therefore comprehensive mechanical, environmental, and user-based testing will be necessary to evaluate its long-term performance, durability, and effectiveness in improving prosthetic comfort and residual-limb health.

The development of the proposed smart prosthetic liner required expertise beyond the knowledge available within our team. One of the main gaps was advanced biomaterials engineering, particularly in designing and fabricating reinforced hydrogel systems incorporating microencapsulated phase-change materials (mPCMs). This expertise was essential to optimize the hydrogel composition, mechanical durability, moisture absorption, and thermal regulation required for long-term prosthetic use. A second area of missing expertise was prosthetics and orthotics, particularly regarding socket design, liner fabrication, and clinical evaluation of prosthetic interfaces. Such expertise is necessary to ensure that the proposed liner is compatible with existing prosthetic systems and meets user comfort and clinical requirements. Finally, expertise in flexible electronics manufacturing was needed to integrate printed sensors into a soft wearable substrate such as PDMS while maintaining reliable electrical performance.

To address these limitations, the design process relied extensively on published scientific literature, commercial prosthetic technologies, and expert consultation. Evidence from recent studies was used to guide the selection of materials and the overall liner architecture, while commercially available systems such as SmartTemp®, Roliner, and Össur AeroFit™ were analyzed to identify current design strategies and limitations. Because of the limited fabrication expertise and available resources, only the flexible sensor was fabricated as a proof-of-concept prototype on a PET substrate using a Voltera printer. The complete three-layer smart liner and sensor integration on PDMS were therefore proposed as future work requiring collaboration with specialists in biomaterials, prosthetics, and flexible electronics.

FINAL 3-LAYER SMART PROSTHETIC LINER DESIGN

Moisture Control • Temperature Regulation • Skin Health Monitoring

1 OUTER LAYER

Perforated medical-grade silicone / polyurethane

- Rigid socket friction
- Durable & wear-resistant
- Ventilation through perforations

2 MIDDLE LAYER

Active tough hydrogel core (Double-network hydrogel) reinforced with knitted nylon mesh

- Hydrophilic & thermoresponsive
- Sweat absorption & vapor release
- Microencapsulated PCM for heat regulation
- High tear strength & durability

3 INNER LAYER

Smooth skin interface

- Soft & biocompatible
- Low friction
- Protects sensitive skin

REINFORCEMENT WITH KEVLAR®

Kevlar (aramid) fibers integrated in hybrid textile reinforcement for maximum durability and controlled stretch.



Tear Resistance & Durability

High-strength Kevlar blended with nylon and spandex for tear resistance and elongation control.



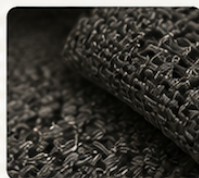
Stretch Control (Anti-Milking)

Allows radial expansion for donning, restricts longitudinal stretch to prevent skin pulling (milking effect).



High-Stress Applications

- Distal mesh matrix secures locking pins
- Wear-resistant patches in high-friction zones (kneecap, tibial crest)
- Woven into protective socks that shield the silicone liner from rigid socket friction

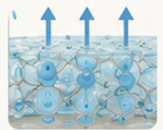


DUAL-ACTION CLIMATE CONTROL



FOR SWEAT (Moisture Management)

Thermoresponsive polymer (PNIPAM) acts as a hydrophilic sponge at baseline, drawing sweat away from skin and opening vapor pathways when temperature rises.



FOR HEAT (Temperature Regulation)

Microencapsulated Phase-Change Materials (mPCMs) absorb, store, and release heat. Paraffin beads melt between 31.5°C – 33°C to keep skin cool.

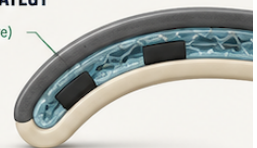


SENSOR PLACEMENT STRATEGY

DEPTH ZONE (Behind Hydrogel Core)

Sensors embedded behind the hydrogel core and anchored to the mesh.

Porous hydrogel passes moisture and heat data while keeping electronics away from skin.



SURFACE ZONE (Targeted Monitoring)



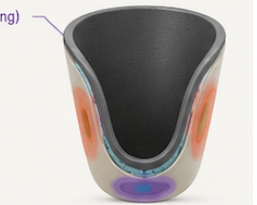
Moisture & Temperature

Sensors at distal end (bottom tip) where sweat pools.



Pressure & Shear

Sensors over bony prominences (tibial crest, fibular head) to monitor friction.

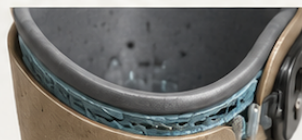


WIRING & DATA FLOW

All data lines run upward through protected channels to a processing unit outside the top rim of the socket.



SYSTEM OVERVIEW (IN SOCKET)



- Outer Layer (Protection & Friction)
- Middle Layer (Active Hydrogel Core)
- Inner Layer (Skin Interface)
- Sensors (Embedded)

References

- [1] A. Albrecht, M. Trautmann, M. Becherer, A. Rivadeneyra, and P. Lugli, "Multi-layer printed shear force sensor on flexible substrates", *ALLSENSORS 2018*, pp. 70–75, 2018.
- [2] A. Albrecht, M. Trautmann, M. Becherer, P. Lugli, and A. Rivadeneyra, "Shear-force sensors on flexible substrates using inkjet printing", *Journal of Sensors*, Art. no. 1864239, 2019. doi:10.1155/2019/1864239.
- [3] Q. J. Liew, A. S. Abd Aziz, H. W. Lee, M. W. Lee, H. F. Hawari, and M. H. Md Khir, "Inkjet-printed flexible temperature sensor based on silver nanoparticles ink", *Engineering Proceedings*, vol. 2, Art. no. 3, 2020. doi:10.3390/ecsa-7-08216.
- [4] J. Jäger, A. Schwenck, D. Walter, A. Bülau, K. Gläser, and A. Zimmermann, "Inkjet-printed temperature sensors characterized according to standards", *Sensors*, vol. 22, no. 21, Art. no. 8145, 2022. doi:10.3390/s22218145.
- [5] W. Ait-Mammar, S. Zrig, N. Bridonneau, V. Noël, E. Stavrinidou, B. Piro, and G. Mattana, "All-inkjet-printed humidity sensors for the detection of relative humidity in air and soil", *MRS Advances*, 2020. doi:10.1557/adv.2020.86.
- [6] A. Albrecht *et al.*, "Over-stretching tolerant conductors on rubber films by inkjet-printing silver nanoparticles for wearables", *Polymers*, vol. 10, Art. no. 1413, 2018. doi:10.3390/polym10121413.
- [7] J. Sun *et al.*, "Fabrication of bendable circuits on a polydimethylsiloxane surface by inkjet printing semi-wrapped structures", *Materials*, vol. 9, Art. no. 253, 2016. doi:10.3390/ma9040253.
- [8] J. M. Abu-Khalaf, L. Al-Ghussain, and A. Al-Halhouli, "Fabrication of stretchable circuits on polydimethylsiloxane (PDMS) pre-stretched substrates by inkjet printing silver nanoparticles", *Materials*, vol. 11, Art. no. 2377, 2018. doi:10.3390/ma11122377.
- [9] J. Abu-Khalaf, L. Al-Ghussain, A. Nadi, R. Saraireh, A. Rabayah, S. Altarazi, and A. Al-Halhouli, "Optimization of geometry parameters of inkjet-printed silver nanoparticle traces on PDMS substrates using response surface methodology", *Materials*, vol. 12, Art. no. 3329, 2019. doi:10.3390/ma12203329.
- [10] F. Adda Cherif *et al.*, "Comparative study by FEM of different liners of a transfemoral amputated lower limb", *Scientific Reports*, vol. 15, 2025. doi:10.1038/s41598-025-15974-x.
- [11] H. Yuk, T. Zhang, G. A. Parada, X. Liu, and X. Zhao, "Skin-inspired hydrogel–elastomer hybrids with robust interfaces and functional microstructures", *Nature Communications*, vol. 7, Art. no. 12028, 2016. doi:10.1038/ncomms12028.
- [12] L. Lu *et al.*, "Sweat-sensing patches with integrated hydrogel interface for resting sweat collection and multi-information detection", *Biosensors*, vol. 15, Art. no. 342, 2025. doi:10.3390/bios15060342.
- [13] L. Paternò, V. Dhokia, A. Menciasci, J. Bilzon, and E. Seminati, "A personalised prosthetic liner with embedded sensor technology: a case study", *BioMedical Engineering OnLine*, vol. 19, Art. no. 71, 2020. doi:10.1186/s12938-020-00814-y.
- [14] L. Armitage, K. Cho, T. L. Sparling, T. Sleeman, G. Alici, and M. Sreenivasa, "The use of residual lower limb-prosthetic socket interface pressure measurements to inform clinical decisions: a systematic review", *Journal of Biomechanics*, Art. no. 113187, 2026. doi:10.1016/j.jbiomech.2026.113187.
- [15] J. E. Sanders, C. H. Daly, and E. M. Burgess, "Interface shear stresses during ambulation with a below-knee prosthetic limb", *Journal of Rehabilitation Research and Development*, vol. 29, no. 4, pp. 1–8, 1992. doi:10.1682/JRRD.1992.10.0001.
- [16] J. Tang, L. Jiang, M. McGrath, D. Bader, P. Laszczak, D. Moser, and S. Zahedi, "Analysis of lower limb prosthetic socket interface based on stress and motion measurements", *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 236, no. 9, pp. 1349–1356, 2022. doi:10.1177/09544119221110712.
- [17] M. Zhang and C. Roberts, "Comparison of computational analysis with clinical measurement of stresses on below-knee residual limb in a prosthetic socket", *Medical Engineering & Physics*, vol. 22, no. 9, pp. 607–612, 2001. doi:10.1016/S1350-4533(00)00079-5.
- [18] K. Woo, J. Tran, and N. Waters, "Novel smart sensor platform for monitoring multiple pressure injury risk factors: a feasibility study in a post-acute care facility", *Surgical Technology International*, vol. 41, pp. 37–44, 2022. doi:10.52198/22.STI.41.WH1602.
- [19] J. W. Kwak *et al.*, "Wireless sensors for continuous, multimodal measurements at the skin interface with lower limb prostheses", *Science Translational Medicine*, vol. 12, no. 574, Art. no. eabc4327, 2020. doi:10.1126/scitranslmed.abc4327.
- [20] A. J. Yu, R. Z. Gao, P. S. Lee, C. Mele, D. Dittmer, A. Schirm, C. L. Ren, and J. Y. Tung, "Soft robotics-inspired sensing system for detecting downward movement and pistoning in prosthetic sockets: a proof-of-concept study", *Prosthetics and Orthotics International*, vol. 48, no. 5, pp. 519–527, 2024. doi:10.1097/PXR.0000000000000302.
- [21] S. Turner, S. Jain, A. Patel, M. O. Hopkins, and A. H. McGregor, "A visual feedback tool for quantitative pressure monitoring in lower-limb prosthetic sockets", *Prosthesis*, vol. 3, no. 4, pp. 394–405, 2021. doi:10.3390/prosthesis3040035.
- [22] J. Quinlan, J. Yohay, V. Subramanian, B. Poziembo, and S. Fatone, "Using mechanical testing to assess the effect of lower-limb prosthetic socket texturing on longitudinal suspension", *PLOS ONE*, vol. 15, no. 9, Art. no. e0237841, 2020. doi:10.1371/journal.pone.0237841.
- [23] J. Tabor, T. Agcayazi, A. Fleming, B. Thompson, A. Kapoor, M. Liu, M. Y. Lee, H. Huang, A. Bozkurt, and T. K. Ghosh, "Textile-based pressure sensors for monitoring prosthetic-socket interfaces", *IEEE Sensors Journal*, vol. 21, no. 7, pp. 9413–9422, 2021. doi:10.1109/JSEN.2021.3053434.
- [24] L. Paternò, A. Z. Zaidi, M. G. Polizzotto, S. Dalmiani, D. Helsloot, S. Heikens, E. Gruppioni, and A. Menciasci, "Smart transfemoral prosthetic socket with motorized cable-driven system", *Advanced Intelligent Systems*, vol. 7, Art. no. 2400995, 2025. doi:10.1002/aisy.202400995.

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- [25] J. Phillips, A. Nagpal, and A. Azhari, "A biofidelic mock residual limb for prosthetic socket testing", *Canadian Prosthetics & Orthotics Journal*, vol. 8, no. 2, pp. 1–9, 2025. doi:10.33137/cpoj.v8i2.45759.
- [26] M. McGrath, K. C. Davies, A. Gallego, P. Laszczak, J. Tang, S. Zahedi, and D. Moser, "Using a sweating residuum/socket interface simulator for the evaluation of sweat management liners in lower limb prosthetics", *Canadian Prosthetics & Orthotics Journal*, vol. 4, no. 1, Art. no. 3, 2021. doi:10.33137/cpoj.v4i1.35213.
- [27] L. Greenspan, "Humidity fixed points of binary saturated aqueous solutions", *Journal of Research of the National Bureau of Standards—A: Physics and Chemistry*, vol. 81A, no. 1, pp. 89–96, 1977. NIST full text.
- [28] A. Noll, O. Rinderknecht, and P. Beckerle, "Systematic experimental assessment of a 2D-motion sensor to detect relative movement between residual limb and prosthetic socket", *Sensors*, vol. 18, no. 7, Art. no. 2170, 2018. doi:10.3390/s18072170.
- [29] L. De Waele, F. Gooijers, and D. Mantini, "Smartphone-based automated photogrammetry for reconstruction of residual limb models in prosthetic design", *Sensors*, vol. 26, no. 4, Art. no. 1251, 2026. doi:10.3390/s26041251.
- [30] novel gmbh, "pliance-PMS prosthesis system", product datasheet, Apr. 2019. Official product datasheet (accessed July 30, 2026).
- [31] Tekscan, Inc., "F-Socket System: pressure mapping in prosthetics", product webpage. Official product page (accessed July 30, 2026).